

Research Methods in Education (EDU 407)

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Lecture No.1

Topic 1: Introduction to Research Methods

Need of Research Methods

Whenever you face a challenge/problem in life or to find answer of a question its natural to search for solution through gathering knowledge or information. We draw conclusion and made decision on the basis of the knowledge or information collected through searching. So research is connected to our life in anyway where we gather information to find solution of a problem and made decision on the basis of that problem. Here are some points to explain the need of research in Education:

- Research is in your instinct.
- Educational Research is a systematic process of inquiry and exploration of issues and problems in the field of education, with an aim to find possible solutions.

- This course will only give exposure regarding use of research in teaching and learning.
- You will be exposed to philosophical, sociological, economic, and political foundations of education in your program to learn about challenges related to students, teachers, the process of teaching and learning, and their outcomes.
- This course aims to provide teachers with essential knowledge for classroom inquiry and importance of practice, hence the focus on action research.

All the above mentioned points explain the need of research methods in education e.g. to solve all the problems teacher and student face in classroom and to make teaching and learning process more successful.

Areas of Research

Typically, in this course following four areas of research are going to be addressed:

1. Foundations of Research

- i. Introduction to Research in Education
- ii. Selecting and Defining a Research Problem
- iii. Reviewing the Literature
- iv. Research Philosophies and Approaches

2. Process of Research

- i. Research Designs/ Strategies of Inquiry
- ii. Sampling Techniques

- iii. Selecting Research Instrument(s)
- iv. Development & Administration of Research Tool(s) Collecting.
- v. Analyzing and Interpreting

3. Data Analysis and Interpretation

In this section you will learn different methods to collect data using different tools, to use that data and to draw results and conclusions from the collected data.

- i. Quantitative Data
- ii. Qualitative data

4. Reporting Research

In this section you will learn how to deliver the findings of your research to people, how to write the Dissertation and Research report. You will also learn to disseminate findings to ordinary people using different formats in a user friendly form.

- i. Research Ethics
- ii. Research Proposal
- iii. Research Report

At the end of this course students will learn all these skills related to research mentioned above in areas of research.

Topic 2: Sources of Knowledge

What do we mean by sources of knowledge? Have you ever think that what are the sources and tools through which we acquire knowledge? This topic is particularly about those sources of knowledge. We need to ponder on the following questions:

- How we know, what we know?
- What tools of knowing are available to us?
- Can we trust our ways of knowing?

Almighty has bestowed us with the five senses which we utilize as tools for acquiring knowledge. These five senses or sources are:

1. **1 Hearing:** It is one of the major source which we utilize to get knowledge through hearing. For example, we learn through listening radio and T.V programs, discussions with friends and teachers.

2. **2. Smell:** It is the second major source of knowledge though which we recognize the characteristics of anything and differentiate as good and bad things based on their smell.

2.3 Taste: Thirst source or tool of knowledge is the sense of taste. Our tongue conveys the information of different tastes e.g sweets, salty, and bitter and we differentiate among them based on their different attributes.

2.4 Touch: The fourth source is the touch which help us to recognize the things based on their texture e.g soft, hard or temperature e.g hot or cold.

Examples: We have water with different temperate in three different tubs. In the first tub, we have iced water. The second tub is full of tap water with normal temperature and the third one is full with hot water. You may feel the tap water hot after putting your water in iced water and you may feel the tab water as cold after putting your hand in hot water although the temperature of tap water is not changing. This proves that our sense of temperature is relative and subjective.

2.5 Vision: The last source of knowledge is our vision. Our eyes see many things and make differences among different types of things and classify them based on those differences for example in the colors, shapes, distance and dimensions.

After discussing the sources now the next question is ‘Can we trust our ways of knowing?’ ‘Do they deceive us?’

Examples:1 Let us suppose you have a glass in-front of you with full of water and a pencil in it . It seems that the pencil in the glass is broken or bent in its center. However, this is not the case. The pencil is not broken nor bent which pointed towards our wrong perceptions.

Examples 2: Same like, when we travel in the hot weather, we feel the water on the road which do not exist in the reality. Actually, the hot air around the road creates such image of water on the surface of the road. These examples reflects how your vision can deceives you.

Examples3: In another such example, you can see three lines on the screen with arrows at their end. You may feel that the center line is longer in the length as compare to other two which is incorrect. Actually all three lines are equal in their length and you can verify by measuring them. It is the illusion of your sight which proves that our senses have few limitations and we may sometimes perceive wrong images due to those limitations which is not correct.

Examples 4: You can see a picture of boiling water with bubbles in the slides. You may also perceive this picture in another way as a pot full of water with small pieces of ice instead of bubbles. Which perception is right and which is wrong in this case?

Examples 6: A picture which I perceived as a plant with full of white spikes. But when I deeply observed the picture, I found those were not actually spikes rather small threads of cotton which gives an impression of spikes. This is another example that how your senses deceives you.

Examples 7: A picture of a piece of citric fruit depicts that it may be a sour fruit due to having green skin. Usually it is perceived that fruits with green skin are not fully prepared and have sour tastes. It can also be a deception as many fruits with sweet and good tastes having green skin exists in different parts of the world.

Examples 8. You have to check different perfumes with different fragrances. When you are asked to differentiate by smelling them in small quantity one after other at the same time , you may loose your sense of differentiation in those fragrances for the time being. You may feel all the fragrances as same and you cannot differentiate among them unless and until you may take a sufficient time in between the two smells.

In short, we conclude that senses are the sources of knowledge but do not provide you with the authentic knowledge in every case due to their limitations.

Topic 3: Sources of Knowing- Experience

In last lecture we discussed about tools of knowing to [acquire](#) knowledge; now we will discuss [about the sources](#) to use these tools to acquire knowledge. In today's lecture we are going to discuss about module three which is sources of knowing. The six sources of knowing are following:

1. Experience
2. Authority
3. Revelation

4. Deductive Reasoning
5. Inductive Reasoning
6. Scientific Method

There are many sources of knowledge but educational researchers choose six sources of knowing as these are most useful and acknowledged sources of knowing. All the other sources of knowledge are either a reworded form or in another way similar to these sources.

We will discuss these resources of knowledge keeping two points in view:

- a. Definition of the source of knowing
- b. shortcoming of the source of knowing.

Lets discuss the sources of knowing in detail.

1. Experience

It is personal experience of human beings. You can find the answers to many of the questions you face. For example you want to know about a question erupts in your mind or a problem you're facing. All the knowledge gathered in this world or all the knowledge we own as individuals is gained through experience. So, the source that use in these two methods is experience. If you recall your knowledge you'll get to know that a lot of knowledge that you gain is from experience either its direct experience or indirect experience. All the knowledge that we have is the result of these experiences.

For example if you put your hand in fire it'll burn that is something you learn by your own personal experience (direct experience) or by listening to someone else (vicarious experience). Much wisdom from generation to generation is the result of experience. So, experiences can be defined in two ways:

I. Direct Experience:

The first method to find answer or solution is to learn from your personal experience or you'll do that thing to know better about problem known as direct experience.

II. Vicarious Experience:

The second method is to observe someone else doing that thing, facing that same problem or finding answer to the same question known as vicarious experience. In this method you learn from someone else experience. For example we don't know how much hot surface of sun is or we can't see atom structure. We only know about the temperature of surface of sun or structure of atom because of people who has knowledge about it.

Challenges of knowledge through experience

There are also some challenges of knowledge gained through experience. It means that if you gain knowledge through experience so you have to be careful about some points.

- Two people will have very different experiences in the same situation. It means when two people experience something in the same situation the outcomes could be different for both of them. So, the subjectivity of the person who is experiencing is a big challenge to judge the knowledge gained through experience.
- You need to know things that you as an individual cannot learn by experience. When you listen to other experience you have a specific use of that knowledge that will affect the understanding you gain through listening which could be different from the outcome of the person that experiences it. So, the subjectivity of the person who is listening to someone else experiencing is a big challenge to judge the knowledge gained through experience.

However, experience is a source of knowledge through which we gain a lot of portion of knowledge through experience.

Topic 4: Sources of Knowing- Authority

What do we mean by authority?

We seek knowledge from someone who has experiences, and we don't have some other source of expertise. For example:

We consult with a Physician with health questions.

I understand that we learn from our own experiences however, sometimes we face such cases in life for that we don't have the capacity to solve so we refer to such sources which are considered authority in those matters who are considered experts. For example, when there is a health problem, instead of learning something by ourselves, we always go to the physician and go away to a doctor who is considered expert in this field.

We consult with a Lawyer for court related questions.

In the same way, when faced with problems related to the courts, I always refer to a lawyer and contact them as he knows the subject.

We consult with Bureau of statistics for population in country.

Authority does not necessarily mean a person, it can be an institution, for example, if I want a information about the population of Pakistan, then instead of contacting a person, I will go after the bureau of statistics and may get exact number. Bureau of statistics is not the name of a person rather it is the name of an institution, so the authority is not a person only, in that case, it is also an institution,

We consult with a Dictionary for accepted pronunciation of word.

If I don't understand the meaning of a word, I don't understand it, I will consult with a dictionary for its pronunciation, spelling and meaning. In this case, dictionary is the authority.

As a beginning teacher, **we consult with an** an experienced one for suggestion.

Authorities can be wrong. People often claim to be expert in a field when they do not really have the knowledge to back up the claim.

You may find authorizes disagree among themselves on issues, indicating that their authoritative statement is often more personal opinion than facts.

In the same way, if you are a beginning teacher, you finish reading here, you start teaching somewhere and you encounter problems or challenges in the classroom, you can contact them directly to your seniors that how they handle such situations and I take them as authority.

Hens you need to keep in mind that whenever we talk about authority It will not be the only person who will tell anything. There may be institutions and a book or any authentic source you may get the relevant information.

What type of challenges we can face while accepting authority as a source of knowledge?

The major shortcoming in this source of knowledge is that anybody who is recognized an authority in may not be the expert in that particular field. And if I contact that particular body to get knowledge, it will harm my basic premises regarding the concept.

For example I have selected the dictionary to look the pronunciation of a word which is not famous to tell the pronunciation then my knowledge will definitely be doubtful. Authority itself is a subjective thing. For example, different authorities have different opinions regarding their related fields e.g. religious scholars, social experts. So we must consider these differences while accepting any authority as a source of knowledge.

Lecture 2 Introducing to Research Methods**Topic 5: Sources of Knowing- Revelation**

Let's continue our talk on sources of knowledge. The third source of knowledge is 'revealed knowledge'. Revealed from such a source which is beyond our intellect. It cannot be part of our observation if you share it and not only from the perspective of Islam.

Definition of Revelation

It is defined as 'knowledge that cannot be attained through philosophical, or the merely human, powers of reason or inquiry, but through God's revelation in scriptures'.

All the things that we have acquired are through our senses. However, the case is different in revealed knowledge. You cannot get it by the skills you've been given. Revealed knowledge is conveyed to us through Books and the authority of the deliverer is of critical importance here. The type of authority described here is different from the ordinary and worldly meaning of authority. The authority involved in this case is of higher level and without the margin of mistake.

Revealed knowledge it is conveyed to us in the form of scripts in the form of revelation in the form of a book and we use this knowledge for our guidance. This knowledge is conveyed to us as authentic and correct and we have to obey it. Sometimes we may face difficulties in understanding and the interpretation of this knowledge for example think over this sentence:

'A conclusion in search of evidence.'

We may not find any evidence, reason or logic related the given knowledge and we rely on our belief which acts as a sources of strength in such case. Because sometimes we are unable to understand the revealed knowledge by employing our rationality or may be our wisdom is limited and insufficient to cater the reasons and evidences related the revealed knowledge. In the case of revealed knowledge, the sender is the ultimate authority to whom we complete trust and we take the given knowledge as it sent to us without any question and without any doubt.

Topic 6: Deductive reasoning

Deductive reasoning is a thinking process in which one proceeds from general to specific and get knowledge through logical arguments. An argument consists of several statements standing in relation to one another. The final statement is the conclusion, and the rest, called premises, offer supporting evidence. It lets researchers deduce from existing theory what phenomena they should observe.

We take a theory or an established premise to apply on a specific situation to draw the conclusions. For example:

There is an agreement that all elders are wise. If I accept this as my major premise than I will deduce that Imran is an old man so, he is wise.

“All elders are wise” (major premise)

“Imran is elder” (minor premise)

“Imran is wise” (conclusion)

To arrive at true conclusions, you must begin with true premises. Deductive reasoning is the method to learn through logic and reasoning. Deductive reasoning can organize what people already know and can point out new relationships. However, one must begin with the true premises to arrive at the conclusion and major premises must be factually proven and undoubtedly correct.

Deductive reasoning can organize what people already know and can point out new relationships. It is not sufficient as a source of new knowledge. Because I can make premise only based on the already existing knowledge. Minor premise is also based on the existing knowledge or information so there is no new starting point to create knowledge. Therefore, deductive reasoning solely cannot serve as a base for research work as the research is conducted to know the unknown things. This method has its limitations as it does not regenerate new knowledge, new relationships and new causation. Hens we conclude that deductive reasoning can be be good and logical way of thinking but do not serve as base for research independently.

Topic7: Inductive reasoning

We have talked about deductive reasoning and concluded with the point that deductive reasoning is a logical process of thinking process which proceeds from already known and ends on the same. So it

can become the base of the research. For its complete understanding, we will talk about a related or connected source of knowledge 'Inductive reasoning'.

We take a general premise and specify it for a particular case in deductive reasoning. Inductive reasoning is the reverse of deductive method in which we make generalization on the bases of or with the combination with so many small experiences.

Example:

We can see the difference between the deductive and inductive reasoning in the following example:

Deductive	Inductive
Every human being has two eyes	All human that I ever saw had two eyes
All men are human being	Therefore every human being has two eyes.
Therefore, every man has two eyes	

What challenge is involved in inductive reasoning?

In the first example of inductive reasoning the word 'All' actually do cater the literal definition of all. Rather 'all' represent the sample in this example which is called an imperfect induction.

Imperfect induction.

Imperfect induction is a system in which you observe a sample of a group and infer from the sample what is characteristic of the entire group group.

Hence, the observations and claims based on the imperfect induction cannot be completely accurate to claim that nothing is possible beyond my observations and claims. Because there is the possibility one has observed many people with one eye where the other is damaged or there may be many people who have loosed vision from both of the eyes.

Specific examples are countless and unlimited and it is impossible to observe all of them for anyone. Therefore, one can form his premise based on few of them which he had gone through and can claim that all 'all human beings that I have ever saw have two eyes, thus I conclude that human beings have

two eyes'. such premises are accurate in general but one cannot guarantee that this premise is always accurate.

Going back to the comparative view of deductive and inductive reasoning, we analyse that deductive reasoning involves the general to specific and the induction is the reverse where one observes enough of specific examples and concludes that the rests of the cases likely to be similar to those he has already watched or experience.

Here we conclude that the combination of the both, deductive and inductive reasoning can be used in research process despite that both has certain limitations.

What is deductive reasoning? (3) (comprehension

We take a general premise and specify it for a particular case in deductive reasoning. Inductive reasoning is the reverse of deductive method in which we make generalization on the bases of or with the combination with so many small experiences.

Topic 8: Sources of Knowing (Scientific Method)

Today we are going to discuss about the scientific method. The scientific method is based on the usage and combination of both (previously discussed sources) inductive and deductive methods. The learning process starts with the inductive and move towards the deductive method is called, the scientific method. Scientific method is based on the defined steps which starts from the identification of the problem and reach to the conclusion.

Steps of scientific method:

1. **Observation:** From where problems arise? Being a teacher I used to come across with the students' problems on daily basis. I try my best in teaching but still there are some students who do not pick it up what I teach them. My such observation becomes the starting point that motivates me to understand their problem and to search for its solution
2. **Hypothesis formation:** Hypothesis formation is the next step after the problem identification in the scientific method. Whenever I used to think about any problem, I also think about its possible solution which is natural. Those possible solution does not necessarily accurate. There is a common example as you think that you have to go somewhere for shopping. Soon as you think about that

specific place, automatically you will also think about the sources to reach that specific location. If there are multiple sources, likely you will think about all those resources and choose any one. That chosen resource may not necessarily be the best. You may think it as the best option based on your previous experiences but it is still an assumption which will be confirmed or rejected after you experience it. It is called hypothesis because it is a supposition to solve any problem.

How you do form this hypothesis or supposition? It is based on your previous experiences or knowledge you already have about the problem in hand. If you do not have any related experience or knowledge, then you may prepare a plan that how the issue can be resolved. These resources become the base for the hypothesis which needs to be tested.

What is meant by hypothesis testing?

3. **Experiment:** Let's remind previous example in which you have decided to go to a market and you think about the shortest route to shortage of time. You may decide to choose the route A by assuming it as the shortest route. After travelling on that route, you may be able to conclude either it was right choice to save your time or not? This testing of hypothesis is mentioned as experiment in slides but experiment is not needed to test the hypothesis in every case. There are other ways to test the hypothesis which we will discuss further. We are discussing about experiment as a source of hypothesis testing. Experiment refers to the methods which is used to test the hypothesis.

4. **Conclusion:** The data generated after using this method will lead you to the conclusion whether the problem is solved or not. At times, it is resolved but at times it may not and you have to get back to theory again.

5. **Theory/ Law:** Theories are made after the rigorous experimentation which ultimately becomes the base of a law.

Difference between theory and law: Scientific process is a cyclic process of a knowledge generation and the end outcome of this theory is the theory and law.

Theory refers to all those major premises that can be proved logically and mathematically but you cannot experiment it. For example, we talk about the theory of Einstein, not about the law of Einstein. Same like, you talk about the Newton's law of movement not the theory.

Einstein theory explains when any object is moved with the square of the light of speed, the mass gets converted to the energy. He had mathematically proved his theory which we understand and accept although we cannot verify it because we are unable to experiment it. We still believe its truth because all the related arguments and logic are true as per best of our knowledge of mathematics.

As compare to theory, laws are verified after practical experimentation and not limited to the logic and reason only.

For example, Newton's third law that every action has its reaction can be experimented. His second law is that $F = ma$, or net force is equal to mass times acceleration. A larger net force acting on an object causes a larger acceleration, and objects with larger mass require more force to accelerate which is quite right and you can test it in the laboratory.

These laws can be proven through experimentation in the laboratories and the theories are theoretically provable and the reasons behind are logically justifiable but not practically testable.

So, the scientific process ended up in any one of these either in the form of theory or in the form of law.

Limitations of the scientific process:

The scientific method has certain limitations which are:

Deliberately falsifying results/ Systematic manipulation bias: Prior confidence in the hypothesis can affect accuracy of observation and interpretation of results. Conclusion in such cases is not the actual conclusion rather, that is the manipulated conclusion. So, the integrity or fairness of the research towards the process is the biggest challenge in the scientific process. Usually people try to manipulate the result to prove their hypothesis however, this threat can be minimized and resolved through training of the researcher.

Data Interpretation: constrained by human ability to interpret the result. Wrong interpretation can lead to wrong conclusions.

It is another situation is when researcher do not manipulate data but tilted the situation while interpreting the data. You might have listened a very common saying that no one can present as much logical lies as a statistician can who cannot be refuted because they are connected with the data. It is because the same set of numbers can be interpreted in many different ways.

Hens interpretation can be another flaw of this method which can be overcome by training.

By concluding our discussion, I must say that all we do not intend to prioritize any of the source to another that we have discussed here. We don't want to define which is bad and which one is good rather these all are important and necessary. But there is only one reason to stress over the scientific which is its verify ability.

If we discuss about other sources, I do believe on revealed knowledge that it is authentic but I cannot verify it. I can learn from others experiences but I cannot justify the accuracy of their experiences. Authority has certain other problems. Scientific method is given preferences in research just because its quality of verify ability. It does not claim the truthfulness but provides you the freedom to verify the conclusion presented by someone through following the same procedure or steps.

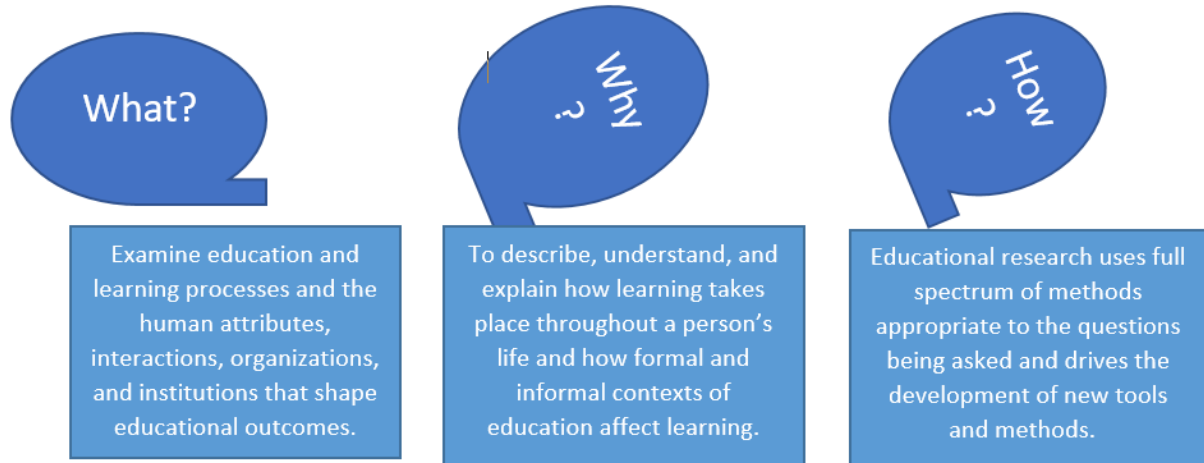
Write the steps of the scientific method. (3)

- 1. Observation, 2. Hypothesis formation, 3. Experimentation, 4. Conclusion, 5. Theory/Lawa**

Lecture 3 Process of the Research

Topic 9: Classification of Educational Research

Today we are going to discuss about the educational research. We take a start from the definition of educational research. All the terms we frequently use in our life are difficult to define because they have countless definitions in the literature and each single of those is different in its scope. I am dividing the definition of this term in the following three parts by ensuring that it should not be too brief to understand nor it be too complicated for you to grasp.

Defining Educational Research:**Source: American Educational Research Association (AERA)**

I try to sum up this definition as:

To study the learning processes in human beings along with all the attributes in order to measure the outcomes is called educational research. Educational research is conducted in both, formal and informal settings. Educational research utilizes multiple scientific methods to in order to ensure the credibility of the results so that we may minimize the learning challenges of the people through the

Topic10: What is not research?

Today, we will analyze that whether all those things for which use the term of research

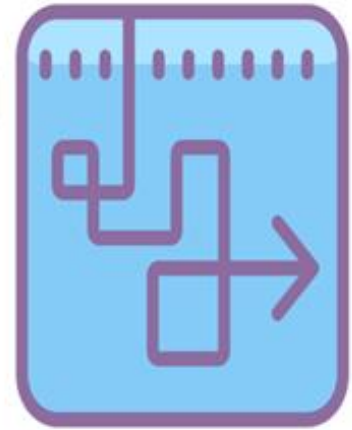
are genuinely falls under this term or not? It is important to differentiate because when we use the term RESEARCH in professional meanings, it is something different. Sometime, people use this term randomly in their day to day life which does not refers to the actual research. In such a scenario, students can confuse and misunderstand the term therefore it is necessary to differentiate it with the things which are not actually research.

What is not research?

gathering
information.



transporting facts from one location
to another



digging around for hard-
to-locate information.

Although all the above mentioned tasks are done by the researcher and it can be a part of his research however, the whole process of research cannot be only based on these three tasks. So, it is important for you to analyse your topic against it while doing research.

Now the question is **What discriminates research?**

Research follows a proper mechanism and based on the following steps:

What Discriminates "Research"?



Research is a cyclic process which starts from the problem, following by the Specifically articulated goals of the research, dividing the goals into more manageable sub problems, formulating hypothesis, developing a research plan and collecting, organizing and analyzing the data against the defined research objectives, only then it is research otherwise it is not research.

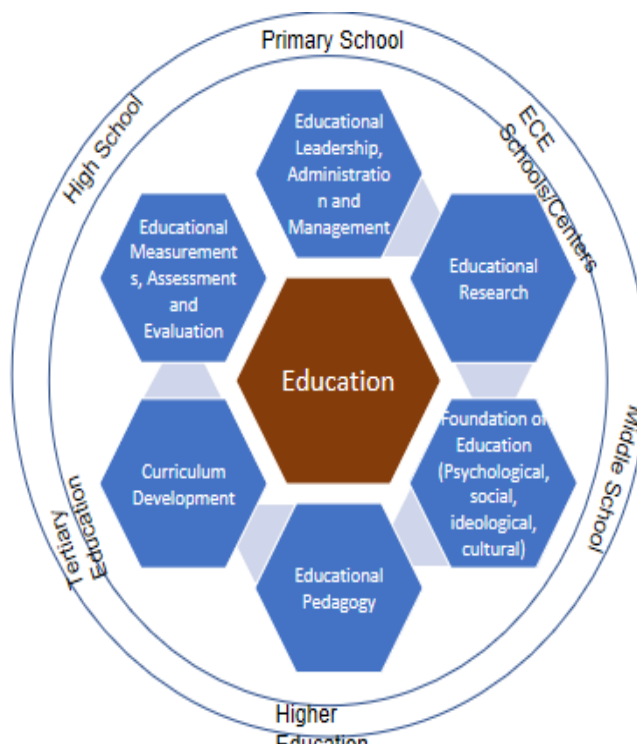
Topic11: Scope of Educational Research

We are going to talk about the scope of educational research especially for those students who are enrolled in any degree program such as B.ed, M.Phil or any other in which research is essential.

For example, when you are enrolled in a B.ed programme, you study all of these subjects shown in the following diagram. Studying these subjects provides you with the basic knowledge in each discipline and it will further help you to define the area of your interest to pursue in research.

Sources of Educational Issues

There can be different ways of looking at the scope of EDUCATION as subject of study

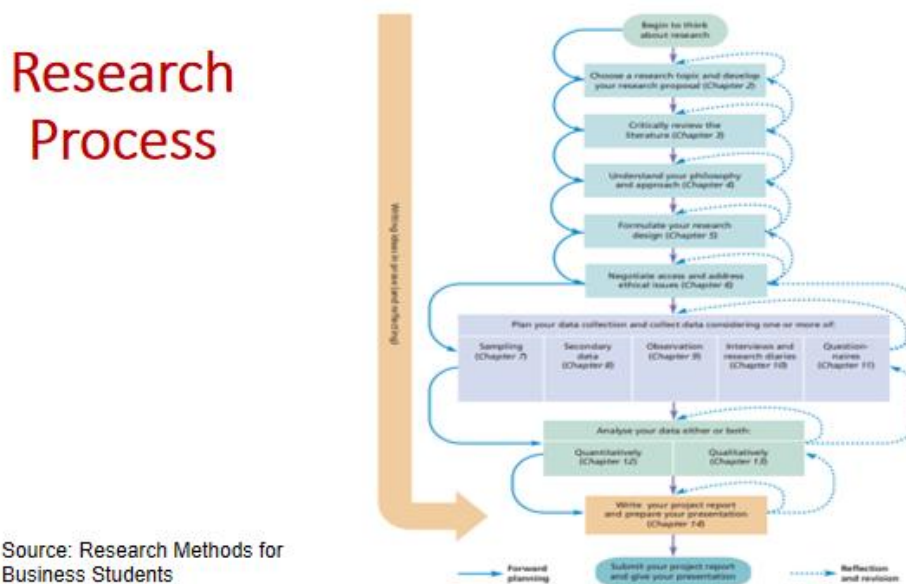


After teaching of all the above mentioned disciplines, it is expected the students must have their world view, their philosophy of education and they are aware about the current challenges in each discipline so that they can define their area of research and generate the idea of research.

All these areas can be studied on different levels because all these six domains are applicable on the primary education, early childhood education, high and higher secondary education, college education as well as university level. The outer boundary of the circle basically is depicting all those levels. It is on the part of researcher who will decide that which of the area and which level they are going to study in his research.

Topic12: Research process

The research process typically follows a series of steps that are similar to the scientific method. Let's explore each step in more detail:



- 1. Identification of the Research Problem:** The first step involves identifying a research problem or question. This is where researchers convert their area of interest into a researchable form. It is important to define the research problem clearly and precisely.
- 2. Literature Review:** The second step is conducting a thorough literature review. This involves reviewing existing research and scholarly articles related to the research problem. The literature review helps researchers understand the current state of knowledge on the topic, identify gaps in the existing research, and inform their own research approach and methodology.

3. **Research Approach:** The third step involves deciding on the research approach, which includes the philosophical orientation and overall strategy for conducting the research. The research approach can be deductive (testing existing theories) or inductive (generating new theories or hypotheses). It also involves selecting the appropriate research design.
4. **Research Design:** In this step, researchers choose a research design that aligns with their research approach. The research design outlines the overall plan and structure of the study, including the methods of data collection and analysis. It is crucial to select an appropriate design to ensure valid and reliable results. Sampling, data access, and tool development are subsequent steps in this phase.
5. **Data Analysis:** Once the data is collected, the next step is data analysis. The specific data analysis techniques depend on the nature of the data and the research design. Quantitative data may be analyzed using statistical methods, while qualitative data may involve thematic analysis or other qualitative analysis techniques. Data cleaning and preprocessing are important steps before analysis.
6. **Writing Research Report:** The final step involves writing the research report. This includes summarizing the research problem, reviewing the literature, describing the research approach and design, presenting the data analysis findings, and drawing conclusions based on the research results. Recommendations for future research may also be included. The research report should be well-organized, clear, and accessible to both technical and non-technical readers.

It's important to note that while these steps provide a general framework, the research process can be iterative and may require adjustments based on the specific research context and requirements. Researchers should also pay attention to refining their language and ensuring the readability and comprehensibility of their research reports for a wider audience.

Lecture 4 Classification of research

Topic13: Approaches of the Research

Let us talk about the research approaches. You have seen while studying that people use certain terms e.g. research philosophy, research approach, design and research method interchangeably which cause confusions. To avoid such confusion, it is important to differentiate among these terms and to understand their hierarchy.

The top most is the **philosophies of science** which serve as guiding principles in the research process. Then comes research approaches associated to those philosophies. Approaches are further divided into certain research designs and different designs further lead to the different research methodologies. We will discuss about the research philosophy later, here we will discuss about the most commonly used following three approaches:

Approaches to Research

1. Quantitative Research
2. Qualitative Research
3. Mix Method Research

1. **Quantitative Research:** Quantitative research is about the test the hypothesis. Quantitative research is the collection and analysis of numerical data to describe, explain, predict, or control phenomena of interest.

Quantitative researcher assumes that we inhabit a relatively stable. Uniform, and coherent world that we can measure, understand, and generalize about.

Example: Effect of teacher feedback on student performance in mathematics in early grades.

2. **Qualitative Research:** Qualitative research is the collection, analysis, and interpretation of comprehensive narrative and visual (i.e., non-numerical) data to gain insights into a particular phenomenon of interest.

Qualitative researchers do not necessarily accept the view of a stable, coherent, uniform world.

Example: Developing Teacher Epistemological Complexity about Multicultural Curriculum: A Case Study

3. **Mix Method Research:** Mixed methods research combines quantitative and qualitative designs by including both quantitative and qualitative data in a single study. The purpose of mixed methods research is to build

on the interaction and strength that exists between quantitative and qualitative research designs to understand a phenomenon more fully than is possible using either quantitative or qualitative approaches alone.

Topic14: Steps in the Process of Research

Let us talk about the research classification of educational research. We will start with the same comparison as we did while discussing about qualitative and quantitative research. If you have ever been studied any research course, you might have listen about the term experimental research, survey, causal comparative, phenomenology and so on. The main purpose is to decide which type of the above mentioned research is aligned with which of the approach.

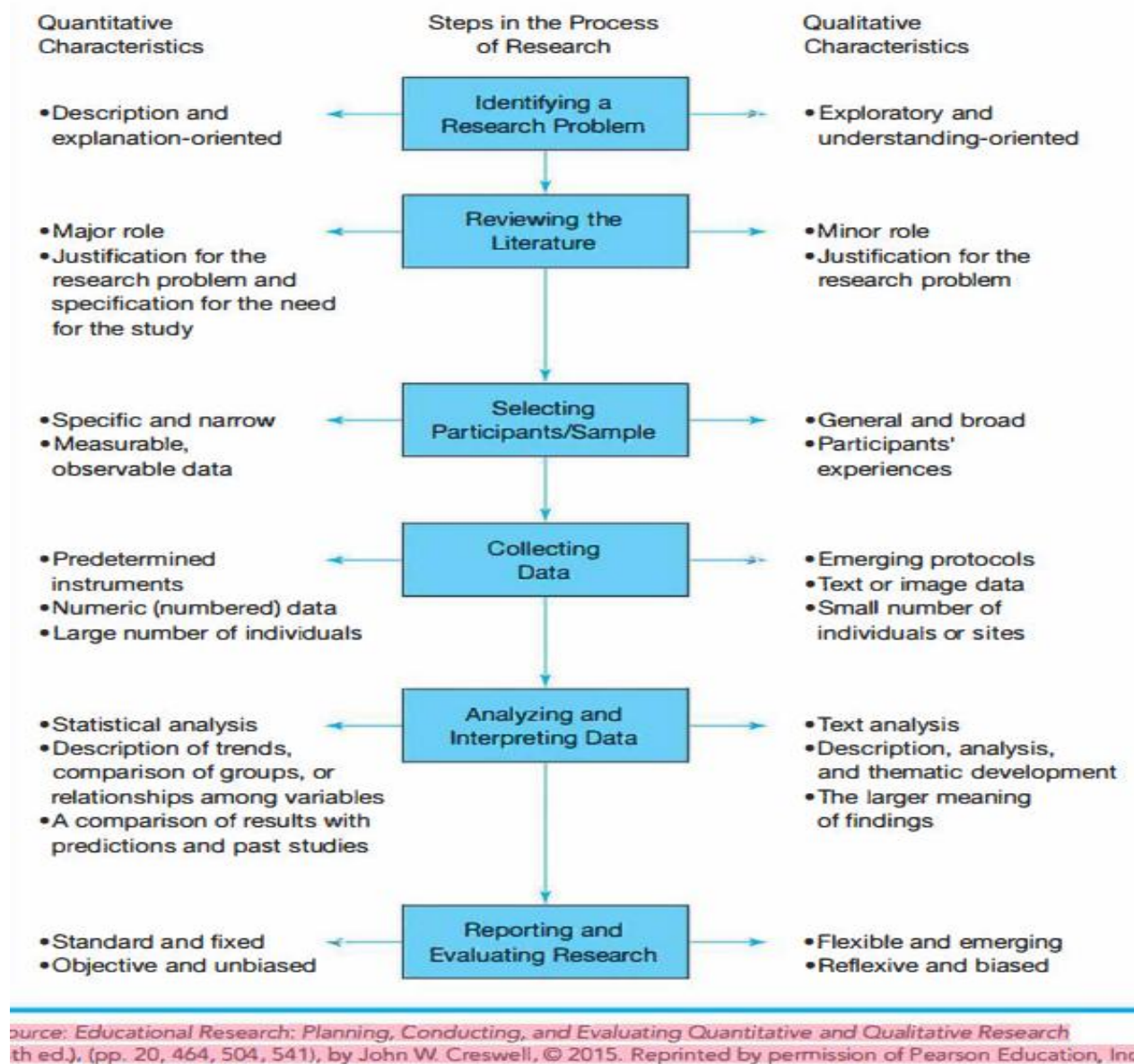
In the following diagram, steps of research process are given with a comparative view of both of the approach (qualitative and quantitative) on each side.

Comparison of Characteristics of Qualitative and Quantitative Research

The research process typically involves several steps, regardless of whether you are using a qualitative or quantitative approach. However, there are some notable differences in how these approaches are applied at each stage. In the following diagram, steps of research process are given with a comparative view of both of the approach (qualitative and quantitative) on each side.

Identifying the research problem: This step involves identifying and clearly defining the research problem or the question you want to address. Qualitative research often aims to explore complex phenomenon, while quantitative research typically focuses on description, evaluation of the effects, relationships relationship, and testing hypothesis or measuring variables. Although phenomenon exist also in the quantitative but how it acts in a particular situation is the focus of the quantitative research. Therefore, the nature of the topic under both (quantitative and qualitative) will be different.

1. **Review the literature:** Conducting a literature review involves has a key importance in quantitative research. This step is equally important for qualitative research as it helps you to understand the existing knowledge, identifying gaps, and refine your research problem. Whereas, the purpose of literature review in qualitative research is to gather the basic information and not the identification of the gap. Therefore, literature review does not have major role in it.



2. **Selecting the Participant/ sample:** In the quantitative research, we have to draw the sample from an identified population and we do utilize sampling techniques to draw a representative sample to ensure that the information we may get from those few selected people will be the same if we get it from the larger group of population. However, there is no concept of population. There is no need of sampling in it because there is no hypothesis testing, no generalization in the qualitative research. We just need to select the research participant based on a set criterion.

3. **Data collection:** Usually close ended tools/instruments are used for data collection in the quantitative research in which question statements are given and respondents need to just provide

their response on the given response options. The quantitative research will gather the data in numeric form. Whereas, the qualitative research is mainly based on the interviews, text, notes, diaries and reflective journals for data collection to understand the phenomenon.

4. Analyzing and interpreting the data is also different in both of the approaches.

Quantitative

Statistical analysis

Description of trends, comparison of groups, or relationship of variables

A comparison of results with predictions and and past studies

Qualitative

Text analysis

Description, analysis and thematic development

The larger meaning of findings

5. Reporting and evaluating research is different in both types as well. In quantitative research, report writing has a fixed style based on the different type of statistical tests and data analysis techniques. In qualitative research, data analysis is not based on statistical techniques instead it is based on the interpretations. Therefore, while utilizing the interpretation, the major focus in the report writing is on the explanation.

Topic15: Quantitative and Qualitative Research

Comparison based on Selected Basic Characteristics of Research

In this lecture, we will do comparison between the key characteristics of the quantitative and qualitative research in the form of a table to differentiate between the two. This comparison may help you out the decide which approach will be relevant to investigate your research topic.

Qualitative vs. Quantitative Research

	Quantitative Research	Qualitative Research
Type of data collected	Numerical data	Non-numerical narrative and visual data
Research problem	Hypothesis and research procedures stated before beginning the study	Research problems and methods evolve as understanding of topic deepens
Manipulation of context	Yes	No
Sample size	Larger	Smaller
Research procedures	Relies on statistical procedures	Relies on categorizing and organizing data into patterns to produce a descriptive, narrative synthesis
Participant interaction	Little interaction	Extensive interaction
Underlying belief	We live in a stable and predictable world that we can measure, understand, and generalize about.	Meaning is situated in a particular perspective or context that is different for people and groups; therefore, the world has many meanings.

1. In the first column, the first parameter to compare is the **type of data collection**. The data is collected in the numeric form in quantitative research because we have to analyse this data with the help of various soft wares and statistical techniques. As the statistical techniques require data in numeric form, therefore we have to convert the date in numbers though coding if it is not the numeric form. Whereas, the qualitative research gathers the data in non-numeric form. Non-numeric means, it can be in the form of text, graphics, visuals, and documentary. If you do collect by yourself, it can be in the form of interview, observation and observation notes.

2. Second parameter for the difference in the **nature of the research problem**.

All of the research topic in quantitative research suggest the research hypothesis. It means, the theories already exist related to the topic we are going to study and about is nature.

For example: “Highly motivated people perform well.”

Many theories exist explain that the increase in the level of motivation increase the level of performance. To test the theory, I will take a group of people to investigate whether their achievement

improves if we motivate them as described in the theory or not? If it is not proved, then I will identify the factors behind to explain them.

In comparison, qualitative research explains the evolving phenomena and we cannot guess how it will affect the related things. For example, what is the nature of social relationships protocols after the emergence of social media usage? What things are the acceptable and non-acceptable?

We cannot predict about the behavioral pattern of people in such a case when the phenomenon is evolving or recently evolved. The purpose of research here is to understand the behavior patterns. There is no hypothesis we just need to explain the phenomenon in such a case.

3. **Manipulation of the context** is the third parameter of comparison. The context does not matter in the quantitative studies as the major aim is to generalize the findings to the larger population based on the data provided by the sample. But the context matters because we do studies the embedded phenomena.

4. **Sample size** is the fourth parameter of comparison. The word sample has some meaning in it. Whenever we think about the sample we may think that there exist population as well. The concept of sample relates to the quantitative studies because whenever we think about the concept of sample/population, there comes a related concept of generalization. But the qualitative studies do not have such concepts of sample and generalization. The concept of participants is applicable in the qualitative studies in comparison. Participants are not selected in a bigger number just like the surveys, rather few participants are taken to collect the in depth information.

5. **Next parameter** is of the research procedures. As we have already discussed in the previous lecture (sources of knowledge) that the quantitative studies are deductive in nature whereas the qualitative studies are inductive which is a 180 degree difference between the procedures of the two.

6. **Participant interaction:** There is no need to personnel interactions with the people in quantitative studies as we do circulate the questionnaires to collect the information. It is because we use structured tools and we don't need further explanations. But in the qualitative studies, in depth information is required from a small audience therefore we need to contact them multiple times. We need to acquire the explanations of their shared information in order to reach the bottom line of the issue.

7. We can discriminate based on the parameter of **Underlying belief** as well. This point is basically the extension of already discussed the second parameter **nature of research problem**. The underlying

belief of quantitative research is that the principles of this universe are already established. We just need to understand and verify those principles in order to make generalizations. On the other side, the underlying belief of qualitative research is that things and behavior get evolved with the passage of time and we need to explore them with the new procedures. Already established rules do not work. Therefore, topics and directions of research change with the passage of time.

These parameters can be used as a yardstick to understand the difference between the quantitative and qualitative approaches.

Topic16: Classification of Research

In quantitative and qualitative research, we use multiple terms as survey research, experimental, correlational, causal comparative, narrative, case studies and many more. There is an endless list of such names/terms. Unless and until we do not classify these names under the quantitative and qualitative approach, there are chances that we may utilize an inappropriate research design for the problem under investigation.

In today's session, we will try to classify or to form such a structure in our minds that may be useful to decide what methods are relevant to which of the research approaches.

Classification of research

Classification of research can be done in two following ways:

- **Design/Method:** Although design and method are two different terms however, both terms are used simultaneously in research books that's why I used both the terms here in similar meaning.
- **Purpose**

The third is mixed method which will be discussed in further chapters.

Classification of research by design/methods is presented in the given table:

Classification of Research by Design/Method

Quantitative Approach

Survey Research
Correlational Research
Causal-Comparative Research
Experimental Research

Qualitative Approach

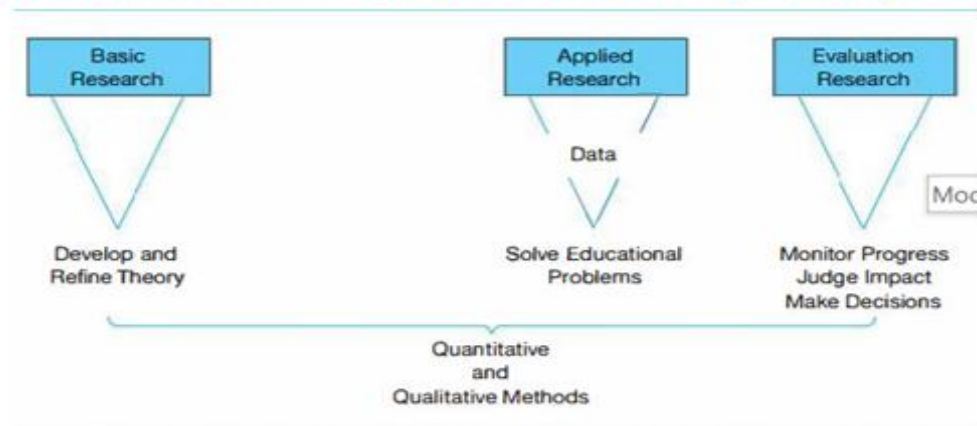
Narrative Research
Ethnographic Research
Case Study Research
Phenomenological Research
Historical Research
Grounded Theory Research

This is not the final list of methods under each category, you may find certain other methods in different research books but after an analysis you will find that those other names ultimately fit into the same methods presented in the table. For example, you may find focused group and observation which are basically sources of data collection but sometimes people use them as methods/design as per their understanding which is wrong.

We will talk about the listed methods in detail in next slides. Here you need to clarify the main and popular methods related to quantitative as well as qualitative approach. This clarity will help you to ensure the right choice of methods under each approach and minimize the chances of mistakes.

Classification of research by Purpose:

Classification of Research by Purpose



All these three categories are different in their purpose for example **Basic research** is conducted to understand all the existing phenomena in this universe. For example, the concepts of gravity, acceleration, evaluation, interaction and movements in cosmos. It is not conducted for other purposes as commercial or to solve daily life problems.

So, to find out the forces in the universe are the key subjects of action research. In short, the action research is conducted to understand the principles of this universe which are further **applied** by the scientists of **applied research**.

Example: finding the characteristics of x-rays was the domain of **basic research**. But the discovery of X-ray machine that is utilized to scan the human body for diagnostic purpose is done by the **applied research**.

Evaluation research is a recently added phenomena in this list. It includes more similar to basic research. Evaluation research aims to check the performance of any program which is also called program evaluation.

For example, a recently developed curriculum is working well or not? Does it need improvement? if needs improvement, in which aspect?

Lecture 5 Research Approaches-I

Topic17: Classification of Research by Design/Method

In this lecture, you will come to know about research methods briefly.

Classification of Research by Design/Method

Quantitative Approach

Survey Research

Correlational Research

Causal-Comparative Research

Experimental Research

Qualitative Approach

Narrative Research

Ethnographic Research

Case Study Research

Phenomenological Research

Historical Research

Grounded Theory Research

We will discuss about the difference among all of the method enlisted under each approach. How survey is different from co-relational research? What is the difference between the co-relational and causal comparative research? In the next chapters, we will study in detail about the methods, design and the way problem is stated in each of the above.

As the focus of our today's lecture is to differentiate among the listed methods, we will start from the methods listed under the category of quantitative research. First comes survey in the list.

Survey research: Survey research determines the current status of the problem understudy e.g. any product, book or curriculum newly launched or implemented. It reports the way things are, how people are reacting towards the products, either they accept or reject it? Survey research is not retrospective or prospective rather explores the current status of the things. So the key word in this type of research is the **current status**.

A very **good example** of survey can be the election polls. Organizations and individuals conduct telephonic and live surveys to know which of the political party or candidate has advantage or disadvantage over the other and what extent?

it involves collecting numerical data to answer questions about the current status of the subject of study.

Example: How do second-grade teachers spend their teaching time? Second-grade teachers are asked to fill out questionnaires, and results are presented as percentages (e.g., teachers spent 50% of their

time lecturing, 20% asking or answering questions, 20% in discussion, and 10% providing individual student help)

Here we conclude, all the things and research problems in which you are aiming at checking the current level is survey research as **“Teachers perception regarding the effects of Covid-19”**.

Topic 18: Classification of Research-- Correlational Research

While continuing the discussion about classification of research, we will move towards the co-relational research. We must keep in mind that how this research design is different from others methods of quantitative research?

Classification of Research by Design/Method

Quantitative Approach

Survey Research
Correlational Research
Causal-Comparative Research
Experimental Research

Qualitative Approach

Narrative Research
Ethnographic Research
Case Study Research
Phenomenological Research
Historical Research
Grounded Theory Research

Correlational research: Correlational research is about checking the association between a two or more variables.

Association between the clouds and chances of the rain.

In correlational research, the key factor to consider is that the variables being studied must exhibit covariance. This means that if one variables changes, the other variable will also change accordingly. Furthermore, if there is a change in the second variable, it must be attributed to the first variable.

For Instance: Let us consider the height and the size of one’s step. The reason for taking larger steps is attributed to having a greater height. Similarly, if an individual has a larger height, it is expected that their step size will also be larger. In short, both of the variables exhibit variations that are influenced by each other.

Variables in correlational research or not independent or dependent.

Let us consider a phrase:

‘Effect of motivation on students learning.’ in this case, **Motivation** is an **independent variable** which will affect students’ learning while increasing or decreasing. Students achievement is the **dependent variable**. This is not the case in correlational studies.

Correlational research involves collecting data to determine whether and to what degree, a relation exists between two or more quantifiable variables.

The purpose of a correlational study may be to establish relations or use existing relations to make predictions.

Example: How do the entry test scores of college students correspond to the first-semester Bed (4 years) grades?

So, regression is the next level of the correlation study which help to make predictions based on the existing nature and quantity of a relationship.

Topic 19: Quantitative Approach - Causal-Comparative Research

While continuing the discussion about classification of research, we will move towards the third research method under the quantitative research, which is **causal comparative**. Causal comparative research is based on natural occurrence that has been done in the past and aim of research is to identify the reasons of that happening. It is a **retrospective research**.

Definition: Causal-comparative research attempts to determine the cause, or reason, for-existing-differences in the behavior or status of groups of individuals.

Example: How does having a working mother affect a child's school absenteeism? The grouping variable is the employment status of the mother (again with two possible values-the mother works or does not work); the dependent variable is absenteeism, measured as number of days absent. The researcher identifies a group of students who have working mothers and a group whose mothers do not work, gathers information about their absenteeism, and compares the groups.

In the above example, **employment status** of the mother is the **grouping variable** which may bring the change in children's behavior. Being a researcher, we will investigate the factors or component in working mother’s actions that are contributing the absenteeism of the children. The outcome of this research may help us out to guide the working mother to plan and adjust their work routine in such a way to control the absenteeism behavior of the children.

In short, after any happening or seeing of such behavior that we need to identify the reasons behind, will lead towards the causal comparative research.

Topic 20: Quantitative Approach - Experimental Research

The last method under the quantitative research, experimental research is our topic of discussion today. Experimental research is considered the most established method of the research. It is also widely used method of the research.

In experimental research, at least one independent variable is manipulated, other relevant variables are controlled, and the effect on one or more dependent variables is observed.

So the most important attribute of experimental research is the **manipulation of the variable** which is absent in the other methods of the research.

Let us say:

“Effect of exercise on physical health of students”

In this example **Exercise** is the **independent variable** because the increase or decrease of exercise (which is my choice) will affect the **health**, which is the **dependent variable**. Hence the quantity of exercise is increasing or decreasing by the choice of the person, is actually the manipulation which will affect the dependent variable. So, in an experimental study, we study that how the change of manipulation of independent variable affect the dependent variable. It is conducted in a controlled environment and all variables other than intervening variable are tried to control. So in the above mentioned examples all variables other than the exercise.e.g. quality of food and quality of sleep must be controlled because they can also affect to improve the physical health.

In short, the characteristics of experimental research we discussed here may help you to differentiate it with the other quantitative methods of research e.g. causal comparative, correlation and survey.

Lecture 6 Research approaches-II

Topic 21: Qualitative Approach - Narrative Research

Today we will discuss about the methods of qualitative research under the major topic Classification of research. The first topic in the list is the narrative research we are going to discuss about.

Narrative research is the study of how different humans experience the world around them; it involves a methodology that allows people to tell the stories of their "storied lives".

The researcher typically focuses on a single person and gathers data by collecting stories about the person's life. The researcher and participant then construct a written account, known as a narrative, about the individual's experiences and the meanings the individual attributes to the experiences.

Important points:

- Focus on an individual
- Collecting the life experiences of individuals related to the research focus.
- Listening of the stories repeatedly to connect their thoughts in order to verify their credibility to draw the meaningful narrative.

Goals of narrative research:

One of the goals of narrative research in education is to increase understanding of central issues related to teaching and learning through the telling and retelling of teachers' stories.

Example: A student in class perceives being targeted due to his short height in game time in school. He shares his story with games teacher in detail and teacher records the story.

Telling, sharing and re-telling the story helps refine the narrative based on the valid narrations. While drawing from the narrations, teachers will plan strategies to tackle the similar situation in better ways.

Topic 22: Qualitative Approach - Ethnographic Research

In Classification of research by method or by design, we are going to discuss about the Ethnography which is the second design in qualitative approach.

Ethnographic research, or ethnography, is the study of the cultural patterns and perspectives of participants in their natural settings. Ethnography focuses on a particular site or sites that provide the researcher with a context in which to study both the setting-and the participants, who inhabit it. The researcher becomes the part of the setting in which the ethnography research is being conducted. This research is context bound.

Ethnographic setting can be defined as an elementary principal's office. Same like, school staffroom can be a setting to observe the relationships among teachers. School staffroom is an informal and natural setting where they sit in and make inform discussion in their free time. The participants are observed as they take part in naturally occurring activities within the setting. The ethnographic researcher creates acceptability by the participants and gaining rapport with them. Then, over time, the researcher collects data in waves, making initial observations and interpretations about the context and participants, then collecting and examining more data in a second wave of refining the initial interpretation, and so on until the researcher has obtained a deep understanding of both the context and its participants' roles in it.

So, whenever you use this procedure to answer your research question, you choose an ethnography research.

Topic 23: Qualitative Approach - Case Study Research

In Classification of research by method we are going to discuss about the third qualitative design which is **case study research**. Let us discuss the context of case study, what is it and how we use it? It is important her to define the word case first.

Case can be an individual, a group of individuals even an event can be taken as a case. But the major consideration while choosing a defining these thing as a case is its uniqueness.

A case study is an in-depth study of one person, group, or event. In a case study, nearly every aspect of the subject's life and history is analyzed to seek patterns and causes of behavior.

case study is that it allows researchers to investigate things that are often difficult to impossible to replicate in a lab. Some other benefits of a case study

Example: Rameez was a railroad employee who experienced a terrible accident in which an explosion sent a metal rod through his skull, damaging important portions of his brain. Rameez recovered from his accident but was left with serious changes in both personality and behavior.

So, an ordinary person or a routine event cannot be taken as a case, it must be unique which is the major consideration. Therefore, case study design cannot be used for any of the random topic.

For example, there is a school who shows an extraordinary permanence without having any usual or special attributes e.g. location or teaching staff. I perceive such a school with extraordinary performance without any unusual set up as a case. One can take it as case to study those factors that are working behind the constant extra ordinary performance of that school in the board exams.

That data gathered from the case study are larger in its-depth as well as its width which is used to infer and to explain how people and organizations can be guided to perform well.

Lecture 7 Research approaches-III

Topic 24: Qualitative Approach - Phenomenological Research

In classification of research by design, we are going to discuss about the next qualitative design phenomenological research. The uniqueness of this design is that it is based on the lived experiences of the people.

A qualitative research approach that helps in describing the lived experiences of an individual is known as phenomenological research. The phenomenological method focuses on studying the phenomena that have impacted an individual. This approach highlights the specifics and identifies a phenomenon as perceived by an individual in a situation. It can also be used to study the commonality in the behaviors of a group of people.

If I talk about the major characteristic of phenomenological research that differentiates it from the other designs of qualitative research, that it is about the **lived experiences**. It means that the participants of the study must have the lived experiences about the topic one intends to investigate.

Although certain other types collect the experience base data but that experiences are not necessarily lived experiences in which the participants have gone through. So, the phenomenological research is based on the lived experiences of the study participant that specific experiences has certain psychological and social impacts on their life.

The research is aimed at taking the lived experiences of people on a same thing to determine that how they perceive the specific experience differently. The commonality and differences in the perceptions and views of the people are taken as source of information in this type of research.

Example: Losing Family Members to Covid-19 Hasn't Been Easy. A Detailed Study of Survivors and People Who've Lost Loved Ones Can Help Understand Coping Mechanisms and Long-Term Traumas.

In short finding out the coping strategies to any phenomena from the lived experiences of the people is called the phenomenological research.

Topic25: Qualitative Approach - Historical Research

In Classification of research by design, we will discuss about the historical research under the qualitative research designs. It is one of the eldest methods of research.

Definition:

Describing and examining past events to better understand the present and to anticipate potential effects on the future. To identify a need for knowledge that requires a historical investigation. Piecing together a history, particularly when there are no people living to tell their story.

Historical research focuses on past events rather than current or future events. Its primary objective is to study and understand events that have already taken place. The purpose of historical research is to draw lessons and insights from past occurrences in order to enhance preparedness for similar situations in the future. By examining historical events, researchers aim to gain valuable knowledge and understanding that can inform decision-making and improve future outcomes.

Example 1: Historical research can shed light on the factors that contributed to the outbreak of covid-19 in the past, which can provide valuable insight into addressing current outbreak. By examining the previous experiences, we can identify the causes, transmission patterns, and response strategies that were effective or ineffective. This knowledge can help us develop more effective measures to treat and prevent the spread of upcoming variants of Covid-19.

Example 2: What caused an outbreak of polio in the past that may contribute to the outbreaks of today?

In essence. Historical research plays a crucial role in guiding future actions across various fields, including education. By studying past events and their outcomes, we can make informed decisions and shape the future course of action. This allow us to learn from past success and failures, implement effective strategies, and improve outcomes in different areas of life, including education.

Topic 26: Qualitative Approach - Grounded Theory Research

Today, we will try to learn about the Grounded theory which is the last design of qualitative research under the classification of research by design. There are two perspectives or usage of grounded theory in the literature. Sometimes it is used as a research type and sometimes as a research design.

We will discuss it as a research type here or type by method.

Definition:

Grounded theory proposes that careful observation of the social world can lead to the construction of theory (Rice & Ezzy, 1999). It is iterative and evolving, aiming to construct new theory from collected data that accounts for those data. It is also known as the “grounded theory method”, although the terms have become interchangeable (Bryant & Charmaz, 2007).

The grounded theory approach employs an inductive methodology. It is important to note that a single study on an evolving phenomenon does not constitute grounded theory research. Multiple researchers gather information on different aspects of the evolving phenomenon simultaneously to enhance understanding and clarity. After collecting rigorous information, philosophers and theorists analyze it to identify causal relationships and develop theories. In essence, theories are not the result of a single research endeavor, although a single study can serve as foundational unit. Therefore, if someone wishes to adopt grounded theory as a research method, it can be utilized in any study lacking theoretical foundations.

For instance, consider the following example:

Example: What are people’s reaction to changes in friendship patterns following the widespread use of social media?

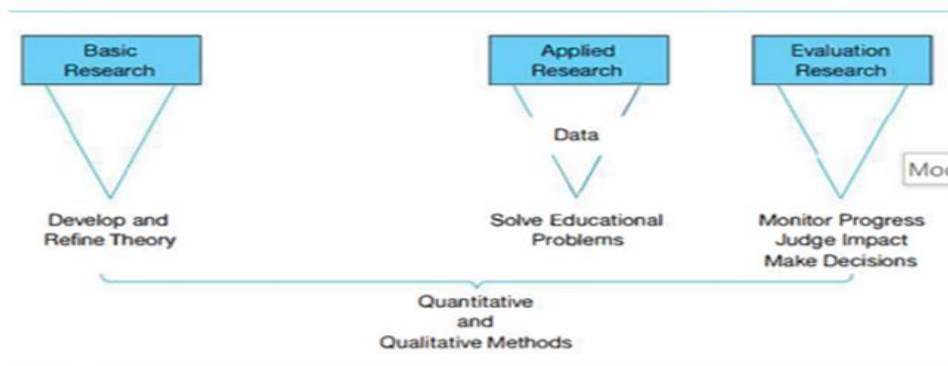
Example: What are the barriers to camp schools for Afghan refugee population in refugee camps?

Lecture 8 Classification of Research by Purpose**Topic 27: Classification of Research by Purpose-Basic Research**

We have already discussed about the classification of research by design, now we will move towards the discussion of classification of research by purpose. We can classify research by purpose in the following three areas:

We will discuss all the three type one by one:

Classification of Research by Purpose



1. Basic Research:

Basic research is a study conducted for the sole purpose of enhancing or growing the knowledge base in a particular field of study. This type of study is used to explore what is not known and is driven by the researcher's interest and curiosity. Basic research is frequently referred to as fundamental research due to it being knowledge specific. This research simply collects and examines data in such a way that the information obtained enhances the quality of knowledge in the subject.

The scope of research is universal as it encompasses a wide range of concepts across various subject matters. Through basic research, researchers can develop theories, generalizations or specific principals to explain concepts that have a universal applicability. A notable discovery is X-ray and radioactive materials. When X-ray were initially discovered, they don't have any immediate practical application. They were initially recognized as byproduct or outcome of research. However, over time, the understanding and knowledge gained from this research led to the development of numerous practical applications and advancements in the field of medicine and technology.

There are very few individuals who invest in this type of research because it is highly time and resources consuming. Furthermore, there is no guarantee of the outcomes, if any findings are obtained, it remains uncertain whether they will be practical or applicable. Some large states allocate significant funds towards the investigation of outer space, but the timeline for when the results of such research will be implemented and how they will be utilized remains unknown. Therefore, basic research is carried out

primarily to satisfy the curiosity of researchers. Newton's curiosity, for instance, led to the discovery of the law of gravity, which was later applied in various fields such as the construction industry to ensure stability. It all began with the researchers' curiosity and interest in exploring the existence of certain phenomena and understanding their underlying reasons.

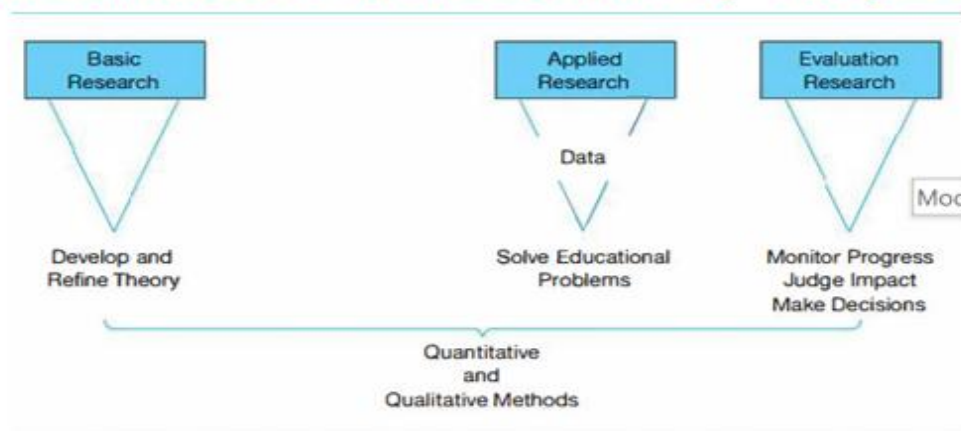
Example: What does anxiety in teenagers look like? This is basic research because it adds to the body of knowledge regarding anxiety, specifically in teenagers. However, these discoveries led towards the invention of x-ray machines and scanners which are being used for diagnosis purpose to cure the diseases and come under the domain of applied research.

In short basic research intends to investigate the properties of the things working in this universe.

Topic 28: Applied Research

In the classification of research by purpose, we will discuss about the second type of **Applied research** here.

Classification of Research by Purpose



There is need to understand this type of research in relevance to the **basic research** to differentiate between the two. Action research apply the principals developed by basic research to solve the real life problems e.g. discovery of X-rays to scan the human body for treatment purposes.

Definition:

Applied research is research conducted to provide a solution to an established problem, issue, or concern. This form of research includes solution-oriented questioning for a particular event, experience, or circumstance. Therefore, it is frequently referred to as "conceptual research." Applied research is often the follow-up to a theme, theory, or question arising from basic research developments. Applied research examines empirical evidence for answers.

Flowing example explains the applied research in educational context:

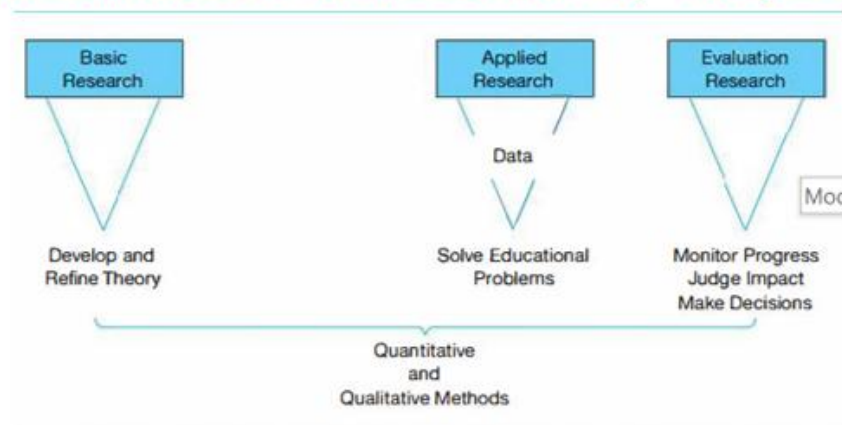
Example: How better classroom interaction with students can improve the memory of students in language learning?

In short, to solve the real life problems by applying the already developed theories and principals by **basic research** is called **applied research**.

Topic 29: Evaluation Research

We have categorized research in following three types by its purpose. we will discuss about the third type of **Evaluation research** in today's lecture.

Classification of Research by Purpose



Evaluation research is a recent development in the field. This type of research cannot be found in older research books. Evaluation research is conducted to monitor the ongoing processes, ensure the right

directions or assess the impact of certain projects. There are two types of studies conducted in the evaluation research:

I. Action type research is not intended to generalize outcomes. It is conducted to solve the immediate problems. For example, as a teacher when we strive to solve any specific issues that hinders the smooth flow of classroom teaching learning, we collect data within that particular situation. The outcomes of such research are only applicable to that specific situation and particular group, not for others.

II. Program evaluation studies are conducted to monitor the process and to assess whether all the planned activities are progressing as intended. The evaluation aims to determine if set targets are being achieved. This type of research is referred to as project or program evaluation.

Example: Is the new geography curriculum meeting the teachers' needs?

In conclusion, it is important to highlight that all three types of research, namely basic, applied, and evaluation research, are distinct in their scope and nature, and they do not overlap. Each type serves as unique purpose and follows specific methodologies. Basic research focuses on expanding knowledge and understanding fundamental concepts, applied research aims to solve practical problems and develop real world applications, and evaluation research is conducted to assess ongoing process, determine effectiveness and make informed judgement. Therefore, while these types of research may intersect in certain contexts, they remain separate and serve different purposes.

Topic 30: Skills Required for Conducting the Research Study

After getting familiar with the basic concept of this course, now we will study all those topics in detail in form of independent units. So, today we are going to discuss about the Skills Required for Conducting the Research Study. You may already have many skills and you just to refine those skill. You need to develop two types of skills to develop in your personalities simultaneously if you want to be succeeded in your life as a researcher. These two types are categorized in the following:

Classification of Skills for Research Success

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Classification of Skills for Research Success

Intellectual skills	Research Management skills.
creativity	Searching information
judgment	Taking notes
communication	Time management
Organization	Communicating results
Persistence	

Intellectual skills are discussed in the following:

Creativity refers to find such way that is different from the existing methods and useful with in the available circumstances and resources. In usual practices, we get aligned with the existing methods along what we have studies or leaned. Instead, one must be ready for being innovative and creative to bring changes and apply new methods if existing methods do not work in a specific situation.

Judgement: You must be able enough to judge that the you are on a right track as you have already planned to proceed with.

Communication: Many students cannot communicate their work as effectively as they did it. It is important to understand this fact that although doing the right job is the part of research, but is very critical to communicate that work with others and if you are unable to do it, it is useless.

Organization: Ones personality must be organized enough to present his/her work in an organized and meaningful manner.

Persistence: One must not throw way the research ideas if finds difficult to respond or answer. Hens it is important to get clear understanding of the topic and its importance. If you are clear in your understanding then try to sort out the rest thing including, methods, tools, sample, literature and data analysis methods rather than jumping on a new topic.

Once research carrier may be successful with the development of these basics intellectual skills.

Research Management skills

Research Management skills are discussed in the following:

Searching: searching the relevant information, and reading it is an important skill for researcher. One cannot do credible work without developing this skill.

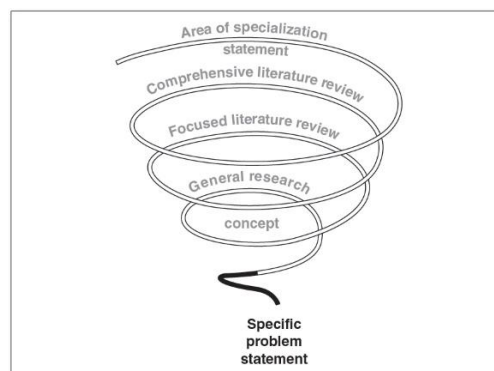
Taking notes: Storing information in heads is not sufficient while searching and studying on any topic. Writing relevant points and taking notes is important.

Time Management: All the activities carried out while conducting research are time bound. So, time management is a very important factor to complete it on time.

Lecture 9 Selecting and Defining a Research Problem

Topic 31: Research Problem

In this module, we will discuss the process of identifying a research problem. When you join a teacher education program, you study various subjects, including foundational, official, and professional courses. Throughout these courses, you are exposed to different types of problems and learn how to solve them. Before enrolling in the program, you may have already observed different aspects of the education system as a student, teacher, observer, or sibling. Combining these experiences with your coursework, you become aware of the issues within your education system. However, not every idea that comes to your mind is suitable for research. Some ideas may not be feasible or researchable. Nevertheless, there are many researchable ideas as well. If the broad idea in your mind is not initially in a researchable form, there is a method to make it researchable through a specific way of thinking and writing.



The spiral shape depicted on the slide represents the first area of specialization. During your coursework, you study a variety of courses. As you progress towards the final years of your degree, you are offered specific areas of interest. In these areas, you deepen your studies, and teachers encourage you to start identifying and shaping research problems that you are genuinely interested in exploring.

For example, some individuals may be interested in conducting research on topics such as educational assessment, measuring student learning, examination-related issues, test development, curriculum development, or educational administration.

You study one of these areas and combine your observations with your knowledge to develop a research topic. You narrow down your broad area of interest, such as educational assessment, to a specific focus. Educational assessment encompasses numerous aspects, including assessment administration, assessment development, and analysis of results for reporting purposes. Since it may not be feasible to research all aspects, you select a specific area, such as student learning achievement. As you delve into the topic, it begins to expand, and you identify further sub-areas. You continue to specify and narrow your focus until you arrive at a specific research area, such as reporting of learning achievement results.

Once you have identified your research area, the next step is to articulate it in a way that is understandable to yourself and others who will read your topic statement. When you write your research topic on paper, it should convey the exact thoughts and ideas that are in your mind. Others cannot read your mind, so it is crucial to write your topic statement clearly and precisely. The more clearly you can convey your intentions through your written statement, the better your supervisors, friends, colleagues, and classmates will be able to understand and assist you in developing your research proposal.

In the upcoming topics, we will discuss how to write a research topic statement effectively, including what should be included and how to convey your ideas accurately. Writing the research topic statement correctly is essential for ensuring that everyone understands your intended research focus.

Topic32: Topic of Research Study

Let us continue our discussion related to research problem and talk about how to write the research topic. It means writing a research problem, problem statement or topic in a way that a reader can understand it well. Following four things are required in a meaningful research topic:

- **Phenomenon**
- **Variables**
- **Subjects**

● Geographical Location

Phenomenon: If you recall, you may be noticed that most of the research topic you may go through starts from the words ‘Effect, impact, relationship or comparison’. these words show the phenomenon. Such words help the research to decide about the type of analysis. Look at the two examples:

Example 1: Impact of organizational commitment on customer services in tourism industry of GB

Example 2: Effect of Test Anxiety on Student Achievement in Grade 9 examination by BISEs

Second one is starting from the word ‘Effect’ which depicts that this study is about investigating the changes occur in a variable due to another variable and the analysis will be based on a test of difference. Likewise, the words relationship etc leads towards the prediction of certain type of tests of analysis.

So, the topic is not just a statement, rather it may be helpful to decide about the analysis, objectives, hypothesis or research questions.

Commonly used words for Phenomenon

- ✧ Effect
- ✧ Impact
- ✧ Analysis
- ✧ Comparison
- ✧ Relationship
- ✧ Descriptive

Unless you are unable to differentiate among the above written words, you may not be able to get the clarity to write the research objective and hypothesis for your research study.

Example: I conducted an experimental followed with a pre-test then intervention and then posttest. The changes occurred in the post test are due to the intervention. Therefore, I must say that I have checked the **effect** of my intervention.

Same like if I conducted another posttest after a longer period of the 1st one without making any further intervention to check either that changes sustained or not, it is call the impact.

In-short, the immediate changes after intervention are called '**effect**' and the sustained changes are called the '**impact**'.

Understanding the difference in the use of certain words is critical to write the research objectives, hypothesis and further steps.

Topic33: Phenomenon Variables, Subjects, Geographical location

We will continue our discussion related to research problem. As we have talked about the four things (Phenomenon, variables, subjects, geographical Location) to consider while writing the research topic. Here we will discuss about the variables. Your research topic must include at least one variable if it is a survey study. If it is another type of research, then the minimum of variables must be two.

Example: Effect of **Test Anxiety and Examination Environment** on **Student Achievement and Self-confidence** in Grade 9 examination by BISEs

In the above example, words written in bold are variables.

If you are conducted a study to check the difference, then one variable must be **Independent variable** and the other must be **Dependent variable**.

But if it is a study of relationship, it must be based on the co-variation concept. Although the variables exist here but these are not the independent and dependent variables. Variables must be two at least but it both of them are co-variant of each other. Hence, it is important to understand that you cannot conduct the study of difference on all types of variables.

Example 1: **Impact of organizational commitment** on **customer services in tourism industry of GB**

In the above example, it is shown that the 'organizational commitment' is the **independent** and the '**customer services**' is the **dependent variable** will may likely be changes with the changes in the independent variable.

It must keep in mind that the topic can include more than one independent and dependent variables. However, the increased or decreased number of variables does not impact on the quality of research.

Same like in the following example text anxiety is the independent and the student achievement depend on the test anxiety. So the student achievement is dependent variable.

Example 2: Effect of **Test Anxiety** on **Student Achievement** in Grade 9 examination by BISE.

Topic 34: Topic of Research Study-Variables

When considering the title of a research study, it is important to include variables that represent the phenomena being studied. In most types of research, there are at least two variables, and they can be **independent** or **dependent variables**. The number of variables included in a study has implications for the research design and analysis.

If you increase the number of variables in a study, whether independent or dependent, it affects the complexity and demands of the research. It is crucial to understand the implications of adding variables and acquire the necessary skills to handle them effectively. Failing to consider the applications of these variables in relation to the research objectives can lead to errors.

For example, let's consider a research topic that initially explores the '**effect of test anxiety on student achievement**'. By adding another independent variable, such as examination environment, the topic becomes "**The Effect of Test Anxiety and Examination Environment on Student Achievement**." Now there are two independent variables: test anxiety and examination environment, and the dependent variable remains student achievement.

The addition of the examination environment variable contributes to the change in student achievement alongside test anxiety. Each variable plays a role in influencing the dependent variable. If we want to determine the specific contributions of test anxiety and examination environment to the change in student achievement, we need to employ statistical techniques to separate their effects. Without the ability to discern the influence of each variable, we cannot do justice to the research process.

Adding variables to a research study increases its complexity. It introduces multiple effects: the effect of test anxiety, the effect of examination environment, and the interaction effect between test anxiety and examination environment. These effects must be properly analyzed and understood. If you possess the necessary statistical techniques and skills to handle this complexity, you can include such variables in your study. However, if a variable is added unnecessarily and does not contribute meaningfully to

the research, it can create unnecessary complexity. In such cases, it becomes challenging to defend the results obtained from the study.

In summary, it is advisable to keep the research topic simple and avoid unnecessary addition of variables. However, if including variables is necessary, it is crucial to acquire the skills and techniques required to handle them effectively. In the context of studying the effect of two independent variables on one dependent variable, understanding the statistical techniques and skills necessary for analyzing and interpreting the results is essential.

If we add another dependent variable, self-confidence, alongside the existing dependent variable, student achievement. This means that the title now includes two independent variables, test anxiety and examination environment, and two dependent variables, student achievement and self-confidence. The total number of variables has increased to four.

With this increased complexity, we can observe that changes in student achievement are influenced by both test anxiety and examination environment. Similarly, changes in self-confidence are influenced by both test anxiety and examination environment. Test anxiety contributes to both dependent variables, while the examination environment also plays a role in influencing both dependent variables. Additionally, there are two interactions: the interaction between test anxiety and examination environment affecting student achievement, and the interaction between student achievement and self-confidence.

By adding just one dependent variable, the number of effects and interactions has significantly increased. It is important to recognize that the addition of variables can greatly impact the complexity of the research. Each variable introduces new dynamics and relationships that need to be carefully analyzed. Understanding these interactions is crucial for justifying the choice of variables within the research design.

It is always advisable to prioritize essential variables in the research topic. Adding variables should be done with a clear understanding of their significance and relevance to the research objectives. Increasing the number of variables does not inherently improve the quality or level of research. The quality of research is determined by the methodological soundness and the ability to justify the chosen variables and their interactions. Therefore, it is important to approach variable selection with a thoughtful and justified rationale.

In summary, the addition of variables should be done sparingly and with careful consideration of their implications. The focus should be on conducting the research in a methodologically sound manner, rather than simply increasing the number of variables.

Lecture 10 Research Aims & Objectives-I

Topic35: Research Problem Subjects

This module focuses on advancing the discussion of the research problem by addressing the third element in the research title, which is the subject or the population under study. Including the subject in the research topic helps to specify the focus of the study and provides clarity to the reader. By identifying the specific group or industry that the research is targeting, it becomes easier to understand the research objectives and the variables involved.

For example, if the research is examining the impact of organizational commitment on customer service, specifying that the study is focused on people in the tourism industry narrows down the scope of variables. Different industries may have varying organizational structures and levels of commitment, so identifying the subject as the tourism industry helps the reader understand the context of the research.

Similarly, when investigating test anxiety, specifying the grade level, such as grade nine students, adds meaning to the variables being studied. Test anxiety may affect students at different levels differently, and by specifying the grade level, it becomes clear which group of students the study is targeting. This specificity aids in understanding how the variables relate to the subjects of the study.

Additionally, mentioning the subject or population in the research title helps in determining the study's population and sample. The title provides an indication of who will be part of the study and enables the researcher to identify the appropriate population to draw samples from. For example, when studying the relationship between test anxiety and achievement in grade nine students, the title suggests that the sample will consist of students from various institutes preparing for the board of intermediate exam.

In summary, including the subject or population in the research title serves multiple purposes. It helps the reader understand the research focus, context, and the specific subjects under study. Furthermore, it guides the researcher in determining the study's population and selecting an appropriate sample.

Topic 36: Topic of Research Study Geographical Location

In the last part of our discussion on the research problem, we will focus on the fourth component of the research title, which is the geographical location. The geographical coverage of the study refers to the specific area or region that the research will cover. It is important to mention the geographical location in the title to provide clarity about the scope and context of the study.

For example, if the research topic is related to the tourism industry, which exists globally, specifying the geographical location in the title becomes crucial. Let's say the study focuses on tourism in Gilgit-Baltistan, a region known for its natural landscapes and mountains. By mentioning Gilgit-Baltistan in the title, it immediately informs the reader about the specific area of the tourism industry that the research will address. This specification is important because the nature of tourism and the services provided may vary across different regions.

Similarly, in another example, if the research is about the effects of test anxiety among students, and it specifically focuses on the Grade Nine exam, it is important to mention this in the title. In different countries, there are various educational systems and different types of exams. By specifying that the research is related to the Grade Nine exam in the context of BISE (Board of Intermediate and Secondary Education) institutions, which are prevalent in Pakistan, it clarifies the target population and the specific geographical area of the study.

By including the geographical location in the title, it helps readers understand the particular type of phenomenon or area of research being studied. It also helps in defining the scope of the study and narrowing down the population or sample. For instance, if the title mentions "BISE Lahore," it indicates that the study will focus on institutions and students falling under the jurisdiction of the Board of Intermediate and Secondary Education Lahore, rather than encompassing all BISE institutions in Pakistan.

By ensuring that the title includes these four elements - research problem, variables, population, and geographical location - it becomes easier to visualize and understand the focus and scope of the research. This clarity at the outset helps in framing the research effectively and guides subsequent steps in the research process.

In summary, the geographical location component of the research title specifies the area or region that the study will cover. It helps in defining the scope of the research, clarifying the target population, and providing context for the phenomenon being studied. Including the geographical location in the title enhances the understanding of the research and facilitates subsequent research planning and execution.

Topic 37: Choosing (characteristics of) a good Research Problem

Now we will continue the discussion of the research problem, leaving the structure of the research title, we discuss the characteristics of a good research problem. When evaluating the characteristics of a good research problem, there are five important aspects to consider:

1. **Interest:** The research problem should be interesting and relevant to both the researcher and potential users. It should address a topic or issue that people find valuable and want to learn more about. For example, researching the discrepancy between students' grades and their intellectual capacity would be of interest to parents and educators who are concerned about educational outcomes.
 2. **Research ability:** The research problem must be feasible and researchable within the constraints of available resources, such as time, data, and access to information. It should be possible to gather the necessary data and conduct the research to address the problem effectively. If a topic is intriguing but lacks the means for investigation, it may not be a suitable research problem at the current time.
 3. **Significance:** The research problem should have significance in terms of its potential impact on people's lives or the field of study. It should address a problem or contribute new knowledge that can lead to improvements, whether theoretical or practical, and have relevance to the social or academic context in which it is being studied. This ensures that the research has broader implications beyond the immediate scope.
- For example,** studying the phenomenon of children's chatting habits may be of great interest. However, if someone were to conduct research on this topic in Sweden, it may not be as relevant because the educational system and exam structure there differ significantly. In Sweden, for instance, students may not have the opportunity to cheat on exams or seek external assistance. Therefore, researching the impact of chatting on academic performance may not yield meaningful results in that specific context due to the minimal occurrence of such practices.
4. **Ethical considerations:** Ethical guidelines and principles should be followed when selecting and conducting research. The research problem should not infringe upon individuals' rights or compromise their privacy. It should be conducted with integrity, transparency, and respect for the well-being and autonomy of research participants. Ethical considerations are crucial for maintaining the trustworthiness and credibility of the research.

5. Manageability: The research problem should be manageable within the available time and resources. It should be realistic in terms of the researcher's skills, expertise, and capacity to conduct the study effectively. Consideration should be given to the scope of the research problem and whether it can be reasonably addressed within the given timeframe and available resources.

After formulating a research title, it is essential to evaluate it against these five criteria to determine if it meets the standards of a good research problem. By analyzing the topic based on interest, research ability, significance, ethical considerations, and manageability, researchers can ensure that their chosen research problem is well-defined and suitable for investigation.

Lecture 11 Research Aims & Objectives-II

Topic 38: Research Aims & Objectives

In this module, we will discuss about research aim and objectives and difference between the two as well as their significance in the research process.

Formulating research aim and objectives in an appropriate manner is one of the most important aspects of your research.

This is because research aim and objectives determine the scope, depth and the overall direction of the research.

Research aim refers to what you hope to achieve. Aims are statements of intent. They are usually written in broad terms. They set out what you hope to achieve at the end of the project. You may not necessarily achieve the research aim, however, in such a scenario, it will guide the future directions for researchers. It is of utmost importance to be very precise while writing the aim of the research to avoid any confusion.

Research objective refers to the action(s) you will take in order to achieve the aim.

As compare to aim, research objective should be specific statements that define measurable outcomes, e.g. what steps will be taken to achieve the desired outcome. When writing your objectives try to use strong positive statements.

Basically, to achieve the research aim, it is breakdown into research objectives.

So, if you become able to write the research aim and objective with clarity and make connections between the two. You may be in a position to define the research scope. Scope mean that what variable or aspects of phenomenon under investigation you are going to cover in your research.

Recognizing and remembering the aim and objectives of the research may help you focus on the defined variables.

Topic 39: Research Aims and Objectives

Let us continue our discussion. As I have told you in the previous module that keeping research objectives precise will make you able to implement the design and scope of your research in an appropriate way. Now the question is,

if want to keep our objective precise, what we need to do practically?

What I need to focus specially while writing the statement?

You might have listen many times that objectives must be smart.

Here we will explain that the word SMART stands for what? What step are involved in it while writing the SMART objectives?

S.M.A.R.T. is a simple acronym that can be used to help you identify and evaluate project objectives.

Specific (What is the specific task?):

- Who is involved?
- What do you want to accomplish? It is directly involved your research aim.

Measurable (What are the concrete criteria for measuring change?):

- How will we know that change has occurred?
- How much change is expected?

Achievable (Is the task feasible?):

- Are the limitations and constraints understood?
- Can you meet this objective with the resources available, and within the proposed timeframe?

Relevant and/or Realistic (Does this relate directly to the aim?):

- Does the objective accurately address the scope of the problem?
- Are there other factors that need to be considered to ensure the plan works?

Time-specific (Within what timeframe are you working?):

- When will this objective be met?
- When can short term outcomes be expected?

So, after writing your research objective, you must check the written statements against above mentioned steps to ensure to complete your research study within the given time frame and by meeting the set criteria.

Topic40: Formulating Research Aims and Objectives S.M.A.R.T

In the previous module, we learned about the **SMART** technique, which helps us keep our objective statements focused and specific. **SMART** stands for:

S: Specific,

M: Measurable,

A: Achievable,

T: Time-bound,

R: Relevant.

It is essential to avoid poorly written objectives that we often encounter in theses and research papers. For example, a poorly stated objective might be

"Analyzing Leadership Practices."

There are different dimensions of this concern. Leadership practices can be of different nature. As part of all leadership practices, I currently have to address all of the practices in leadership practice that are not easily quantifiable in any single research unless I specify that which was the first point. There is a

specific point in the mind of the writer of the objective statement, but when he writes it, he writes it generic, which then makes its scope limited. If we use the word analysis instead of leadership practices, let's see, analysis is a big broad word, which we call them as verbs. There are different angles to analyze. This is that the words you choose, the verbs you use, the more specific, the more to the point, the easier it will be for you to convey it to others and to understand it yourself.

To improve this objective, we can start by converting the research title into a problem statement or an objective statement. People often write objectives in a generic manner, limiting their scope. By using specific and precise verbs, we can convey our intentions more clearly to others and enhance our own understanding. Let's consider the following objective statement:

"There is a study into the impact of ethical behavior of school leaders." In this statement, we have made the word **"leaders"** more specific by specifying **"school leaders."** We have also chosen one aspect of leadership, which is ethical behavior, to focus on.

- **What do teachers think about at the level of teachers' motivation?**
- **What is teachers' motivation?**

It is apparent that the objective statement of a study will not be one but multiple. Sub-variables will be from among the variables of organizational culture we took leadership practices which is one of the set of sub-variables and on the other hand we took a sub-variable of teacher commitment teacher motivation and the first objective is focus on it. We have chosen to focus on one aspect of leadership, namely ethical behavior. Additionally, we aim to explore teachers' motivation as a sub-variable of teacher commitment. By utilizing strong and clear verbs, our objective is to investigate the influence of ethical behavior on both school leaders and teachers' motivation. This specificity also extends to the variables we need to collect data on, which include ethical behaviors and teacher motivation.

When analyzing our objective statements against the **SMART** criteria, we ensure that they are measurable, achievable, time-bound, and relevant. This practice enables us to write objectives that are clearer for readers and more aligned with the purpose of our study or problem statement. Furthermore, it is crucial to pay attention to proper English grammar when formulating objective statements. By following these guidelines, we can write objectives that are more concise, effective, and grammatically correct.

Topic41: Connecting Objectives and Research Questions

After discussing the statement of objectives and the SMART ad it 5principle, the next step is to break down each objective into research questions or hypotheses. However, let's make a small distinction here. Whenever the objectives are breakdown, it must be written in the form of research questions or research hypothesis. Research questions are made usually in such cases where inferential states are not involved, instead, we will be collecting data and assessing the extent of each variable in a descriptive manner.

Regarding the research questions or hypotheses, let's consider our objective of studying the **impact of ethical behavior exhibited by school leaders on teachers' motivation**. This objective is derived from our previous model, where we explored the relationship between organizational culture in public schools and teacher commitment. Now, we have narrowed down the scope to ethical behaviors exhibited by school leaders, specifically head teachers, and their impact on teacher motivation (a sub-variable of teacher commitment).

To further address this objective, we need to break it down even further. For example, we need to define what we mean by ethical behavior and identify indicators that determine whether a behavior is ethical or not. I have listed three indicators related to head teachers, but it's important to note that there could be more indicators, which can be found through a literature review. These indicators form the basis for three research questions, although there may be more than three indicators in total.

Let's take the first research question as an example:

"What language does the head teacher use when addressing teachers?"

This question focuses on the language used by head teachers towards teachers and its relevance to ethical behavior. Another research question could be:

"Which appreciation strategies does the head teacher use?"

Appreciation strategies can be considered as an indicator of ethical behavior. Lastly, a research question could explore how well the head teacher handles issues that arise among teachers in the workplace. This also serves as an indicator of ethical behavior.

By collecting data on these research questions, we can gain insights into the ethical behavior of head teachers and its impact on teacher motivation. This information can also provide an understanding of the head teacher's leadership style and the overall organizational culture.

It's important to note that not all objectives require multiple research questions or hypotheses. Depending on the nature of the objective, you can choose to write either research questions or hypotheses or both. If there is a relationship or difference to be investigated, hypotheses can be formulated. Therefore, each objective may be accompanied by its own statement of research questions or hypotheses.

Remember to write your statements in a clear and correct manner, aligning them with the nature and objectives of your study.

Lecture 12: Research Hypothesis

Topic42: Formulating and Stating Hypothesis

Let's talk about formulating and stating hypotheses.

Hypotheses play a significant role in the research process. When formulating research objectives, they are further divided into research questions or research hypotheses.

But what exactly are hypotheses, and how are they written?

Whenever a problem occurs in human mind, it is very natural the solution will also come in mind irrespective it may be right or wrong, but if you think about something and don't think of a solution, it doesn't happen.

For example, let me say to you that you have bought bread from the market, then immediately a way is formed in your mind that from where to go, from where to buy, how to get back home, ego is a natural thing. The person said that this question was put in front of me, this task was put in front of me, but there was no solution in front of me for this task. There is no guarantee that that way will be the most correct way to go, but it will definitely be one of the tentative ways to address the problem.

A **hypothesis** can be defined as a tentative statement or a proposed solution to a research problem. It is based on prior knowledge, reading, and experience. When faced with a research question or problem, individuals naturally start thinking about possible solutions. While these solutions are not guaranteed to be the most accurate or effective, they represent tentative ways to address the problem at hand.

Hypotheses are derived from the researcher's predictions regarding the research question. Each hypothesis represents the researcher's estimated answer or solution to the problem being investigated.

However, it is important to note that a hypothesis is not the final solution. It is a tentative statement based on the researcher's knowledge, observations, and experiences.

A hypothesis is not some random or haphazard idea. It is formulated based on relevant information and is a thoughtful prediction by the researcher. It serves as a starting point and guides the rest of the research process, including data collection, analysis, and decision-making.

Throughout the research process, the researcher collects data and analyzes it to determine whether the hypothesis holds true or not. The researcher does not commit to proving the hypotheses correct but rather seeks evidence to support them. If the evidence does not align with the hypothesis and it is rejected, it still provides valuable information and becomes a finding in itself.

It is essential to understand that writing a hypothesis does not obligate the researcher to prove it correct. Even if the hypothesis is disproven, it contributes to the body of knowledge by highlighting the reasons for its rejection. These reasons and findings help in better understanding the research problem.

Therefore, when writing hypotheses, it is crucial to acknowledge that they are tentative statements and not definitive solutions. They are valuable tools for guiding research and generating meaningful insights, regardless of whether they are proven or rejected.

Topic 43: Inductive & Deductive Hypothesis

Now let's talk about types of hypotheses. Hypotheses as I discussed earlier have a tentative solution, can be written in the classification of inductive or deductive and the second way to classify is to segregate them by the way they write their statements. We discuss what kind of hypothesis statements are written as a result of the classification we do in these two ways.

Let's delve into the differences between inductive and deductive hypotheses and explore how they are written.

Inductive Hypotheses:

Inductive hypotheses are formulated based on specific observations or experiences rather than pre-existing theories. They are derived from ground-level, personal, and specific experiences. These

hypotheses are typically used when addressing problems that are not directly related to theory testing. Inductive hypotheses move from specific instances towards generalizations.

Example:

"I have observed that students in the eighth grade experience less anxiety when taking essay-based tests compared to multiple-choice tests. Based on this observation, I hypothesize that essay-based tests are associated with lower levels of anxiety in eighth-grade students compared to multiple-choice tests."

Note that the inductive hypothesis is driven by the specific observation and does not rely on a pre-established theory.

Deductive Hypotheses:

Deductive hypotheses, on the other hand, are formulated based on pre-established theories or generalizations. They are used when researchers aim to test a specific situation against the backdrop of existing theoretical conditions. Deductive hypotheses move from generalizations to specific cases.

In this deductive hypothesis, the researcher starts from a generalization and tests it in a specific situation.

It's important to note that regardless of the type of hypothesis (inductive or deductive), it should be testable, specific, and falsifiable. Hypotheses provide a tentative solution to a research question and guide the research process.

Module 44: Non-directional, Directional and Null Hypothesis

Hypothesis are classified in terms of statements. Statements refer to their write-up. Hypotheses are written in three ways: In research, hypotheses are classified based on the statements they make. There are three types of hypotheses: non-directional hypotheses, directional hypotheses, and null hypotheses. Let's examine each of them individually.

1. Non-directional Hypothesis:

A non-directional hypothesis acknowledges the existence of a relationship between variables but does not specify the direction of that relationship. It does not state which variable is greater or lesser, better or worse, or ahead or behind. It simply suggests that a difference or relationship exists without indicating the specific direction. For example, consider the hypothesis: "There will be a difference in

achievement between students who receive interactive multimedia instruction and those who receive traditional instruction." In this case, the hypothesis acknowledges that there will be a difference in achievement but does not specify whether the students receiving multimedia instruction will perform better or worse than those receiving traditional instruction.

2. Directional Hypothesis:

A directional hypothesis, on the other hand, not only acknowledges the existence of a relationship but also specifies the direction of that relationship. It predicts which group or variable will perform better or worse than the other. Using the previous example, a directional hypothesis could be: "Students who receive interactive multimedia instruction will achieve higher performance compared to those who receive traditional instruction." Here, the hypothesis not only states that there will be a difference but also predicts that the group receiving multimedia instruction will perform better than the group receiving traditional instruction.

3. Null Hypothesis:

A null hypothesis states that there is no significant difference or relationship between variables. It assumes that any observed difference or relationship is due to chance or random variation. In the context of the previous examples, a null hypothesis would be: "There will be no difference in achievement between students who receive interactive multimedia instruction and those who receive traditional instruction." This hypothesis assumes that there is no significant difference in achievement between the two instructional methods.

The choice of which type of hypothesis to use depends on the research question and the specific goals of the study. Non-directional hypotheses are useful when the researcher wants to explore the presence of a relationship without making specific predictions about the direction. Directional hypotheses are appropriate when the researcher wants to make specific predictions about which group or variable will perform better or worse. Null hypotheses are used to test the absence of a significant difference or relationship.

Topic 45: Characteristics of stating Hypothesis

Statement of Hypothesis is our topic, earlier we have written hypotheses in three different forms which are directional, non-directional, null hypothesis, now you have to write a hypothesis in any form of this

hypothesis. What things should be taken into account while writing a statement. By writing a statement, we mean the language or the write-up of English or Urdu in which I write the statements.

Use the Following format to write a clear and concise statement of hypothesis, you should consider the following guidelines:

1. Eliminate unnecessary words: Use only the words that are essential to convey the meaning of your statement. Remove any words that do not add clarity or are redundant.

2. Use clear language: Avoid using ambiguous or vague terms. Choose words that have precise meanings and convey your intended message accurately.

3. Novelty in language: While clarity is important, try to use language that is engaging and avoids excessive repetition. Use appropriate vocabulary and sentence structures to make your statement interesting to read.

4. Choose the appropriate type of hypothesis: Determine whether your hypothesis is directional, non-directional, or a null hypothesis. Clearly indicate the relationship or difference you expect to observe between the variables.

5. Specify the variables: Clearly identify and define the independent and dependent variables in your hypothesis. Ensure that the variables are measurable and there are appropriate methods available to measure them accurately.

For example

Hypothesis: Among low-achieving tenth-grade students, the presence of a tutor will significantly improve academic achievement compared to the absence of a tutor.

In this hypothesis statement, we have considered the three key elements:

Subject of the research: Low-achieving tenth-grade students.

Independent variable: Presence or absence of a tutor to mentor 12th-grade students.

Dependent variable: Academic achievement.

6. Maintain consistency: If your research aims to investigate differences, use appropriate terms related to differences. If it focuses on relationships, use terms related to relationships. Avoid mixing

both types of terms in a single hypothesis. For example, relationship or difference will not be used together rather keeping your phenomenon in mind, be clear whether it is to see the relationship, or the difference.

By following these guidelines, you can create a clear and concise statement of hypothesis that effectively communicates your research question and the expected relationship or difference between variables.

Topic 46: Testing the Hypothesis

After writing a clear and concise statement of hypothesis, it provides several benefits for the subsequent research process. A well-defined hypothesis helps in various aspects, such as sample selection, tool development, research design, and determining research procedures.

For instance, if the hypothesis focuses on low-achieving students, it guides the researcher in selecting the appropriate sample from this specific category. This simplifies the sampling procedures to be followed.

Similarly, if the hypothesis states a difference between two variables, it aligns with tests of difference in statistics. The hypothesis indicates which inferential tests will be relevant for analysis and which ones will not. The statement of hypothesis serves as a commitment or a proposition to be tested, and it informs the data collection procedures required to verify or refute the hypothesis.

Any result obtained from hypothesis testing falls into two categories: either supporting or rejecting the hypothesis. Both outcomes are considered findings. The responsibility of the researcher lies in ensuring a correct research design, proper sampling, accurate data analysis, and using appropriate tests based on the demands of the research context.

The researcher is not burdened with proving the hypothesis itself. The focus is on executing the research process correctly. The result, whether accepting or rejecting the hypothesis, leads to the dissemination of the reasons behind its acceptance or rejection, contributing to the body of knowledge. Furthermore, the outcomes may generate new hypotheses and provide directions for future research.

Therefore, the significance of a hypothesis lies not solely in the results but also in the rigorous procedures employed during the research process.

Topic 47: Characteristics Qualitative Quantitative Research

Hypothesis is more commonly associated with quantitative research. In quantitative research, hypotheses are formulated and tested using empirical data, and inferential statistics are often employed for data analysis.

On the other hand, **qualitative research focuses** on generating hypotheses or theories by exploring and understanding phenomena. It is a process of inquiry where the researcher aims to gain insights, understand the nature of different occurrences, and explore relationships between various phenomena. In qualitative research, the emphasis is on developing an understanding rather than testing specific hypotheses.

While **qualitative research** does not typically involve hypothesis statements, it is not uncommon for researchers to have guiding hypotheses or research questions. These hypotheses or questions serve as a starting point for exploration and understanding. They provide a framework for inquiry and direct the researcher's attention to specific aspects of the phenomenon under study.

However, in qualitative research, these guiding hypotheses or questions are not treated as formal hypotheses in the deductive sense. They are not subjected to hypothesis testing or statistical verification. Instead, the focus is on exploring and interpreting the data to gain a deeper understanding of the phenomenon.

Therefore, in **qualitative research, the role of hypotheses is different** from that in quantitative research. They serve as exploratory statements or assumptions that guide the research process rather than being tested or validated through statistical analysis. It is important to use appropriate language to describe the nature and purpose of hypotheses in qualitative research, emphasizing their guiding role rather than their traditional hypothesis-testing function.

Lecture 13 Literature review-I

Topic 48: Difference in Literature Review and Introduction

Earlier, we discussed how to write the statement of research objectives, statement of the problem, research question, and hypotheses. Now, logically, the next step in the research process is the literature review. First, let's discuss the difference between the introduction and the literature review.

Generally, we have observed that when people write the literature review in their theses, the introduction and literature review are often mixed up, making it difficult to distinguish between the

two. Without a clear understanding of the differences and how they should be presented, it becomes challenging to write them in a coherent manner. In addition to the literature review and introduction, the term "rationale" is also mentioned. To further emphasize its importance, I will read a text excerpt from a published research paper, and you can determine whether it belongs to the introduction or the literature review section.

Approximately one-third of American women are obese (Hedley, Ogden, Johnson, Carroll, Curtin, & Flegal, 2004). Most are attempting to lose weight through either caloric restriction (dieting) alone or caloric restriction combined with increased physical activity (Powell, Calvin, & Calvin, 2007). Although the association of weight loss and reduction in health risk is acknowledged, a primary reason for women attempting weight loss is an improvement in satisfaction with their bodies (Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999). Analysis of body image as a changing process has been advocated (Gleeson, 2006), however research on the association of weight changes and changes in body satisfaction has been unclear (Houlihan, Dickson-Parnell, Jackson, & Zeichner, 1987; Foster, Wadden, & Vogt, 1997).

While obvious markers of one's body such as weight and waist circumference are readily available (through, for example, self-weighing and fit of clothes), some research suggests that feelings of competence and self-efficacy, associated with participation in an exercise program, predicts improved satisfaction with one's body even when little physiological change actually occurs (Annesi, 2000, 2006). Research also suggests ethnic differences in what is acceptable to women regarding the shapes and sizes of their bodies (Rodin & Larson, 1992; Roberts, Cash, Feingold, & Johnson, 2006; Powell, et al., 2007). For example, Euro-American women have been described as being more critical of their bodies than African-American women (Miller, Gleaves, Hirsch, Green, & Snow, 2000). Research on psychological responses to weight loss behaviors have only rarely accounted for ethnic differences. This is exemplified in some of our recent research with obese women (Annesi & Whitaker, 2008).

Source: Annesi, J. J. (2009). Correlations of changes in weight and body satisfaction for obese women initiating exercise: Assessing effects of ethnicity. *Psychological Reports*, 105(3), 1072–1076.

Now, let's break down this text. The initial point of this text is the problem being addressed, which is women's satisfaction with their physical appearance. If you recall the lines on the slide, the first two or three lines mention the prevalence of this problem. It provides information about the extent of the problem. After discussing the problem's size, it is mentioned that previous researchers have also addressed this issue. It states that some people try to solve it through dieting, while others attempt a combination of dieting and exercise. This indicates the methods people currently use to address the problem.

Next, it is mentioned that if the problem were solely related to weight and waist measurements, there are charts available on the internet to calculate the ideal weight based on one's height. However, it is observed that even individuals who meet these criteria, meaning their weight aligns with their height,

are still dissatisfied with their bodies. Conversely, there are people who may be considered overweight but are satisfied with their appearance and enjoy life. From this, the argument is built that weight and waist ratio alone do not determine satisfaction with one's appearance.

Following this, it is stated that the problem is not purely physical, focusing on weight and waist measurements. The problem is identified as psychological. Many individuals register and pay fees for weight loss programs but never follow through, yet they are mentally satisfied with their appearance despite no physiological changes in their bodies. This argument suggests that the issue is psychological rather than purely physical. The author then states their intention to study it as a psychological construct and explores the psychological factors that contribute to satisfaction with one's appearance. Additionally, they mention their own research to further support their argument.

It's important to note that this text is not a literature review; instead, it serves as a rationale or description of the study. It provides an argument for the research being conducted and explains how the author intends to approach their study. This text should be part of the introduction section, as it provides a rational description of the study. With this understanding, we can now focus on the literature review in the upcoming modules.

Topic 49: What is Literature Review?

Now let's see about the literature review what we do in the process of research that we call that we are doing the literature review or what are the main purposes we want to achieve from it. Generally, there are four things that in the literature review. Let's break them down:

1. Surveying Existing Literature: The first objective is to thoroughly explore the existing literature in your field of study. Researchers search for relevant sources, often in digitized form, to gain a comprehensive understanding of what has already been written on their research topic. It is essential to review a sufficient amount of literature to grasp the current state of knowledge in the field. While it is impossible to read every single piece of literature on a topic, researchers aim to read a representative and diverse sample to inform their work.

Reflecting on personal experience: "I can share my personal approach to conducting a literature review. There is no standard rule or method, but when I begin researching a topic, I initially use the internet to search for material specifically related to that topic. I enter relevant keywords into Google search and download the papers or documents that appear to be useful or relevant. Once I have

downloaded several papers, I read them and take notes on the points that are helpful for my understanding of the topic.

After reading the initial set of papers, I examine their reference lists. From those references, I select additional papers that seem relevant to my research. I perform another search to find and download these selected papers. I then repeat this process, examining the reference lists of the newly downloaded papers to find even more sources. Sometimes, I may not be able to find all the papers I initially selected, but I continue the process until I start encountering a significant amount of repetition in the references.

When I notice that the same works are being referenced repeatedly, typically around 70-80% of the time, I consider it an indication that I have covered a substantial part of the published literature on the topic. At this point, I feel confident in starting to analyze and synthesize the information I have gathered. It is worth mentioning that during the literature review process, I may occasionally identify the need for information on other related topics, prompting me to conduct additional searches.

This approach helps me gauge the extent of the published literature I have reviewed and ensures that I have a comprehensive understanding of my research domain.

2. Synthesizing Information: Once relevant literature is gathered; the next step is to synthesize the findings. This involves summarizing and organizing the information obtained from different sources. By synthesizing the literature, researchers gain a comprehensive understanding of the topic, identify common themes, and discover gaps in the existing research. Synthesizing the information allows researchers to connect the findings to their own research objectives.

3. Critically Analyzing the Literature: Critical analysis involves evaluating the quality and validity of the reviewed literature. Researchers assess whether the conducted research was well-designed, if the methodology was sound, and if the conclusions drawn are supported by the evidence. By critically analyzing the literature, researchers identify strengths and weaknesses, potential biases, and areas for improvement. This analysis helps them develop a strong foundation for their own research and identify gaps in the current knowledge that their study can address.

4. Writing and Organizing the Literature Review: The final objective is to write a coherent and logical literature review. Researchers must present the reviewed literature in a structured manner, conveying their understanding, synthesis, critical analysis, and the inferences they have drawn. This involves organizing the literature review in a logical sequence, determining the order of topics,

allocating appropriate space to each aspect, and effectively communicating the importance and relevance of the literature to the reader.

By accomplishing these objectives, researchers can provide a comprehensive overview of the existing knowledge, identify research gaps, and lay the foundation for their own study. The literature review serves as a critical component of any research project, providing context, supporting the research rationale, and guiding the direction of the study.

Topic 50: Why to do Literature Review?

After understanding the process of the literature review now, we need to understand its need and significance. Key questions in this regard can be:

- What if we do not review the literature?
- How it impacts on the research process?
- What are the reasons of doing a literature review?

There are several reasons to review the literature:

1. **Identify the developments in the field of study:** This is important to know in order to avoid re-doing of research already conducted. Usually, we assure it before the finalization of our topic however, we have to do it again after the finalization of our research topic.
2. **Learn about the information sources and the research methodologies:** It may help us to learn about the methodology employed by different research and the results and inferences drawn were effective or not? What type of improvements are needed to do effective research?
3. **Find gaps in the literature that can become research questions:** Usually, whenever we talk about the literature review, we used to say that it is for the sake of finding the research gaps. That is true however, finding the research gap is one of the components of literature review. There are certain other components as well. We are doing research from many years and suggesting solutions but why the problem is not being sorted out? Either the problem is changing its nature or may be our methods to solving the problems are not that much effective. Studying these two aspect direct us to find the research gaps.

4. **Validate the originality of a research project:** Originality means the newness of my idea, methods, approach, design, and my suggested analysis if it is compared it with the other previous researchers. So, in the same context, the methods are being evaluated to check that by combining certain components of certain methods can bring the effective solutions to my research problem.

5. **Evaluate the methods:** literature review informs me about the appropriate theoretical as well as the conceptual framework for my study. The research may be able to decide it after analysis the successes and failures of already conducted research.

6. **Identify errors to avoid:** The drawbacks, problems and errors in the previously conducted research may be taken as learning to avoid in research.

7. **Highlight the strengths, weaknesses and controversies in the field of study:** Recognizing the weakness and controversies may also help to repeat those mistakes in our research.

8. **Identify the subject experts:** Literature review will also help to know connect with the expert of the subject and about aspects they are working.

So, these are the major reasons to conduct the literature review.

Topic 51: What is the Outcome of Literature Review?

Whenever we do literature review, we read a lot more than we write because we have to write the synthesized information relevant to our topic in an organized way.

The outcome of a literature review is multifaceted and serves several objectives. When presenting a literature review, it is important to keep these objectives in mind to ensure that the chapter or write-up effectively reflects these outcomes. The following five points highlight the key outcomes of a literature review:

1. **Inform the audience of developments in the field:** The literature review should present relevant information that informs the audience about the current state of knowledge and developments in the field. It is crucial to selectively choose and present information that directly relates to the research topic. While there may be numerous research studies similar to the topic, their context and purpose may differ. By presenting relevant information, the reader can better understand the researcher's specific research topic.

2. **Establish credibility:** A comprehensive and well-read literature review demonstrates the researcher's credibility and expertise in the field. By showcasing a deep understanding of the topic through correct and updated references, the researcher establishes themselves as a reliable and knowledgeable researcher. It is important to demonstrate both horizontal and vertical reading of the literature. **Horizontal** reading refers to exploring a variety of literature on the topic, while **vertical** reading involves delving into the details and nuances of the reviewed studies. Demonstrating these aspects in the write-up helps establish credibility.

3. **Discuss the relevance and significance of research questions:** A literature review aims to establish the relevance and significance of the research questions being investigated. The connection between the research questions and the literature review needs to be explicitly made. By aligning the research questions with the existing literature, the researcher demonstrates the importance of investigating the chosen research area. It is essential to be aware of the connectivity between the research questions and research objectives.

4. **Provide context for the methodological approach:** In a research study conducted across different contexts and cultures, the methodological approach may vary. The literature review should discuss the context-specific arguments that establish the relevance of the chosen methodological approach. Additionally, it should explain why previously employed methods were not suitable for the specific study. This provides a contextual framework for understanding the chosen research methodology.

5. **Discuss the relevance and appropriateness of the approach:** It is critical to discuss the relevance and appropriateness of the research approach employed in the study. This involves proving that the chosen method was the most suitable for the research and aligns with the research objectives. Simply cutting and pasting information from the literature without considering these factors will not effectively address these points. Describing the chosen approach in an appropriate and well-thought-out manner is an art in itself.

By considering these five outcomes, a literature review can effectively inform the audience, establish the researcher's credibility, discuss the relevance of research questions, provide context for the methodological approach, and demonstrate the appropriateness of the chosen research approach. It is important to avoid relying solely on copying and pasting literature without establishing connections and coherence between the different pieces of information.

Lecture 14 Literature review-II**Topic 52: Organization of Literature Review?**

Usually, students perceive the literature review as a synthesis of all previous research, which is true to some extent. However, the scope of the literature review extends beyond just synthesis. It typically consists of three elements that can be organized into separate sections within the chapter.

1. Explanation of variables and operationalization: In this section, the researcher explains the variables under study and how they will be measured or operationalized. For instance, if the study aims to measure the effect of something, there will be one or more independent variables and a dependent variable. The same applies to correlational studies. In the context of social sciences, variables may have multiple interpretations, which can lead to confusion when reviewing the literature. For example, the concept of "motivation" may have further types such as intrinsic and extrinsic, with meanings and scopes that may differ from one's understanding. It's important to describe the scope of relevant variables as explained in the literature and highlight the indicators associated with those variables. Additionally, researchers should explain why they have chosen specific indicators and why alternative indicators are not being used.

2. Theoretical justification of the study: The second section focuses on providing a theoretical justification for the research. Here, the researcher explains the logical connection between the phenomenon under investigation and relevant theories related to the research topic. It is crucial to accurately represent these theories and their key concepts. By doing so, researchers demonstrate that their research questions are grounded in existing knowledge and contribute to the theoretical understanding of the topic.

3. Synthesis of previous research: The third part involves synthesizing previous research studies, examining their methods, tools, analysis procedures, and findings. This section is often considered the core of the literature review. However, it is important to note that the literature review chapter remains incomplete without the aforementioned two sections. The synthesis of previous research helps provides a comprehensive overview of the existing knowledge in the field and identifies gaps or areas for further investigation.

The most commonly asked question is ‘**what is the length of this chapter (literature review)?**’

Regarding the length of the literature review chapter, there are no strict rules. However, a common guideline suggested by teachers and professors is to make it approximately one-third of the total thesis length. It's important not to use this chapter solely to increase the volume of the thesis but rather to serve the purpose of the literature review, which is to provide a critical analysis of existing research and support the rationale for the current study.

Topic 53: Explaining Variables

In the previous module, we discussed about the three sections a chapter of literature review must include. The first section discussed was 'explaining variables'. In the current module, we will discuss about the points that may help you to write the section in a comprehensive way.

1. Encompass the meanings of variables: It is important to approach the literature with an open mind rather than being preoccupied with your own thoughts and biases. Engage in extensive reading to gather a wide range of definitions and meanings associated with the variables of interest. By encompassing all given definitions and meanings, you demonstrate the breadth of your reading and showcase a comprehensive understanding of the variable.

2. Identify commonalities and differences: Analyze the various explanations of the variables found in the literature and identify commonalities and differences among them. List the points of agreement across different definitions to highlight the commonly accepted aspects of the variable. Additionally, acknowledge and document any differences in views or interpretations. This demonstrates a thorough review of the literature and provides a comprehensive overview to the reader.

3. Operationally define the variables with justification: Once you have reviewed and analyzed the different definitions, you can proceed to operationally define the variables. Operational definitions specify how the variables will be measured or observed in the research study. It is important to justify your choice of operational definitions based on the literature you have reviewed. By doing so, you provide a clear understanding of how the variables will be applied in your study. Operational definitions are typically presented at the end of the first section of the literature review.

Rely on books for operational definitions: When forming operational definitions, it is advisable to rely on books rather than solely on published papers. Books often provide more comprehensive and foundational explanations of concepts. While published papers may contain definitions from specific

authors, books serve as original sources that can provide a broader understanding of the variables. Therefore, consult books to form your operational definitions.

By following these points, you can write the section explaining variables in a comprehensive manner. Remember to approach the literature with an open mind, identify commonalities and differences, justify your operational definitions, and rely on books as primary sources for conceptual clarity.

Topic 54: Theoretical Justification

Theoretical justification, also known as the theoretical framework, is an essential part of research. It helps researchers understand and explain the relationships between variables or phenomena they are studying. The theoretical framework provides the logical reasoning and theoretical basis for the selection of variables and the examination of their relationships.

When conducting a study on the effects or relationships between variables, researchers rely on theories in their field of study. Theories explain how variables are connected and how they influence each other. **For example**, in the field of education, there are theories of motivation, personality, and learning that provide explanations for the relationships between variables in that domain.

To determine which variables to study and the relationships to explore, researchers consult relevant theories. The theories guide them in identifying variables that are logically connected and likely to exhibit a cause-and-effect relationship. **For instance**, if researchers want to investigate the relationship between people's morning exercise and their health, they would turn to theories that explain the impact of physical activity on health outcomes.

Theoretical justification involves connecting variables based on established theories and logical reasoning. It ensures that the chosen variables and their relationships are grounded in existing knowledge and make sense theoretically. Without a theoretical basis, a relationship between variables may lack logical coherence and justification.

Let's consider an example for illustration purposes, although it may not represent an actual relationship. Suppose a researcher collects data on the daily duration of the prime minister's morning walks over the past year and the wheat production in the country during the same period. They calculate the average walk time for the past five years and compare it to the wheat production levels. Although statistical analysis may reveal a correlation, the relationship lacks a logical explanation and

theoretical justification. It is important to establish a theoretical basis for the relationship between variables.

It is advised not to study relationships or effects that lack theoretical justification or causation. Theoretical justifications are typically based on theories found in books and scholarly sources rather than other research papers. It is crucial to thoroughly review the literature in the area of study to identify relevant theories and understand their implications for the research topic.

In summary, the theoretical justification, or theoretical framework, provides a logical and theoretical basis for studying the relationships between variables. It involves drawing on established theories to support the selection of variables and the examination of their relationships. Theoretical justifications are important to ensure that the research is grounded in existing knowledge and contributes to the theoretical understanding of the field.

Topic 55: Synthesis of Previous Research

The third section of the literature review chapter, known as "Synthesis," serves the purpose of reviewing previous research relevant to your proposed research topic. However, it is crucial to approach this section with specific considerations in mind. The primary goal is not to create an exhaustive inventory of all the findings you have come across. Instead, the focus should be on the connection between these previous studies and your research questions.

When discussing the developments in your field, it is important to situate them within the context of your research questions. It is impossible to review all the existing research in your field, and it is not necessary to claim that you have summarized every single study. The aim is to present a summary that is relevant and directly connected to your research question. Through this process, you can identify gaps in the literature, highlighting the problem you intend to address.

As you review other researchers' work, their experiments, data collection methods, and results, you should consider them as pieces of a puzzle. By analyzing these components, you can infer the gaps that exist and understand how your research will contribute to filling those gaps. This section should demonstrate your understanding of the research methodology employed by others and how it informs your own study.

To write this section more effectively, consider converting your research question into a heading. Then, examine the previous literature on that topic and identify the gaps that exist. List these gaps and explain how your research will contribute to filling them. In your methodology chapter, you can provide a detailed demonstration of how you will address these gaps.

It is important to note that the synthesis of previous research in this section should not solely rely on the book you have read. Instead, it should draw from various scholarly sources and research papers. The aim is to extract information and identify gaps in the literature related to the phenomenon you are studying—whether it involves relationships, differences, causal effects, or other logical conclusions.

By reading, synthesizing, and identifying gaps in previous research, you lay the foundation for the subsequent sections of your study. The reader should be able to understand the methodology you will employ based on the conclusions drawn from these three sections. Therefore, the conclusions of each section should contribute to the reader's understanding of the overall research design.

In summary, the synthesis section of the literature review chapter aims to review previous research in relation to your research questions. It is important to focus on relevant studies and identify gaps in the literature. This section should connect with the methodology chapter, where you will provide a detailed explanation of how you will address these gaps.

Lecture 15. Literature review-III (56-61)

Topic 56: Identifying Sources for Literature Review

In current times, access to the information is enormous. So, here is the need to decide that which type of literature is important to review and we can utilize it. We will discuss in detail specially to differentiate among the different resources of literature review.

Usually, we divide literature sources in two types:

- **Primary sources**
- **Secondary sources**

Primary sources: Original documents that contain firsthand information

Publication, or description in which researchers report the results of their own studies

The collected information is factual rather than interpretive. **One example** can be the constitution of Pakistan which is the original and a primary source but the books that are written for the explanation of constitution are the secondary sources.

Other Examples of Primary sources:

Surveys/Interviews

Original Documents (e.g., letters, reports, policies, government records)

Raw Data (e.g., scientific data, market data)

Artifacts (e.g., archaeological finds, historical objects)

Audio/Visual recordings (e.g., videos, audio recordings)

In short, taking information from the original sources is called primary or firsthand information. In research, information from primary sources is proffered over others in terms of its authenticity, and its originality.

Topic 57: Secondary Sources of Information

As we have already discussed that material for literature review is divided into two types:

Primary sources of Information (First hand and original documents e.g. Constitution of Pakistan)

Secondary Sources of Information (Books written to explain the Constitution of Pakistan)

Secondary sources are materials that are based on primary sources and provide interpretation, analysis, or synthesis of the primary sources.

Although the secondary source of information is not as much authentic as the primary source however, it is a also credible source of information. It is based on the people's point of view based on their analysis, synthesis or logic. So we can't deny the importance of secondary sources of information because usually we can access to secondary sources most of the time. However, we are supposed to prefer primary sources over secondary when we have choice between the two.

Examples of secondary sources are:

1. Books (But not all books, only those that are written as interpretation of the primary source)
2. Newspaper articles (Based on opinions of the people)
3. Encyclopedia articles (Provides you the information on a specific topic)
4. Documentaries (For explanations)
5. Review articles (Provide the synthesis of the research on any topic)
6. Statistical compiled (For example reports issued from the bureau of statistics)
7. Commentaries (Audio/visual)

Topic 58: Primary Sources Vs Secondary Sources of Information

In this lecture, we will present you a comparative view of primary and secondary sources of literature review so that you may be able understand the difference and decide when and which resources could be appropriately utilizing.

It is preferable to use primary resource to the possible extent. We use five following points as yard stick to analyze the importance of primary sources over the secondary.

- Originality
- Authenticity
- Purpose
- Evidence
- Type of Information

The following table presents a comparison of the primary and secondary sources of information based on the above mentioned points:

Criterion	Primary Sources	Secondary Sources

Origin	They come directly from the people, events or time period being studied.	They are created later, often by someone not directly involved in the events.
Authenticity	They are more authentic because they provide direct account of the events	They may contain bias or interpretation.
Purpose	They record events as they happened	They provide context, analysis, or interpretation of the events.
Evidence	They are used as evidence in research	They are used to support or challenge the evidence provided by primary sources.
Type of Information	They often provide detailed and specific information about events.	They provide a broader overview or context for the events.

There is no problem or restriction to use secondary sources in the literature review but the condition is to check that source as per the above given criteria.

Topic 59: Evaluating Sources for Literature Review

Evaluating the sources for literature review is critical in current times when you have wide access to the multiple information sources due to computer and internet facility. It is the big challenge to identify credible information among these huge collections of bundle of information.

Seven criteria are proposed to evaluate any of the document (book, web page, paper) either it is authentic or not?

1. **Author:** Is the author an expert in the field? What are their qualifications? Multi-disciplinary research does not allow to jump into an entirely different field in which the author have any qualification. Multiple authors must have some common area of interest or expertise.

2. **Bias:** Does the source have a clear bias or agenda that could affect the accuracy of the information? If the authors have conflict of interest in his/her opinion, it will affect the credibility of the written material.
3. **Consistency:** Does the source present evidence to support its claims and various parts of the publication are consistent within? For example, tool must be consistent to the research objective and the same is for analysis and other steps involved in the research.
4. **Credibility of Publisher:** Is the source published by a reputable publisher or academic institution? It is of critical importance in current times when you have many choices available in the market. Adding fee component made the quality of certain publishers compromised.
5. **Review process:** Has the source been peer-reviewed by experts in the field? The review process is usually blind review which is more professional.
6. **Reliability:** Can the information be verified by other sources?

Purpose: What is the purpose of the source? Is it informative, persuasive, or promotional? Publicity or persuasive material is not considered credible.

Rubric for Evaluating Print and Internet Sources

Dimension	1 Poor	2 Below Average	3 Average	4 Above Average	5 Excellent
Relevancy	The source does not address the research interests of your study.	The source addresses one of the research interests of your study.	The source addresses most of the research interests of your study.	The source meets all of the research interests of your study.	The source meets all of the research interests of your study and provides a conceptual framework for a study that is replicable.
Author	Unclear who authored the study.	Author name and contact information is provided.	Author name, contact information, and some credentials	Author name, contact information, and full credentials	Author is a well-known researcher in the research area and provides link

			ntials are included.	are included in the source.	ks to other related research.
Source	Source is a non-refereed and is an author's opinion.	Source is non-refereed and must be closely examined for objectivity, accuracy, and reliability.	Source is a scholarly or peer reviewed journal, an education related magazine.	Source is a scholarly or peer reviewed journal.	Source is a scholarly or peer-reviewed journal with links to related literature by same author/s and downloadable.
Methodology	It is not possible to determine whether or not an appropriate methodology was used.	The description does not include sufficient information to determine research problem and the appropriateness of different components of methodology.	The source includes a full description of the research problem and the appropriate methodology to investigate the problem.	The source includes a full description of the research problem and appropriate methodology. The results are presented objectively and can be connected to the data presented in the study.	The source includes a full description of the research problem and appropriate methodology. Issues of validity, reliability, and limitations are discussed. There is sufficient information to replicate the study.
Date	No date of publication is included in the source.	Date of publication is included but is too old to be helpful for the current research problem.	Current date of publication.	Current date of publication with a list of references consulted by the author.	Current date of publication with a list of references consulted by the author including links to locate the references.

In short, the credibility of any source can be checked against the seven points discussed in the above.

Topic 60: Literature Matrix

After selecting the relevant literature, one needs to make connections among the reviewed literature documents which is not an easy job. Certain techniques are developed to make this job easier and

literature matrix is one among those techniques. Literature matrix is a two-way chart. The matrix is a powerful organizer of literature.

- Along the Y-axis list the authors' names and year of publication are written.
- Along the X-axis list the kinds of variables/themes/issues addressed by the studies.
- The matrix provides a mental map of what you are reading and what the studies share.

These matrices are formed in different styles. Following is the one example of a literature matrix:

Matrix-1

Author/s	Year	Academic Achievement	Social Adjustment	Attitude to Change	School Counselors	Solution Provided
Hussain, W., & Abid, S.	2022	●	●			
Zahida, M. & Khan, A.	2018	●	●		●	●
Shahid, F. M.	2020		●		●	
Fatima, K. A. & Bilal, A.	2020	●		●	●	●
Shah, A., Rabia, N. & Adnan, M.	2021	●	●	●		
Ikram, H. & Ramzan, M.	2019		●	●		●
Misbah, R. & Saima, W.	2022	●			●	
Zafar, M. I.	2022	●	●	●		●

By looking into the Matrix-I, you can find the author's name in the vertical form. The next column provides the year of publication so that you may be aware that the specific information or source is updated or old. In the next 5 columns, you will find five variables, which I choose by keeping the key words of my research are in my mind. The black dot in columns show that the specific variables is present in the relevant literature which was reviewed and the absence of dot tells that the variable was not studies in the specific paper. As the matrix-1 shows that the variable **academic achievement** was present in all the reviewed papers.

So, there is **another example of matrix** which provides the details of studied variables.

Author/ Date	Theoretical/ Conceptual Framework	Research Question(s)/ Hypotheses	Methodology	Analysis & Results	Conclusions	Implications for Future research	Implications For practice
Maisto Pollock Lynch Martin Ammerman (2001)	Coping factors in relationship to decreasing substance abuse with adolescents one year post drug treatment	What factors contribute to the variability in adolescent functioning regarding substance abuse one-year post treatment?	Quasi-experimental design involving 166 subjects in Pittsburgh adolescent research center. Initial baseline assessment and 1 year later. Pre and posttest measures included ACQ, ISE, CTI, LEQA, SCQ, and DUSI.	First set of analysis involved one-way ANOVA. Four independent t-tests conducted to determine specific group differences. The final set utilized ANOVA with repeated measures 1 year later. 36% of subjects discontinued alcohol use.	All clinical groups demonstrated improvement at one year.	Stress and coping model useful for examining clinical course of alcohol use disorders in adolescents.	Differences between participants at baseline regarding coping factors indicate significance of acquisition of such skills as part of treatment intervention.
De Anda Bradley (1997)	Stress, stressors, and coping strategies among middle school adolescents	Adolescents' perceptions of their stress use of coping strategies and the adolescents' evaluation of degree of success regarding	54 middle school students 12-14 years old completed ASCM and STAI.	A four point Likert scale was used for analysis. Internal consistency was .95. Results indicated female students report increased degree of stress.	School related stressors rated highest thus schools are a good place for intervention/prevention. Gender differences need to be considered.	Gender and developmental differences in coping need to be examined.	Adolescents might be amenable to treatment which teaches positive coping strategies, schools can help with this process

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This type of matrix provides a generic picture about the presence or absence of the variables but it does not explain the further details about those variables.

In this matrix, further details are provided regarding the variables studied in the reviewed literature e.g. theoretical/conceptual frame work, research questions, methodology, analysis, conclusion, implications etc. This review matrix provides the detail of the research components as well. Two examples are presented here, there are certain other examples of matrix as well.

Review matrix helps you to define the research gaps and avoid to redo the things which have already established by the existing literature.

Topic 61: Literature Matrix

(Analyzing, Organizing, and Reporting the Literature)

We have already discussed about the literature matrix in the previous module. You have already learned about analyzing the literature review by incorporating different components in the literature-matrix and drawing gaps in the previous module in the present module, we will discuss that how can we report literature in the research once it is organized. Organization of literature means sequencing and making logical connections while writing the literature and making its report. If you follow the following ten steps, it will help you in organizing and reporting the literature review.

1. Make an outline – Precise, Ordered, Sorted,

The chapter of literature is based on three sections, Operational definitions of the concepts, Theoretical framework and conceptual framework, and Synthesis of the existing research. You need to think about the sequence and order of the information in each section without disturbing those sections by making an outline in each of the section.

2. Identify and define topics and variables

The outline in each section may be helpful to decide about the order and connections among different topics or variable to be discussed.

3. Identify sources – Print and online sources with their links (DOI and web links)

We do ignore to write about the link of sources in the literature review usually. Incorporating sources link will help you to re-locate those sources in future whenever you need it.

4. **Evaluate sources** - Use the evaluation rubric

Evaluation of the sources is critical to check the authenticity and credibility of that source and a seven step rubric that we have already discussed in the **module 59** can be used for this purpose.

5. **Analyze each source in terms of outline** – Inclusion or exclusion of sources

To decide about the relevancy of the context of the variables and phenomenon studied in a specific document.

6. Start with the **least relevant sources and proceed to the most relevant** sources

Build your arguments with the least relevant sources and proceed to the most relevant sources while strengthening the argument. It is a common observation that people try to avoid the least relevant sources which is not right, instead making connections to build the argument is the appropriate approach.

7. Give a **meaningful overview** of past research

Meaningful review refers to cover all the important aspects or components of the reviewed literature in an integrated form to present a coherent picture.

8. **Follow academic writing style** and formats

Usually organization and universities decide about the format for example APA style is commonly used for social sciences at university level. A report written in a fixed or structured style will be easy to read for the reader.

9. **Divide report into sections** – Headings/Subheadings

Research thesis are organized into different chapters and chapters are sub-divided into further parts or sections. Long paragraphs without headings may be difficult for the readers.

10. **Conclude** with a brief summary

Write a concluding paragraph at the end of the chapter of literature review that what you inferred from this review. This will make the reader to understand the crux of this whole effort.

Writing and reporting literature review in this way will be useful and easy to write as well as to read and understand.

Lecture 16 Theoretical and Conceptual Frameworks

Topic 62: Approaches to Literature Review

People utilize different approaches to do literature review in relevance to the research questions, type of research and the context of the specific research. If we maintain an exhaustive list of that approaches, it may be based on 10-12 approaches of literature review. We will discuss here just four popular approaches that are frequently utilized.

These are some of the commonly used approaches to conducting a literature review. The choice of approach will depend on the research question, the type of study, and the context of the research.

Narrative review: If you look into the meaning of word 'Narrative', it shows that one need to develop a narration of the literature on the relevant topic that is under investigation. A chronological summary of research on a particular topic is maintained by mentioning the key variables and major research findings of all previously conducted research related to the topic. Usually the word 'literature review's used is typically referred to the narrative review of the literature.

Systematic review: A systematic review is basically a review around the questions and more structured. No chronological history is required in this type of literature review. Rather, one needs to build the literature around the themes or the research questions of the undergoing study.

, exhaustive, and structured search and evaluation of relevant literature.

Meta-analysis: A statistical approach that combines the results of multiple studies to arrive at a more precise estimate of an effect. It is time bound which mean many studies that are conducted during a specific time period (1980-2020) to look into to develop and decide consensus outcome of all those

study on a specific topic. The data of all studies of meta-analysis is accumulated in numeric form to calculate the affect size, variation points and discuss the findings.

Thematic review: A type of review that organizes and synthesizes the literature around a specific theme or concept. In simple words, the information from the literature is gathered and presented under the defined themes to synthesize.

Topic 63: Narrative Literature Review

We have discussed for different approaches of literature review. One of those is the 'Narrative literature review'. we have also discussed that usually the word 'literature review's used is typically referred to the narrative review of the literature. The specific type is based a chronological summary of research on a particular topic is maintained by mentioning the key variables and major research findings of all previously conducted research related to the topic. Additional is the critique which you add in it.

The three step involved in it are:

- 1. Gathering the relevant literature**
- 2. Developing the critique**
- 3. Summarize**

The first step, selection of research paper which is the part of all type of literature review. All the criterion is applied in this selection of literature which we have already discussed (related to credible sources of literature)

The second step involves the arrangement of gathered literature in chronological order. Narrative does not mean to add your view point on it, rather you just developed the narration of the key findings in a chronological order.

Critique is the third step. As per my experience, many researcher findings may support your suppositions. Some studies may support your hypothesis and some may be contradicting as well. Drawing conclusion may be difficult in such situation. Therefore, you prepare a critique while

examining the provided information and determining the value and credibility. You will then decide which findings are useful to prepare a final summary.

It is a common observation that novice researcher may be subject to bias, because researchers tend to pick up only the studies (mostly with positive results) that support their arguments. Here the learning points is that those researcher's contradictory to the researcher's suppositions may be utilized to develop a critique mechanism based on their flaws and deficiencies. In-short, while reviewing both type of studies (supportive or contradictory) for making interpretations. Element of bias is not acceptable in research therefore, one needs to learn how to resolve it.

Developing a well ground critique and summary is the most significant step in the narrative type of literature review which makes it credible and dependable.

Topic 64: Systematic Literature Review

In this module, we will discuss the approach of conducting a systematic literature review. The systematic literature review follows similar steps to any other literature review, such as clearly defining research questions, collecting relevant literature, and analyzing or synthesizing it. However, the starting point for a systematic literature review is the specific definition of research questions. Without clearly defined research questions, a literature review cannot be conducted effectively. Therefore, it is essential to begin by clearly defining the research questions, as they provide a logical structure for gathering and analyzing information.

To conduct a systematic literature review, the first step is to define the research questions precisely. This will allow for the collection of related literature, which can then be synthesized and analyzed. Defining research questions is crucial because it sets the direction for the entire review process. Without well-defined research objectives, conducting a literature review can lead to ambiguity or incorrect task execution. To ensure accuracy, it is important to make the research objectives specific and then narrow them down. This will serve as the starting point for the literature review.

The literature for the review is accumulated by searching databases using appropriate search methods. The same criteria for judging the credibility of the literature, as discussed in previous modules, should be applied during the database search. Afterward, studies are selected based on specific criteria. For example, if the review has four research questions or hypotheses, studies relevant to each objective should be identified and categorized accordingly. It is possible for a study to be applicable to one

objective and relevant to another. In this case, the relevant literature is gathered and organized around each objective.

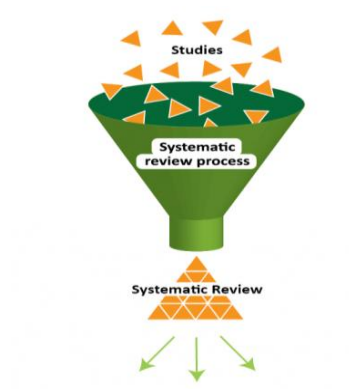
Once the literature is selected, data extraction is conducted. This involves extracting relevant information from each paper using a literature matrix, as mentioned in previous modules. The extracted data should align with the search criteria and be organized in a standardized format, as demonstrated in the literature matrix. At times, a detailed reading of the selected material may be necessary to assess its relevance, consistency, and quality. It should align with the variables and information that are sought after in the predefined set.

After organizing the literature, the synthesis process begins, where the synthesized information is reported using headings and subheadings aligned with the research questions or objectives. In the literature review chapter, this section provides a synthesis of previous research studies. It is important to understand the differences between studies and how they align with the research questions. By following these seven steps and adhering to the sequence of studying the literature around the research questions or problems, a systematic literature review can be conducted effectively.

In summary, a systematic literature review follows a specific approach that includes defining research questions, collecting relevant literature, extracting data, and synthesizing information. The key is to align the literature review with the research questions and objectives, ensuring a systematic and comprehensive review of the existing literature.

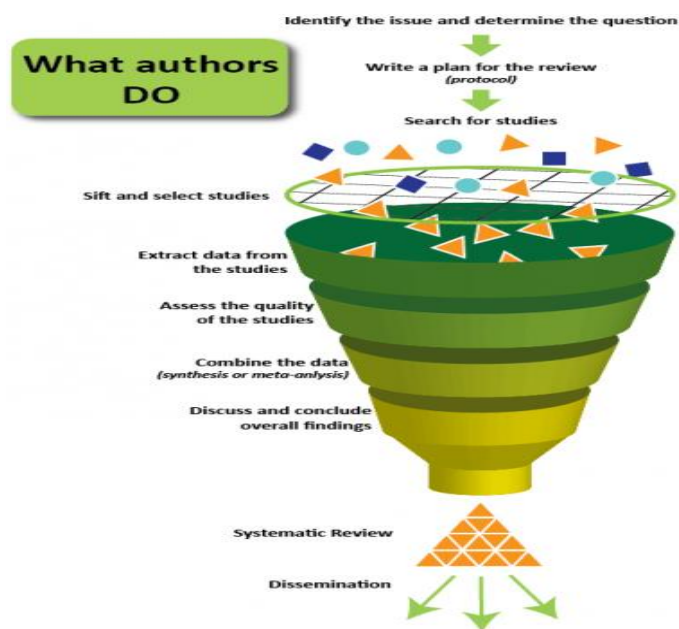
Topic 65: Steps of Systematic Literature Review

In the previous module, we discussed the concept and structure of a systematic literature review.



Figure

If you look in this figure, it looks like you will first collect studies from different places, whatever studies are related to your research topic that you have formulated the questions. Conceptually, the process involves collecting relevant studies related to formulated research questions. These research questions serve as the foundation for the review. By following a systematic approach, various types of studies are organized in a meaningful and understandable way to readers.



Moving forward, let's delve into the steps of the systematic review process. The first step is to identify the research questions that will guide the review. These questions are formulated based on the specific objectives of the review. Once the research questions are identified, the next step is to develop protocols. These protocols outline the main topics and sub-topics that will be covered in relation to each research question, providing a structured framework for the review.

After establishing the protocols, the process of collecting material begins. The collected material may vary in relevance to the research questions. Some material directly addresses the questions, while others require reshaping and interpretation to fit within the desired context. This material can be visualized as triangular shapes, with some being directly relevant and others needing further refinement.

Next, relevant data is extracted from the collected material, focusing on the information that aligns with the research questions. Once the data is extracted, it is evaluated based on predetermined

credibility criteria. This evaluation ensures that the information meets quality standards, considering aspects such as the credibility of the sources and the rigor of the research.

Following the credibility assessment, the collected information can be synthesized using different methods, such as narrative synthesis or meta-analysis. By synthesizing the data, patterns and relationships between variables can be identified across multiple studies.

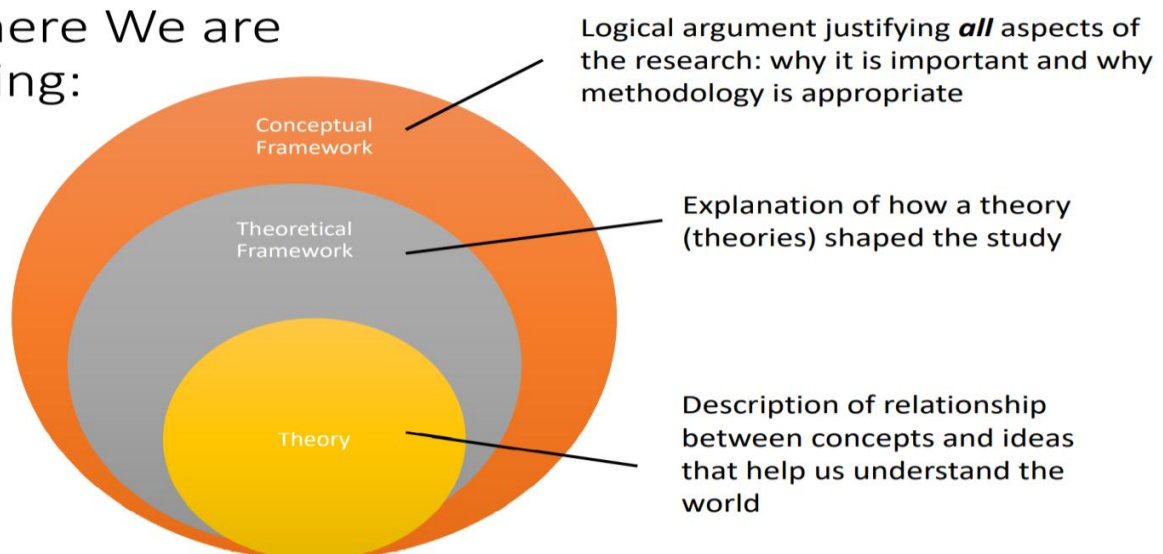
Upon completing these steps, the output of the systematic review is presented in an organized and coherent manner. The relevant papers that passed through the filtering process are synthesized to provide a comprehensive overview of the information gathered. This synthesized information then informs the methodology, tool selection, sampling, and analysis for the broader research project.

It's important to emphasize that conducting a systematic literature review requires careful planning, adherence to established guidelines, and a rigorous methodology. The process aims to provide an unbiased and comprehensive summary of the available evidence on a specific research topic.

Topic 66: Theoretical and Conceptual Frameworks

Differentiating between work-of-study, theory, theoretical framework, and conceptual framework can often be challenging for many students. Let's clarify these concepts.

Where We are Going:



When conducting research, we often aim to study a phenomenon or explore the relationship between variables. This phenomenon or relationship is the focus of our research. Before proceeding, it is important to ensure that the phenomenon we are studying is logically valid and supported by existing theories. Theories connect concepts and ideas and are based on a body of evidence accumulated over time. By connecting our research to theory, we establish the logical validity of the phenomenon we are studying.

The theoretical framework is the connection between our research and relevant theories. It demonstrates how our study aligns with and is supported by the theoretical premises that connect the variables or behaviors we are investigating. The theoretical framework may involve multiple theories, as long as they are relevant to our research topic. Creating a theoretical framework ensures that our research is grounded in existing knowledge and that our findings are logically connected to established theories.

In addition to the theoretical framework, we also develop a conceptual framework. The conceptual framework focuses on the methodology and design of the study. It relates to the variables, their relationships, the study's methodology, and the sequence of events. The conceptual framework helps us develop a logical argument for why our chosen variables, methodology, and sequence of events are appropriate for our study. It establishes the connections between the variables, methodology, and study design, and demonstrates the most suitable approach for conducting the research.

It is crucial to differentiate between the theoretical framework and the conceptual framework. The theoretical framework focuses on the logical connection between the research variables and existing theories, while the conceptual framework focuses on the methodology, design, and logical arrangement of variables and events within the study. Both frameworks play important roles in ensuring the validity and coherence of the research.

In subsequent modules, we will delve into the components of both frameworks to provide a more comprehensive understanding.

Topic 67: Difference between Theoretical and Conceptual Frameworks

In previous modules, we discussed theoretical frameworks and conceptual frameworks separately. To further understand the differences between the two, let's compare them based on several key points:

<u>Theoretical Framework</u>	<u>Conceptual Framework</u>
Broader Scope	Narrower Scope
General, Less structured, Loosely defined	Very Specific, Definite structure, no ambiguity in definition
Derived from theories that already exist	Derived from concepts, specific variables that maybe identified in the study itself.
Provide the basic foundation for a research study	Provide the structure, paradigm for a research analysis.
It is the starting point of the discussion, makes readers aware of what follows	It lies in the entire commencement of the study

Scope and Purpose:

Theoretical Framework: The theoretical framework has a broader scope and aims to ensure that the phenomenon being studied is logically grounded in existing theories. It seeks to contribute to the existing body of knowledge and extend it further.

Conceptual Framework: The conceptual framework is focused on the specific research study and does not directly contribute to the overall body of knowledge. Its purpose is to provide a roadmap for conducting the study and establish the validity of the research process.

Nature:

Theoretical Framework: Theoretical frameworks are general in nature and are often loosely defined. They connect the research topic logically to existing theories while maintaining the uniqueness and originality of the study.

Conceptual Framework: Conceptual frameworks are more specific and structured. They outline the sequence of tasks and components within the study, ensuring that the research is conducted in the best possible way. Ambiguity is minimized, and the framework is designed to guide the study.

Connection to Theory:

Theoretical Framework: Theoretical frameworks are directly connected to existing theories. They establish the logical connection between the research topic and relevant theories, but they also maintain the uniqueness of the research question.

Conceptual Framework: Conceptual frameworks do not necessarily draw directly from theories. They focus on identifying and defining the variables within the study, including independent, dependent, mediator, and moderator variables.

Contribution to Knowledge:

Theoretical Framework: Theoretical frameworks are concerned with contributing to the existing body of knowledge. They ensure that the research study aligns with relevant theories and aims to produce new insights or solutions to a specific research problem.

Conceptual Framework: Conceptual frameworks are not primarily focused on contributing to knowledge. They establish the research paradigm and structure, ensuring that the study aligns with the chosen research approach (positivist, interpretivist, etc.) and the sequence of activities within the study.

Timing:

Theoretical Framework: Theoretical frameworks are developed at the early stages of the research process, often when deciding the research topic or formulating the research question. They provide a foundation for the study.

Conceptual Framework: Conceptual frameworks are created after the theoretical framework has been established. They come into play when the research topic, question, and theoretical justifications have been finalized. Conceptual frameworks outline the specific sequence and methodology for conducting the study.

In summary, theoretical frameworks have a broader scope and focus on the logical connection between the research topic and existing theories, aiming to contribute to the body of knowledge. On the other hand, conceptual frameworks are narrower in scope, focusing on the specific research study's methodology and sequence of tasks. They establish the validity of the research process and ensure alignment with the chosen research paradigm. Remember to revise and improve both the theoretical

and conceptual frameworks to ensure your research is conducted properly and contributes to knowledge in a meaningful way.

Lecture 17 Using APA format

Topic 68: Using APA Styles Manual

In this module, we will discuss different styles used in report writing. Writing style refers to the formatting and presentation of the report to ensure clarity and understanding for readers worldwide. Various institutions have developed style manuals, and some popular ones are mentioned on the slide.

APA – American Psychological Association

MLA – Modern Language Association

Chicago Style – Chicago Manual of Style

Turabian Style – based on Chicago Style

Harvard Referencing System

ASA – American Sociological Association

CBE - Council of Biology Editors

One widely used style in the social sciences is the APA (American Psychological Association) style manual. It provides guidelines for formatting and citing sources in research reports. Another style commonly used in language-related fields is the MLA (Modern Language Association) style manual. It offers guidelines for writing and citing sources in language-focused reports.

For research reports written in languages such as Arabic or Urdu, which are written from right to left, the Turabian style manual is often employed. It was created by individuals associated with the University of Chicago.

Among these manuals, we primarily focus on the APA style because it is widely used in social sciences and education. The latest edition of the APA style manual, as shown in the slide, is the seventh edition published in 2019.

These style manuals serve the purpose of standardizing and representing reports consistently. They aim to establish a common set of guidelines for formatting, citations, and other stylistic elements. It is important to note that these manuals do not rank one style as better or worse than another. Instead, their goal is to establish uniformity and standardization in report writing.

Please ensure that you consult the appropriate style manual, such as the current edition of the APA manual, to adhere to the specific style guidelines when writing your reports.

Topic 69: General Guidelines-I

Since in the previous module we have read different styles or different manuals that discuss or describe style of report and I asked you why we use APA more in social sciences. So I will share with you the details of APA style so that you can design it accordingly when you are writing your research report. In APA style, research reports, including theses, are typically written on white paper resembling normal printing paper. The paper should be letter-sized (8.5 by 11 inches), also known as A4 size. The text should be double-spaced, with a one-inch margin on all sides of the page. If you are using Times New Roman or a similar font, the font size should be 12; for Arial, the font size should be 11. It is recommended to print the text in black ink using a standard printer.

In APA style, each new paragraph should be indented by five to seven spaces. A single page should contain no more than 27 lines. Tables or figures inserted in the report may use single spacing for the text within them, while the main body of the text should always be double-spaced.

It is important to note that while these guidelines represent the standard APA style, some institutions may have slight modifications to suit their specific requirements.

Topic 70: Reference list

"We have previously discussed and studied various forms of text citations in earlier modules. Now, let's focus on creating a reference list for all the citations used in your thesis or report. The reference list follows specific features and principles.

Firstly, the reference list should always be placed at the end of your thesis or report, just before any appendices or acknowledgments. It should start on a new page, and the heading 'References' should be centered at the top of the page as a level one heading. Each entry in the reference list should begin

flush left, with a hanging indent. This means that the first line of each entry starts at the left margin, while subsequent lines are indented by five spaces.

The entries in the reference list should be double-spaced, both within and between the entries. If a reference entry spans multiple lines, only the first line is flush left, while the subsequent lines are indented. Additionally, certain elements such as book titles, journal/magazine names, and newspaper names should be italicized in the reference list.

When arranging the reference list, alphabetical order is followed based on the authors' last names. If multiple authors share the same last name, their entries should be sorted according to their first names. If authors have the same first and last names, the publication year of their works should be considered, with the earliest publication listed first. If the authors' names and publication years are the same, the titles of their research papers are used to differentiate and arrange the entries. In case the titles also start with the same word, the subsequent elements such as the name of the journal or magazine come into play for differentiation.

If an author's name is not available, the entry is alphabetically arranged based on the first significant word of the title. If no author or title is available, the entry is treated as anonymous and placed alphabetically according to the first significant word.

It's important to note that in the reference list, titles should not be underlined or in title case. Instead, they should be written in sentence case, with only the first letter of the first word and proper nouns capitalized. Additionally, no additional formatting, such as bolding or resizing, is required for the titles. Quotation marks should be used for titles of articles or chapters within larger works.

These are the basic rules to follow when creating a reference list in APA style. While the examples in the subsequent modules will provide a better understanding, it's essential to adhere to these guidelines for accurate formatting."

Topic 71: Examples – Reference List Order

"In the previous modules, we have discussed the characteristics of a reference list in APA style. When creating a reference list for your thesis or report, there are several basic principles to follow. One important principle is the order of the list, specifically how authors are referenced. The reference list

should be arranged in alphabetical order based on the authors' last names. If multiple publications have the same author, they should be organized chronologically, with the earliest publication listed first.

Baheti, J. R. (2001a). Control ...

Baheti, J. R. (2001b). Roles of ...

Kumpfer, K. L. (1999). Factors ...

Kumpfer, K. L. (2002). Prevention ...

Kumpfer, K. L., Alvarado, R., Smith, P., ...

Yoshikawa, H. (1994). Preventions ...

Let's consider an example to illustrate this principle. Suppose we have two publications by the same author, A. Both publications happen to have titles starting with the words 'Control' and 'Role.' In this case, we would look at the first significant word after these initial common words to determine the order. If the first word after 'Control' starts with the letter 'C,' and the first word after 'Role' starts with the letter 'R,' then the publication with the title starting with 'Control' would be listed first, followed by the publication with the title starting with 'Role.' This ensures alphabetical order when considering the first significant word after any common words.

Additionally, if the authors' names and publication years are the same, we look at the first significant word of the title of the article to differentiate and arrange the entries. Let's say we have three publications by the same author, A, with publication years of 1999, 2002, and 2005. In this case, the publication from 1999 would be listed first, followed by the publication from 2002, and then the publication from 2005. The alphabetical order of the first significant word in the titles is not necessary to consider in this example since the publication years are different.

Lecture 18 Research Philosophies (72 - 77)

Topic72: What is Research Philosophy

In this module, we will discuss how our underlying assumptions about the world influence our research, specifically in the context of philosophy of science. These assumptions play a significant role in various stages of the research process, starting from selecting a research problem to conducting a literature

review, collecting and analyzing data, and drawing conclusions. Let's explore how these assumptions are connected to our concepts of reality, the generation of knowledge, and value systems, and how they shape our research process.

Concept of Reality: Our beliefs about reality influence how we perceive and understand the world. This includes our understanding of the nature of knowledge, its origins, and how we view reality. For example, positivism, one of the oldest philosophical perspectives, holds that knowledge is obtained through objective, empirical observation and that reality exists independently of our subjective experiences. On the other hand, interpretivism emphasizes the importance of subjective interpretation, social constructs, and the role of language and context in shaping our understanding of reality.

Generation of Knowledge: Our philosophical assumptions also influence how we view the generation of knowledge. Positivism emphasizes the use of scientific methods to uncover universal laws and causal relationships. It values objectivity, quantifiable data, and generalizability. Interpretivism, on the other hand, focuses on understanding and interpreting human experiences, meanings, and social phenomena. It values subjective perspectives, qualitative data, and context-specific understanding.

Value Systems: Our beliefs and values shape our approach to research and influence the selection of research problems, methodologies, and interpretations of findings. Different philosophical perspectives have varying stances on the importance of values in research. For instance, criticalism or critical theory examines power dynamics, social structures, and the impact of ideologies on research. Fragmentism acknowledges the existence of multiple perspectives and seeks to include marginalized voices. Realism emphasizes the significance of objective reality and the search for truth.

It is important to note that these philosophical perspectives on research do not exist in isolation. They form a spectrum, with positivism and interpretivism representing two major understandings. Other classifications, such as criticalism, Pragmatism, and realism, emerge as variations within this spectrum. Each perspective offers unique insights and approaches to research.

As you delve into the literature, you will encounter various philosophers and their perspectives, which will help you understand and differentiate these philosophical approaches. Keep in mind that these classifications are not exhaustive, as there are more than just five philosophies. However, understanding the broad spectrum and the major philosophical distinctions will provide a solid foundation for comprehending and navigating the diverse landscape of research philosophies.

Topic 73: Research Philosophies

In the previous module, I mentioned five research philosophies that we will discuss further. While these five classifications provide a starting point, it is important to note that there are more philosophies with different nuances and perspectives. Each philosopher may have their own unique classification and understanding of these philosophies. However, for the purpose of our discussion, we will focus on the five essential ones you mentioned. Let's briefly explore each of them and highlight their differences:

Positivism: Positivists believe that the world is objective and exists empirically. According to positivism, knowledge and reality are objective, independent of individual experiences. They argue that the scientific method is the most suitable way to study and understand the objective world. Positivists emphasize objectivity, empirical observation, and the search for universal laws and causal relationships.

Interpretivism: Interpretivists hold a contrasting view to positivists. They contend that multiple versions or interpretations exist for phenomena in the world. They emphasize subjectivity, meaning that interpretations vary depending on the observer. Interpretivists argue that understanding and explaining phenomena require considering the subjective perspectives, backgrounds, cultures, and experiences of individuals. They highlight the importance of studying people as subjects and exploring power structures and inequalities.

Criticalism (Critical Theory): Criticalism focuses on examining power dynamics, social structures, and the impact of ideologies on research and society. Critical theorists aim to challenge and transform oppressive structures and advocate for social justice. They critique existing systems and emphasize the need to give voice to marginalized groups. Criticalism encourages researchers to address real-world problems and explore research problems from the perspective of those affected by the issues.

Pragmatism is a philosophy that emphasizes the practical application of research findings to solve real-world problems. Pragmatism acknowledges the existence of multiple perspectives and encourages inclusivity in research. It recognizes that different individuals and groups have distinct viewpoints and experiences. Pragmatism seeks to address the Pragmatism of knowledge and perspectives, aiming to create a more comprehensive understanding by bringing together diverse voices.

Realism: Realists believe in the existence of an objective reality external to our subjective experiences. They argue that empirical methods, similar to positivists, are essential for understanding reality.

However, realists also emphasize the importance of studying reality within its social and cultural contexts, considering value systems and the social implications of research findings.

While positivism and interpretivism represent the core perspectives, the other three philosophies can be seen as variations within the spectrum. Criticalism emphasizes power dynamics and social structures, fragmentism emphasizes inclusivity and multiple perspectives, and realism combines empirical methods with an understanding of social and cultural contexts.

In short, these are the major five classifications of research philosophies.

Topic 74: Why Do I need a Research Philosophy?

Understanding one's worldview and its relationship to research is crucial for researchers. While conducting research without explicit awareness of one's worldview is possible, it may lead to unintended consequences. Researchers may unknowingly incorporate their worldview into their work, which can affect the research process, findings, and subsequent publications. Therefore, it is important to comprehend the philosophy of research, as it essentially encompasses one's worldview.

The concept of a worldview encompasses a combination of intellectual, epistemological, and exegetical views and beliefs held by an individual. Within the context of research, understanding one's worldview allows for more conscious utilization of it. Although many researchers may not be aware of this concept, they may have already conducted extensive research and published their findings. In such cases, it becomes necessary to reflect on the significance of understanding one's research philosophy.

To illustrate the importance of understanding worldviews, let's examine two extreme perspectives. Some individuals hold a worldview that asserts the existence of inherent qualities within the objects and phenomena of the world, independent of human influence. According to this perspective, the meaning of these objects remains unchanged, regardless of an individual's existence. Researchers subscribing to this view strive for objectivity in their work. They believe that by exploring phenomena

as they are, without personal biases or influences, they can uncover universal truths and establish causal relationships. This approach emphasizes objectivity and the existence of a singular reality.

In contrast, others hold a constructivist or interpretivist worldview. They argue that meanings are not inherent in objects but are constructed by individuals who interpret them. These meanings are influenced by social, cultural, and contextual factors. Consequently, the interpretation of phenomena can vary among researchers. According to this perspective, multiple realities exist, and truth is not an absolute concept. Researchers following this view acknowledge the subjectivity of interpretations and the role of the researcher in shaping meaning.

It is important to note that these two perspectives represent the extremes on a spectrum, and most researchers fall somewhere in between, incorporating elements from both viewpoints. By understanding their own worldview and the philosophical underpinnings of their research, researchers can approach their work more consciously. They can critically analyze their assumptions and biases, make informed decisions about research design and methods, and interpret their findings more accurately.

Understanding one's worldview and the broader philosophical foundations of research enables researchers to critically reflect on the influence of their perspectives and experiences. This self-awareness helps researchers avoid unwarranted generalizations, acknowledge the limitations of their findings, and consider alternative viewpoints.

Topic 75: Researcher's World view; The Nature of the Worldview

In the previous module, we discussed worldviews and the two spectrums between which they can exist. Now, let's delve into the concept of Intuitive Beliefs and explore your ideas about the nature of the universe. By asking yourself certain questions, you can gain insight into your ontological beliefs and better understand your worldview. Additionally, we will touch upon your epistemological view of the

world and examine your axiological beliefs—what values you consider acceptable or negotiable, the sources of power, and how these beliefs shape your research.

To begin, contemplate the nature of reality. Do you believe that reality is created by you or transmitted to you by a higher authority? Is reality constant or does it change over time? If you were to define reality, what would its essential characteristics be, and how would you determine if something belongs to the realm of reality? Consider how you categorize different aspects of reality and the criteria you use for this categorization. Reflect on the changes that have occurred throughout your life and how they have influenced your perception of reality.

For instance, let's take the example of the 16th century. During that time, the prevailing belief was that the Earth was at the center of the universe, with the sun revolving around it. However, with scientific advancements, we now understand that the reality is different—the Earth revolves around the sun. This shift in understanding demonstrates that realities can change over time. It prompts us to question the certainty of our current beliefs and raises doubts about whether today's reality will remain the same in the future.

Considering these aspects will help you examine the nature of realities and the criteria you use to define them. It is essential to critically evaluate what you consider as reality and whether it aligns with the questions posed earlier. By answering these questions, you will develop a more comprehensive understanding of your worldview and how it influences your research.

Topic 76: Researcher's World view -Epistemology

In the previous module, we discussed the importance of understanding our worldview by asking ourselves certain questions. These questions can help us identify our epistemological beliefs and their role in shaping our worldview. Rather than focusing on the nature of knowledge itself, we consider questions such as: What is true? What do I know? What are my sources of knowledge?

By reflecting on these questions and noting our answers, we can gain insights into our worldview. We contemplate the nature of truth, how knowledge is acquired, and the sources of knowledge. These fundamental questions revolve around the idea of intellectual balance and the resources and mental faculties we possess to create new knowledge.

When considering the nature of knowledge, we encounter two perspectives commonly found in literature: positivism and interpretivism/constructivism. Positivism asserts that truths are absolute and that knowledge is conveyed to us through an external authority or revealed knowledge. Interpretivism or constructivism, on the other hand, emphasizes the existence of multiple truths that can coexist. In this perspective, knowledge is dynamic and subject to alignment or synchronization within our minds.

To illustrate the concept of alignment or synchronization of knowledge, let's take the example of the atomic model. Initially, we might believe in Rutherford's atomic model, where electrons orbit the nucleus in a cloud-like manner. However, when confronted with the knowledge that moving objects expend energy, we experience a cognitive dissonance. Our minds seek equilibrium, and we strive to reconcile these conflicting pieces of knowledge. Scientists, through their research, propose a revised atomic model where electrons occupy specific orbits and release or absorb energy when transitioning between them. This new understanding aligns the pieces of knowledge and resolves the dissonance.

It is important to recognize that our understanding of reality and knowledge evolves over time. Our knowledge repository expands as we encounter new information and refine our understanding. We consider this refined knowledge as truth or ultimate truth, but it is important to acknowledge that it remains viable and subject to potential changes. Our minds lack the ability to determine the ultimate truth definitively. Instead, we rely on the alignment of knowledge within a contemporary context and its efficacy in solving the problems we encounter in life. This alignment forms our current vibration of truth.

Therefore, it is crucial to contemplate our beliefs about the nature of knowledge and understand that our worldview is shaped by these beliefs. We seek to attain a state of intellectual balance by continuously exploring and refining our understanding, always striving towards ultimate viability.

Topic77: Researcher's World view; The Nature of the Worldview- Axiology

In the previous modules, we discussed the importance of understanding our worldview and how it influences our approach to conducting research. We mentioned two important aspects to consider: epistemology and the nature of knowledge, as well as the role of axiology in shaping our values and ethics.

Epistemology focuses on how we perceive knowledge and truth. It involves questioning what is true, how we acquire knowledge, and what sources of knowledge we rely on. By reflecting on these questions and noting our answers, we can gain insights into our worldview and our beliefs about the nature of knowledge.

Axiology, on the other hand, deals with the study of values, ethics, and aesthetics. It involves contemplating what is right and wrong, determining the values that are important to us and society, and understanding our appreciation and perception of beauty. These aspects shape different aspects of our lives and influence our decision-making processes.

For example, when it comes to research ethics, our worldview plays a crucial role in determining our stance. Let's consider a research example: conducting a longitudinal study in which participants are asked to be part of the research for a specified period, say four years. If someone wants to withdraw from the study midway, the decision on whether to allow them to leave or not depends on our worldview.

If we prioritize individual autonomy and believe that each person has the right to make decisions about their own life, then we may allow them to withdraw from the research. However, if we emphasize the societal responsibility of individuals and view participation in research as a duty towards the betterment of society, we may decide that they should not be allowed to withdraw midway, as it may compromise the integrity and validity of the research.

These examples demonstrate how our worldview shapes our research design, participant engagement, and decision-making processes. By examining our beliefs and values regarding ethics and values, we can better understand how they influence our research practices.

By reflecting on our worldview and understanding our beliefs about knowledge, values, and ethics, we can align our research practices with our guiding principles. This self-awareness allows us to conduct research that is consistent with our worldview and ethical framework, ensuring the integrity and validity of our findings.

Lecture 19: Types of Reasoning:

Topic 78: Research Approaches: Difference among Research approach

In the context of research, it is important to understand the distinctions between research philosophies, research approaches, research methodologies, and research methods. These terms are interconnected but have different meanings and purposes.

Research philosophies refer to the overarching frameworks or beliefs that guide the research process. **Examples** of research philosophies include positivism, interpretivism, and realism. These philosophies shape the researcher's worldview, assumptions, and the way they approach the study of knowledge and reality.

Research approaches, on the other hand, are specific methods of reasoning that researchers employ within a chosen research philosophy. The two commonly discussed research approaches are **inductive** and **deductive reasoning**. Inductive reasoning involves drawing general conclusions from specific observations, while deductive reasoning involves deriving specific predictions from general principles or theories. Another **less common** approach is **abduction**, which involves generating plausible explanations or hypotheses based on limited information.

Once the research approach is determined, researchers need to select a research methodology or strategy. Research methodologies encompass the overall plans and procedures that researchers adopt to answer their research questions. Examples of research methodologies include quantitative research, qualitative research, and mixed methods research. Quantitative research focuses on numerical data and statistical analysis, while qualitative research emphasizes exploring and understanding phenomena in-depth through interviews, observations, and textual analysis. Mixed methods research combines both quantitative and qualitative approaches.

Topic79: Inductive Reasoning

In this module, we will explore the three types of reasoning in detail: inductive reasoning, deductive reasoning, and abductive reasoning. Inductive reasoning is associated with qualitative research, while deductive reasoning is associated with quantitative research. First one is inductive reasoning.

Inductive reasoning is a type of reasoning commonly used in qualitative research. It involves collecting specific observations or instances and using them to reach a generalization or conclusion. The process entails moving from specific cases towards a broader understanding or generalization.

For example, let's consider a scenario where a researcher reads multiple research books on various topics. Upon examining these books, they notice that each book contains a chapter dedicated to sampling. These specific observations from different sources lead the researcher to make a generalization that sampling is a common topic addressed in research books.

The generalization made through inductive reasoning is based on the specific instances or observations gathered. However, it is important to note that the accuracy and validity of the generalization depend on the representativeness of the specific observations. Researchers should ensure that their specific observations are diverse and adequately cover the relevant phenomenon to draw meaningful generalizations.

Overall, inductive reasoning in qualitative research involves using specific instances or observations to develop a broader understanding or generalization about a particular subject or phenomenon.

Topic 80: Abductive Reasoning

In this module, we will continue our discussion on types of reasoning and explore abductive reasoning, in addition to inductive and deductive reasoning.

Abductive reasoning involves starting with a conclusion or an observation and then working backward to find an explanation for it. **For instance**, let's consider a scenario where I'm walking down the street and I see smoke coming from a building. The observation is that smoke is emanating from the building. Using abductive reasoning, I might deduce that there is a fire inside the building, as fire is a common cause of smoke. In this case, I have observed the smoke and reasoned backward to infer the presence of fire as its probable explanation.

Another example could be when I see a student's exam result and based on that, I conclude that they didn't work hard. In this case, the conclusion is derived from the observation of the low marks. Using abductive reasoning, I reason that the student's lack of effort or insufficient studying may have led to their low marks.

Abductive reasoning works by forming a conclusion and then developing an explanation based on prior experiences or knowledge. It is important to note that abductive reasoning is not foolproof and can be subject to errors or inaccuracies.

Let's expand on the example of a student's low exam marks. I may conclude that the student didn't study well, but it's possible that there could be other factors at play. **For instance**, the student may have been unwell during the exam, which affected their performance. The soundness or correctness of the explanation depends on the strength of the observations and the breadth of experiences supporting it. If my conclusion is based on a single observation or limited data, the likelihood of the explanation being accurate decreases. However, if my reasoning is based on a broader range of experiences and observations, the chances of arriving at a correct explanation increase.

To summarize, abductive reasoning involves forming a conclusion and then developing an explanation based on prior experiences or observations. It is important to consider the breadth and scope of observations when evaluating the soundness of the explanation.

Topic 81: Deductive Reasoning

Deductive reasoning is a type of reasoning where you start with generalizations and move towards specific cases to explain or draw conclusions. The process involves taking a general idea and arriving at a specific conclusion.

For example, let's consider the generalization that all humans have two eyes. Based on this generalization, we can then take a specific case, such as Ahmed being a human, and conclude that Ahmed will also have two eyes. The generalization precedes the specific case, and we deduce that since Ahmed falls under the category of humans, he will possess the characteristic of having two eyes.

Another example can be seen in the context of research books. If we generalize that all research books have a chapter on sampling, and we have a specific book in our hand that is a research book, we can deduce that this particular book will also have a chapter on sampling. Here, we start with the generalization that research books include a sampling chapter, apply it to the specific case of the book we are reading, and conclude that it will have a chapter on sampling.

In deductive reasoning, the process involves using existing generalizations or theories to explain or draw conclusions about specific cases. Generalizations are already established, and we use them as the logical basis to interpret specific instances. This deductive reasoning is commonly employed in quantitative research, where researchers gather empirical evidence to verify or support the generalizations derived from established theories.

In summary, deductive reasoning entails starting with generalizations and applying them to specific cases to draw conclusions. It is a reasoning mechanism utilized in research to explain specific instances by using the logic of existing generalizations or theories.

Topic 82: Comparison of Inductive, Deductive, and Abductive Reasoning

In the previous modules, we discussed inductive, deductive, and abductive reasoning and provided examples to illustrate each type. Now, let's compare these three types of reasoning side by side to better understand their differences.

Let's begin with **deductive reasoning**, which starts with a generalization. **For example**, if we have the generalization that all parrots are green, we can use deductive reasoning to conclude that a parrot in my house is also green. Deductive reasoning relies on established generalizations or theories, which may change over time, and it involves starting with a generalization and explaining a specific case based on that.

On the other hand, **inductive reasoning** starts with specific observations and leads to a generalization. If I observe multiple parrots in different places and find them all to be green, I can develop the generalization that all parrots are green. Inductive reasoning involves gathering specific observations and then inferring a general rule or pattern from them.

Now, let's discuss **abductive reasoning**, also known as abduction. Abductive reasoning starts with a conclusion or observation and seeks to find the best explanation or cause for that observation. For instance, if I see smoke coming from a building, I may conclude that there is a fire inside based on the observation that smoke is typically associated with fire. Abductive reasoning involves formulating a tentative explanation or hypothesis based on available evidence.

These three types of reasoning—deductive, inductive, and abductive—can be connected to different research approaches. Qualitative research often relies on inductive reasoning, where researchers gather a substantial amount of evidence and draw generalizations from it. Quantitative research, on the other hand, may involve deductive reasoning, where established theories or hypotheses are tested against specific cases. Mixed-method studies can employ a combination of deductive and inductive reasoning depending on the research questions and objectives.

Understanding which type of reasoning aligns with a particular research approach is crucial. Researchers must select the appropriate method of reasoning that suits their research goals and helps them draw valid and reliable conclusions.

In summary, deductive reasoning starts with a generalization and explains specific cases, inductive reasoning moves from specific cases to generalizations, and abductive reasoning starts with a conclusion or observation and seeks the best explanation. Different research approaches may utilize these types of reasoning in various ways.

Lecture 20: Research Designs/Strategies of Inquiry-I

Topic 83: Methodological Choice

In the field of research, there are three main methodological choices: **quantitative**, **qualitative**, and **mixed methods**. These choices have a long-standing tradition and have evolved over time.

Quantitative research

Quantitative research is characterized by the collection of data in numerical form. Some data sets naturally consist of numerical values, such as height, while other non-numerical data is coded into numerical form. **For example**, gender can be coded as 1 for male and 2 for female. By processing this data in numerical form, researchers can describe and explain phenomena, make predictions, and examine the relationships between variables. Quantitative research focuses on verifying existing phenomena rather than inventing or creating new ones. The emphasis is on quantifying data and using statistical analysis to draw conclusions and identify common elements. One crucial aspect of quantitative research is verifiability, ensuring that other researchers following the same procedures can obtain the same numbers, aliases, and conclusions. Another assumption is that the principles governing the world are relatively stable and predictable, allowing for generalizability. The underlying understanding is that these phenomena operate based on fundamental principles that remain consistent. Quantitative research follows a scientific method, enabling reproducibility of results for different researchers following the same research process. Objectivity is a key feature of quantitative research, as observations are not influenced by the researcher's perspective. Examples of research types commonly associated with quantitative research include descriptive, correlational, experimental, causal, and comparative studies.

Topic 84: Methodological Choice in Research- Qualitative research**Qualitative research**

Qualitative research is based on several fundamental assumptions. Unlike quantitative research, qualitative research deals with non-numeric data, such as verbatim transcripts, visuals, or pictures. It recognizes subjectivity as an inherent element of data and its interpretation. The focus of qualitative research is on comprehensive understanding rather than generalization. Consequently, qualitative research does not aim to be representative, and sampling is not a central aspect. Instead, qualitative researchers seek in-depth information and employ interpretive data analysis.

Qualitative researchers collect data in various forms, including words, pictures, or graphics. They aim to understand concepts, opinions, and experiences from the perspectives of the phenomena being studied, rather than seeking generalizable results. Data collection methods in qualitative research involve open-ended observations, semi-structured interviews, or focus group discussions. These methods allow for follow-up questions and the gathering of detailed information that goes beyond the scope of quantitative research tools, such as fixed-choice items on a survey.

Data analysis in qualitative research does not involve numerical or statistical calculations. Instead, researchers divide the data into themes and extract emerging understandings of the phenomenon. Qualitative researchers acknowledge that their own opinions and reflections can influence the research process and findings, as they are actively involved in the research. This aspect differentiates qualitative research from the perspective of an outsider.

Qualitative researchers embrace interpretivism as their underlying philosophy. They contend that interpretations of evidence are not true or false but contextualized and meaningful within a particular research case. They recognize that multiple interpretations can exist for the same information, depending on the context. Qualitative researchers also emphasize the collection of data from multiple sources to triangulate and enhance the accuracy and authenticity of the information.

Furthermore, qualitative researchers advocate for collecting data in naturalistic settings, where the phenomenon naturally occurs. They believe that studying phenomena within artificial experimental setups can lead to data manipulation and artificial effects. By examining phenomena in their natural settings, researchers gain deeper insights and a more comprehensive understanding.

These assumptions form the methodological stance of qualitative research. Understanding and comparing these assumptions with those of quantitative research, as well as connecting them with the underlying research approach and philosophy, can reveal an alignment between the two approaches.

Topic 85: Comparison of Quantitative and Qualitative Research

In the previous modules, we discussed qualitative and quantitative research as methodological choices that align with different research approaches and philosophies. These choices have implications for the research process. By considering these alignments, researchers can conduct their research effectively. In the slide, we compared qualitative and quantitative research on five different aspects: type of data, scope of inquiry, primary advantages, limitations, and analysis techniques.

Characteristic	Quantitative research	Qualitative research
<i>Type of data</i>	Phenomena are described numerically	Phenomena are described in a narrative fashion
<i>Analysis</i>	Descriptive and inferential statistics	Identification of major schemes
<i>Scope of inquiry</i>	Specific questions or hypotheses	Broad, thematic concerns
<i>Primary advantage</i>	Large sample, statistical validity, accurately reflects the population	Rich, in-depth, narrative description of sample
<i>Primary disadvantage</i>	Superficial understanding of participants' thoughts and feelings	Small sample, not generalizable to the population at large

Type of Data:

Quantitative research deals with data that is numeric and can be analyzed using statistical tools. Qualitative research, on the other hand, collects data in the form of narratives, words, pictures, videos, or graphics. The analysis of qualitative data involves identifying themes and analyzing descriptions rather than statistical calculations.

Analysis Techniques:

In quantitative research, data analysis involves numerical computations, such as calculating averages and standard deviations. The results can be represented graphically. Qualitative analysis focuses on identifying themes and explaining data within those themes. It involves reducing the data from broad to specific categories and exploring common bases.

Scope of Inquiry:

Quantitative studies typically focus on specific questions or hypotheses that are tested with data collected from a representative sample of the population. The analysis is structured, and statistical analysis is used to summarize data, draw inferences, and make generalizations. Qualitative studies have a broader scope and aim to understand different phenomena. They collect evidence to explain variations and do not rely on pre-defined hypotheses.

Advantages and Limitations:

Quantitative research excels in generalizability, as findings can be extended to a larger population. It relies on large representative samples and allows for statistical summaries and generalizations. Qualitative research, on the other hand, values the richness and depth of data. It provides detailed explanations and insights but does not prioritize generalizability. Qualitative studies often have smaller sample sizes and focus on narrative explanations.

By considering these strengths and weaknesses, researchers can determine which approach aligns best with their research goals. It is important to note that quantitative and qualitative research have distinct characteristics and do not overlap. Understanding this alignment is crucial to avoid confusion and ensure a clear research direction.

The aim of this module is to help researchers establish a clear understanding of the alignment between research approaches, philosophies, and the choice between qualitative and quantitative research.

Topic 86: Methodological Choice: Mixed Method Research

In Module 129, we will continue our discussion on methodological choices, specifically focusing on mixed methods research. In recent years, mixed methods research has gained significant popularity among researchers in various disciplines. They have recognized that certain research problems cannot be effectively addressed solely through qualitative or quantitative approaches. In such cases, a combination of both approaches is needed to fully understand and tackle the complexity of the problem. This integrated approach is referred to as mixed methods research.

Mixed methods research allows researchers to utilize the strengths of both qualitative and quantitative methods to gain a more comprehensive understanding of a research problem. Depending on the nature of the problem, researchers may require quantitative data to capture certain aspects or trends, while qualitative data may be needed to explore in-depth perspectives and gain a richer understanding. By

combining these approaches, researchers can leverage a wider range of techniques to enhance their insights and understanding of the problem.

To illustrate this, **let's consider an example**. Imagine a research problem that involves structured questions and requires quantitative data to collect relevant information. However, this data alone may not provide a sufficient understanding of the problem. In order to gain deeper insights, researchers may also need to conduct qualitative research, such as interviews or observations, to gather additional information and perspectives from selected individuals. By combining quantitative and qualitative methods in this way, researchers can conduct a mixed methods study that addresses the complexity of the research problem.

It is important to note that mixed methods research is not tied to a specific philosophical stance like positivism or interpretivism. Instead, the choice of methodological approach should be aligned with the research problem and the underlying research philosophy. While mixed methods research has been evolving for several decades, there are still ongoing discussions and debates regarding its methodological and philosophical foundations.

In terms of design, there are various approaches commonly used in mixed methods research, such as convergent designs, divergent designs, and explanatory sequential designs. These design choices reflect the nature of the research problem and how the qualitative and quantitative components are integrated. **For example**, an explanatory sequential design indicates that the research problem requires an explanation, and the qualitative and quantitative components are sequenced rather than conducted simultaneously.

The essence of mixed methods research lies in the complementary nature of the two sets of data, qualitative and quantitative. They should not be treated as separate and independent studies. Instead, the information obtained from both approaches should complement and strengthen each other, enhancing the evidence and conclusions drawn from the study. When the integration of these methods effectively combines and supports the findings, it can be considered as a converged mixed methods research design.

In summary, mixed methods research utilizes both qualitative and quantitative approaches in combination to address research problems that require a multifaceted understanding. The order and integration of these methods are determined by the research design, and the information obtained

from both approaches should complement and supplement each other. By following these principles, researchers can conduct rigorous and insightful mixed methods studies.

Topic 87: Difference Between Quantitative, Qualitative and Mixed Method Research

In the previous modules, we discussed three methodological choices: qualitative, quantitative, and mixed methods. These choices are closely connected to research approach and research philosophy. The purpose of the previous modules was to understand these connections and summarize them in a coherent framework.

	Quantitative	Qualitative	Mixed Methods
Philosophical Assumptions	• Postpositive knowledge claims	• Constructivist, advocacy or participatory knowledge claims	• Pragmatic knowledge claims
Strategies of Enquiry	• Experimental designs • Non-experimental designs e.g. surveys	• Narratives • Phenomenology • Ethnographies • Grounded Theory • Case Studies	• Sequential • Concurrent • Transformative
Specific Research Methods	• Predetermined • Closed, instrument based questions • Performance, attitude, observational and census data • Statistical analysis	• Emerging methods • Open questions • Interview, observation, document, audiovisual data • Text and image analysis	• Both predetermined and emerging methods • Both open and closed questions • Multiple forms of data drawing on all possibilities • Statistical and text analysis
Motivations for selection	• Test a theory or explanation • Identify factors that influence an outcome • Understand the best predictors of an outcome	• Understand a concept or phenomenon due to insufficient or new research • Identify unknown variables	• Generalise findings to a population whilst developing a detailed explanation of the concept or phenomenon

The philosophical assumption behind quantitative research is positivism. Positivists believe that there is a fixed reality, objective knowledge exists, and values are timeless and universal. They seek to discover truths and consider everything else as false. Quantitative research is aligned with this positivist paradigm.

On the other hand, interpretivism, which aligns with qualitative research, recognizes that there are multiple social realities. It emphasizes understanding subjective meanings and interpretations. In

qualitative research, we explore in-depth information and value the richness of data. Qualitative researchers do not solely rely on numbers; they seek to understand phenomena from various perspectives.

The mixed methods approach is linked to the philosophical perspective of pragmatism. It combines both qualitative and quantitative methods to take advantage of their respective strengths. By using a mixed methods approach, researchers can draw from both qualitative and quantitative research to gain a more comprehensive understanding of their research topic.

Quantitative researchers are interested in any form of inquiry that can generate numerical data. They often rely on experimental designs to create controlled environments for their studies. However, experiments are not the only method used in quantitative research. Surveys, relational studies, and causal comparative studies are also common.

Qualitative researchers, on the other hand, prefer to collect data in naturalistic settings where the phenomenon of interest occurs. They use methods such as narratives, phenomenology, ethnography, case studies, and grounded theory to gain insights into the perspectives of individuals and understand the variations in their experiences.

Mixed methods researchers design their studies by combining qualitative and quantitative elements. They decide on the order in which these elements will be used, which data will support or verify other data, and how the connectivity of the data can enhance their research. The characteristics of mixed methods research include using both qualitative and quantitative data, collecting data in non-numeric forms, and interpreting the data to provide a comprehensive understanding of the research topic.

It is important to note that not every method is suitable for all methodological choices. Some methods are more aligned with specific approaches. For example, quantitative research primarily focuses on testing theories and establishing causal relationships. Qualitative research, on the other hand, is more suitable for exploring phenomena where existing explanations are lacking.

When researchers need to understand a phenomenon for which there are no pre-existing explanations, qualitative research is the appropriate choice. Mixed methods research is used when a combination of qualitative and quantitative methods can provide a better explanation of the research topic.

By considering the research philosophy, research approach, and the nature of the research problem, researchers can make informed choices about the most suitable methodological approach for their study.

Topic 88: Research Approaches: Criteria

In this Module, we will discuss the importance of considering the nature of the research problem when selecting a research approach. It is not advisable to choose or use a research approach without carefully considering the specific characteristics of the problem being investigated. The research approach should be aligned with the nature of the problem to ensure that the study can be conducted effectively and produce meaningful results.

The first criterion to consider when choosing a research approach is the nature of the research problem itself. If the problem requires verification or testing of an existing theory or law, and there is already a theoretical base to build upon, a quantitative approach may be more appropriate. In such cases, the study aims to verify or confirm existing theories and generalizations, making a quantitative approach necessary. On the other hand, if the problem requires explanation and understanding without any existing theorization, a qualitative approach is more suitable. Qualitative research allows for in-depth exploration and understanding of phenomena that do not require verification.

It is essential to avoid selecting a research approach based solely on personal inclination or prior training. Researchers should prioritize the nature of the research problem over their personal preferences or training. If a problem requires a specific approach that differs from the researcher's expertise, it is crucial to seek guidance or collaborate with someone experienced in that particular approach. Researchers should not limit themselves to their comfort zones but instead choose the approach that best aligns with the problem at hand.

Another important consideration is the researcher's experiences and expertise. Researchers should assess their own training and prior work to determine if they have the necessary skills and knowledge to conduct the chosen research approach effectively. If a researcher has significant experience and expertise in quantitative research, it may be more appropriate to select a problem that can be addressed using quantitative methods. Similarly, if a researcher has expertise in qualitative research, they should focus on problems that require qualitative inquiry.

Finally, researchers should consider the feasibility and practicality of data collection methods in relation to the phenomenon and participants involved. Researchers should choose data collection formats that are suitable and comfortable for the participants. For example, if interviews are the preferred data collection method and the participants are not comfortable with interviews, alternative methods should be considered. It is important to ensure that the chosen approach aligns with the availability and preferences of the participants and that the data collected can be effectively utilized.

In summary, when selecting a research approach, it is crucial to consider the nature of the research problem, the researcher's experiences and expertise, and the feasibility of data collection methods. By keeping these factors in mind and aligning the research approach with the problem at hand, researchers can conduct their studies more effectively and produce valuable insights.

Lecture 21: Research Designs/Strategies of Inquiry -II

Topic 89: Research Design: Introduction to Research Methodology, Design, and Method

In this Module, we will explore the relationship between research methodology, research design, and research methods. Understanding this relationship is crucial for comprehending the concept of research design.

Research methodology is the broad framework or approach that guides your research. It encompasses the philosophical underpinnings of your study, such as whether you adopt a positivist, interpretivist, or critical paradigm. It also includes the theoretical background and conceptual structure of your research.

Within research methodology, research design comes into play. Research design involves making decisions about the structure of your study. This includes determining the objectives of your research, the type of data you will collect (qualitative, quantitative, or both), the sampling strategy, and the timeline for data collection. Research design shapes the overall structure and organization of your study.

Research methods, on the other hand, pertain to the specific techniques and tools you employ to collect and analyze data. It involves selecting appropriate data collection methods, defining data collection procedures, choosing data analysis techniques, and interpreting the results. Research methods allow you to address different research questions or hypotheses within your study, often utilizing multiple tools and analyses.

To summarize, research methodology is the broadest term, encompassing the overall framework and approach of your research. Within research methodology, research design focuses on the structure and decision-making aspects of your study. Research methods, on the other hand, deal with the specific

techniques and procedures employed for data collection, analysis, and interpretation. Understanding this hierarchy will help us delve deeper into the concept of research design in the subsequent modules.

Topic 90: Quantitative Research Designs: Survey Research Design

In this Module and the subsequent modules, we will be exploring various research designs commonly employed in the field of education. One prominent design in this realm is **survey research**, which is utilized to investigate the current status and existence of phenomena. Survey research proves particularly valuable when seeking to gauge public opinion, attitudes, and trends. It involves the collection of quantifiable data through surveys from a representative sample of the population, providing insights into their levels of satisfaction and perspectives on specific issues.

Survey research generally encompasses **two primary design types: cross-sectional studies and longitudinal studies**. In a cross-sectional study, data is gathered from different groups at a single point in time, enabling comparisons across various grades or stages. For **example**, a cross-sectional survey could assess the **satisfaction levels of students across different grade levels with regard to library services**.

Conversely, longitudinal studies involve collecting data from the same group of individuals over an extended period. This allows researchers to track changes and trends over time. **For instance**, one could **measure the satisfaction of a specific group of students with library services at the end of each academic year, from grade 5 to grade 10**.

Both cross-sectional and longitudinal designs offer distinct advantages. Cross-sectional studies provide simultaneous data from different levels, while longitudinal studies offer insights into changes over time. The choice between the two designs hinges on the research objectives and available resources.

Conducting survey research entails several key steps. These include defining survey objectives, formulating relevant questions, developing appropriate survey instruments, selecting a representative sample, designing an effective data collection strategy, administering the survey, and analyzing the acquired data. The specific sequence of these steps may vary depending on the chosen design.

Survey research is widely utilized due to its practicality and ability to gather valuable insights from diverse populations. By adhering to a systematic approach and adapting the methodology to the selected design, researchers can effectively collect and analyze data to address their research objectives.

Topic 91: Quantitative Research Designs: Correlational Research

In Module 91, we will continue our discussion on different research designs and focus on co-relational research. Co-relational studies aim to examine the relationship or effects between two variables. It is important to understand the distinction between Co-relational research and difference studies, which explore the differences between variables.

In Co-relational research, the focus is on variables that covariate. This means that when one variable changes, the other variable is also likely to change. In these setups, the relationship between the variables is such that if one variable is the cause, the other variable will be affected accordingly. **For instance, consider the relationship between height and step size.** Generally, as a person's height increases, their step size also tends to increase. When there are no other factors at play, a change in one variable leads to a change in the other, and any subsequent change in the second variable is necessarily due to the first variable. However, it is important to note that causation cannot be inferred solely based on the relationship between independent and dependent variables. While there is a causal relationship between them, one variable being larger does not necessarily mean that it caused the other variable to change. **For example, if you work hard, it does not guarantee good results.** There are multiple factors that can influence your results, so there is a unidirectional relationship here. Increasing effort can lead to good results, but good results may not always be a result of increased effort. In this context, the focus is on one-way causation; if effort is increased, good results may follow, but the relationship should not be interpreted in reverse.

Co- relational research is commonly used to determine the degree of relationship between covariates. **For example,** the relationship between cloud formations and rain can indicate the likelihood of rainfall even before it occurs. When these relationships become strong, one variable can serve as a predictor of the other. Hence, both types of studies fall under the category of relational research.

In terms of the research process, relational studies follow a similar structure to other types of research. They begin with selecting a research problem and formulating objectives or propositions. The subsequent steps include sample selection, instrument development for data collection, conducting the study, and finally, analyzing and interpreting the data. The main difference lies in the design timeline and the sequence in which these steps are carried out.

Topic 92: Causal-Comparative Research

In this Module, we will discuss causal-comparative research in terms of research design. Causal-comparative research examines the relationship between a consequence and its causes. In this type of relationship, you observe the consequence first and then investigate its causes. **For example**, you might study the **effects of an earthquake on people's psychological health**. The earthquake is the **independent variable** that caused the effects, but it was not manipulated by you. The effects on psychological health can be considered the dependent variable. This type of research is called causal-comparative research because you want to connect the cause (earthquake) to the effects (psychological health) that occurred after the cause has passed.

However, it's important to note that causation is not always directly and confidently attributable in such research. The relationship between the earthquake and changes in psychological health may not be solely due to the earthquake itself. It's possible that some psychological health issues already existed before the earthquake. Therefore, it's not possible to attribute the entire change in psychological health solely to the earthquake. Causal-comparative research aims to connect the cause and effects, but the causation is not always straightforward.

The procedure for conducting causal-comparative studies follows the same steps as other research designs. You start with research questions, select a group, identify variables (some of which have already occurred), and examine the influences you want to investigate. You then sample and develop instruments accordingly. Finally, you analyze and interpret the data. These steps are similar to those in previous designs, but adjustments are made in terms of controls, tools, and interpretation techniques to suit the nature of causal-comparative research. The analysis techniques and interpretations are adapted to fit the design, but the sequential steps remain the same as in previous models or designs.

Topic 93: Experimental Research

In this Module, we will discuss experimental research, which is one of the most established and popular research designs. If you are a beginner and want to understand research designs easily, it is recommended to start by studying experimental research designs. Experimental research involves manipulating the independent variable, which is the variable you have control over as a researcher. You decide how to manipulate or interact with this variable in your study.

For example, let's consider **the relationship between physical exercise and the improvement of mental health**. As the researcher, I decide to have a group of students engage in one hour of exercise

in the morning and another session of exercise in the afternoon, every day for three months. The details of the exercise, such as duration and frequency, are determined by me, the researcher. This manipulation of the independent variable, physical exercise, is within my control.

Once I have decided on the manipulation, I observe and measure the effects it has on the dependent variable, which is mental health. However, the response of each subject to the exercise is not under my control. It will vary among individuals, and the effects on their mental health will differ. This aspect is beyond my control but can still be observed and analyzed.

In experimental research, the researcher manipulates the independent variable and examines its effects on one or more dependent variables. It is important to note that these effects are not predetermined or known in advance. The researcher's role is to control the manipulation and observe the resulting impact on the dependent variable.

The steps involved in experimental research are similar to those in previous research designs. You define the research problem, formulate hypotheses, select participants, measure variables before and after the manipulation, create research tools, plan and execute the study, collect and analyze data, and interpret the findings. The main difference lies in the implementation phase, where the researcher manipulates the independent variable. Other than that, the overall process is similar to previous research designs.

Topic 94: Types of Experimental Research

In this Module, we will discuss different types of experimental research designs. Experimental research can be classified into three main groups: **pre-experimental designs**, **true experimental designs**, and **factorial designs**.

1. **Pre-experimental designs** are considered to be preliminary or less rigorous versions of experimental research. They usually exhibit two main deficiencies. First, the participants are not randomly assigned to groups. Instead of random assignment, researchers may use methods like volunteer-based selection or convenience sampling. Random assignment ensures that each participant has an equal chance of being assigned to either the experimental or control group. In pre-experimental designs, this random assignment does not occur.
2. The second deficiency in pre-experimental designs is the absence of a control group. Only the experimental group, which receives the treatment or intervention, is present. Without a control

group, it becomes challenging to compare the effects of the intervention and determine its true impact. Therefore, pre-experimental designs lack a control group, making it difficult to draw definitive conclusions about the effectiveness of the intervention.

To address these deficiencies, researchers employ true experimental designs. In true experimental designs, participants are randomly assigned to both the experimental and control groups. This random assignment ensures that the groups are comparable and representative of the larger population. By having a control group, researchers can compare the outcomes and measure the true effect of the intervention.

Semi-experimental designs and quasi-experimental designs are variations of true experimental designs. They also involve random assignment of participants to groups, including both experimental and control groups. By using random assignment, researchers minimize personal biases and ensure that the groups are representative. The presence of a control group allows for a clear comparison of the treatment's effect and the measured outcomes. These designs are considered to be more rigorous than pre-experimental designs.

3. Factorial designs come into play when there are multiple independent variables, usually more than two. In contrast to single-variable designs, factorial designs involve manipulating and studying the effects of two or more independent variables simultaneously. This complexity arises as there are multiple independent variables and multiple dependent variables, which could each be influenced by the independent variables. Factorial designs allow researchers to examine the interactions and effects of these variables in a comprehensive manner.

In summary, experimental research designs can be classified into pre-experimental, true experimental, and factorial designs. Pre-experimental designs lack random assignment and a control group. True experimental designs address these deficiencies by incorporating random assignment and a control group. Factorial designs are employed when there are multiple independent and dependent variables.

Lecture 22: Research Designs/Strategies of Inquiry- III

Topic 95: Types of Pre-Experimental Research Designs

In the previous module, we discussed three types of experimental designs: pre-experimental designs, causal experimental designs, and true experimental designs. Additionally, factorial designs were

mentioned. Now let's focus on pre-experimental designs, which lack random allocation of individuals to groups and do not include a control group. These designs are used in certain situations when conducting research, particularly when random assignment is not feasible.

There are three common pre-experimental designs:

- One-Shot Case Study
- the One-Group Pretest-Posttest Design
- the Static Group Comparison Design.

These designs serve specific purposes and help address certain limitations. It is important to understand why pre-experimental designs are used, even though true experimental designs are considered superior.

1. **One-Shot Case Study:** This design involves a single group of participants exposed to a treatment or intervention, followed by measurement of the outcome. For example, a researcher might introduce a new teaching method to a class and then assess the students' learning outcomes. However, without a control group or pre-intervention measurements, it is challenging to establish causal relationships or rule out alternative explanations.
2. **The One-Group Pretest-Posttest Design** involves a single group of participants. A pretest is conducted before administering the intervention, and a posttest is conducted afterward. However, this design lacks a control group for comparison. Without a control group, it becomes challenging to attribute the observed changes solely to the intervention. Other factors or circumstances may have influenced the outcomes, making it difficult to establish a causal relationship.
3. Similarly, the **Static Group Comparison Design** involves pre-existing groups. Researchers do not have control over the assignment of participants to these groups. This lack of random assignment creates uncertainty about the comparability of the groups before the study begins. The absence of random assignment and the lack of a control group limit the design's ability to draw conclusive and causal inferences.

Pre-experimental designs are employed when researchers face practical constraints that prevent random assignment or the inclusion of a control group. Although these designs have limitations, they can still provide valuable insights and preliminary evidence. Researchers should acknowledge the

limitations and interpret the findings cautiously, recognizing the potential for confounding factors or alternative explanations.

In summary, pre-experimental designs, such as the One-Group Pretest-Posttest Design and the Static Group Comparison Design, are utilized in situations where random assignment or control group inclusion is not feasible. While these designs have inherent limitations, they can offer initial insights and serve as a starting point for further research.

Topic 96: Threats to Validity of Research Findings

In this module, we will explore the extent to which pre-experimental designs can address threats to the validity of research findings. As mentioned earlier, research designs are used to control for potential threats to validity. We will examine three pre-experimental designs: One-Group Pretest-Posttest Design, Static Group Comparison Design, and Case Study Design. By analyzing these designs, we can determine which threats they can and cannot control.

Validity threats can be categorized as **internal** and **external**. In the headings above, you will find the names of the three pre-experimental designs: One-Group Pretest-Posttest Design, Static Group Comparison Design, and One Shot Case Study Design.

Designs	Sources of Invalidity									
	Internal								External	
	History	Maturation	Testing	Instrumentation	Regression	Selection	Mortality	Selection Interactions	Pretest-X Interaction	Multiple-X Interference
One-Shot Case Study $X O$	-	-	(+)	(+)	(+)	(+)	-	(+)	(+)	(+)
One-Group Pretest- Posttest Design $O X O$	-	-	-	-	-	(+)	+	(+)	-	(+)
Static-Group Comparison $X_1 O$ $X_2 O$	+	-	(+)	(+)	(+)	-	-	-	(+)	(+)

Each line of Xs and Os represents a group.

Note: Symbols: X or X_1 = unusual treatment; X_2 = control treatment; O = test, pretest, or posttest; + = factor controlled for; (+) factor controlled for because not relevant; and - = factor not controlled for.

The presence of a "-" sign indicates that the design does not control those threats. Conversely, a "+" sign in parentheses indicates that the threat is not relevant to that particular design and therefore does not need to be controlled. In other words, if you see a "+", it means that the threat is not applicable to the design, and if you see a "-", it means that the design does not control that threat.

The number of negative signs indicates the severity of the design's limitations. Most studies have a "+" sign in parentheses, indicating that the design is not affected by those threats. However, there may be other threats that are not controlled by the design.

Let's look at the Static Group Comparison Design. It controls for the history effect because it includes a control group. However, it does not control for other threats or they are not relevant to this design.

In the One-Group Pretest-Posttest Design, it can control for one factor, maturation. Other threats are either uncontrolled or not relevant to this design.

It is important to note that pre-experimental designs, while having some controls, may not address all validity threats. These designs are considered less rigorous compared to true experimental designs, which incorporate randomization and control groups to address a wider range of threats. Therefore, pre-experimental designs are not preferred when evaluating the effectiveness of a design based on its ability to control threats.

In summary, pre-experimental designs have limited control over validity threats. While they can address some threats, they are not as comprehensive as true experimental designs. Researchers should carefully consider the limitations of pre-experimental designs when choosing an appropriate research design for their study.

Topic 97: Types of Quasi- Experimental Research Designs

In this module, we will discuss different research designs known as semi-experimental designs, starting with the Quasi- Experimental Design. First, I will explain what is common among all four designs and why they are referred to quasi-experimental designs. These designs lack random assignment of individuals to groups, resulting in the formation of an experimental group and a control group. The participants are not randomly assigned to these groups, hence the term "semi-experimental design."

The term "**non-equivalent**" in the title signifies that the groups created are not a result of random assignment. Therefore, before the start of the experiment, there is no guarantee that the participants in the experimental group and the control group are equal in terms of the variable that the intervention seeks to address. In other words, I did not randomly distribute people into the two groups as described in a randomized controlled trial. In this design, both a control group and an experimental group exist, which allows for comparison of the outcomes of the intervention.

Next, let's consider the **Pretest-Posttest Design**. In this design, I conducted a pretest and a posttest for the participants in the experimental group. Again, random assignment of individuals to the groups did not occur. By observing changes in the outcome variable as a result of the intervention, I could identify some changes within the group. However, I couldn't solely attribute these changes to my intervention due to the lack of random selection. One weakness of this design is the absence of a control group, as random assignment is not employed.

Similarly, a **Time Series Design** also lacks a control group. In this design, I provide an intervention to one group and conduct multiple pretests and posttests, allowing for a long-duration study. However,

I lack a control group to compare with and attribute the outcomes solely to my intervention. This design is suitable for situations where the intervention is conducted in multiple segments over time.

Moving on to the **Non-Equivalent** Control Group Design, the term "non-equivalent" refers to the lack of group equivalence due to non-random assignment. In this design, there is both an experimental group and a control group. The challenge with multiple outcome variables is that it becomes difficult to attribute changes in each outcome variable solely to the intervention, as there may be unmeasured interactions among these variables. The control group serves as a comparison group, and the design accounts for potentially influencing variables, making it a non-equivalent control group design.

Topic 98: Validity Threats and Causal Experimental Designs

In this module, we will examine a table that displays the threats to internal and external validity of findings, along with the causal experimental designs that can control or introduce these threats. Controlling these threats is crucial for establishing confidence in the research findings.

QUASI-EXPERIMENTAL DESIGNS										
4. Nonequivalent Control Group Design $O \quad X_1 \quad O$ $O \quad X_2 \quad O$	+	+	+	+	-	+	+	-	-	(+)
5. Time-Series Design $O O O O X O O O O$	-	+	+	-	+	(+)	+	(+)	-	(+)
6. Counterbalanced Design $X_1 O \quad X_2 O \quad X_3 O$ $X_3 O \quad X_1 O \quad X_2 O$ $X_2 O \quad X_3 O \quad X_1 O$	+	+	+	+	+	+	+	-	-	-

Note: Symbols: X or X_1 = unusual treatment; X_2 = control treatment; O = test, pretest, or posttest; R = random assignment of subjects to groups; + = factor controlled for; (+) = factor controlled for because not relevant; and - = factor not controlled for. This figure is intended to be a supplement to, not substitute for, textual discussions. See note that accompanies Figure 10.1.

In the table, the "+" signs indicate the threats that a specific design can control. If a design is used, it will address those threats. The "+" signs in brackets indicate threats that are not applicable or relevant to the studies utilizing that particular design. On the other hand, the "-" signs or dashes indicate that those threats are not controlled by the design and may affect the results.

Comparing the table with the previous module on pre-experimental designs, you will notice that there are fewer negative signs here. This implies that the causal experimental designs have more control over threats, making the results more reliable. However, it's important to note that there are still some uncontrolled threats associated with these designs.

In many cases, researchers have no alternative but to use these designs due to practical or ethical constraints. When utilizing these designs, it is essential to be aware of the threats specific to the design being used. Researchers can then discuss these threats and their potential impact on the interpretation of results. By acknowledging the presence of these threats and describing how they might affect the

results, researchers can provide a more comprehensive understanding of the limitations and uncertainties associated with their findings.

For example, if a design does not control for regression to the mean, researchers can mention this threat in the interpretation of their results. They can explain the expected uncertainty in the results due to this threat and provide insights on whether or not the results should be trusted considering this uncertainty.

In summary, the table illustrates the threats to validity associated with different causal experimental designs. While these designs offer more control over threats compared to pre-experimental designs, there are still potential uncontrolled threats. Researchers should carefully consider these threats and address them in the interpretation and write-up of their results to provide a comprehensive understanding of the limitations and uncertainties associated with their findings.

Topic 99: Types of True Experimental Research Designs

In this section, we will discuss three types of true experimental designs.

The first design is the **Pretest-Posttest Control Group Design**. In this design, both an experimental group and a control group are created. The subjects in both groups are randomly assigned. Prior to the treatment, pretests are administered to both groups. The experimental group receives the treatment, while the control group does not. After the treatment period, posttests are conducted for both groups. By following this design, researchers can control various threats to the validity of their findings. It is important to note that randomization is a key feature of all experimental designs, ensuring the unbiased allocation of subjects to groups.

The second design is the **Posttest Only Control Group Design**. In this design, there is no pretest conducted. Subjects are randomly assigned to either the experimental group or the control group. The experimental group receives the treatment, while the control group does not. Posttests are then administered to both groups. The absence of a pretest is a distinguishing feature of this design. Researchers need to be aware of the threats to validity that may arise due to the lack of a pretest when using this design. We will explore these threats in the next module.

The third design is the **Solomon Four-Group Design**, which combines elements of the previous two designs. Subjects are randomly assigned to four groups: two experimental groups and two control

groups. One experimental group and one control group undergo a pretest, while the other groups do not. The experimental groups receive the treatment, while the control groups do not. Posttests are conducted for all four groups. This design allows for a more comprehensive analysis by considering the effects of pretesting and treatment on the outcomes.

These three categories represent broad experimental research designs, known as true experimental designs. The random allocation of subjects to groups is a defining characteristic of true experimental designs. These designs provide researchers with more control over threats to validity. It is important to acknowledge the specific design being used and its implications for the validity of the research findings.

Topic 100: Threats to Validity of Research Findings in True Experimental Research Designs

In the previous module, we discussed three types of true experimental designs: The Pretest-Posttest Control Group Design, the Posttest Only Control Group Design, and the Solomon Four-Group Design. The purpose of these designs is to control threats to the validity of research findings, both internal and external.

Now, let's examine the table in front of us.

Designs	Sources of Invalidity									
	Internal								External	
	History	Maturation	Testing	Instrumentation	Regression	Selection	Mortality	Selection Interactions	Pretest-X Interaction	Multiple-X Interference
TRUE EXPERIMENTAL DESIGNS										
1. Pretest–Posttest Control Group Design $R \ O \ X_1 \ O$ $R \ O \ X_2 \ O$	+	+	+	+	+	+	+	+	–	(+)
2. Posttest-Only Control Group Design $R \ X_1 \ O$ $R \ X_2 \ O$	+	+	(+)	(+)	(+)	+	–	+	(+)	(+)
3. Solomon Four-Group Design $R \ O \ X_1 \ O$ $R \ O \ X_2 \ O$ $R \ X_1 \ O$ $R \ X_2 \ O$	+	+	+	+	+	+	+	+	+	(+)

It lists the internal and external threats to validity horizontally, and symbolically represents whether each threat is controlled or not for each design. If we compare this table with the previous tables, we can observe that in the true experimental design tables, the design names are preceded by a capital "R." This signifies the main difference between true experimental designs and other research designs. Random allocation of subjects to groups is a key feature of true experimental designs. This randomization ensures that subjects are assigned to groups in an unbiased manner.

In the table, each box contains either a plus sign, a plus sign in brackets, or a negative sign. A plus sign indicates that the design controls the associated threat, meaning that the threat does not affect the research findings when using that design. A plus sign in brackets indicates that the threat is not relevant in studies utilizing that specific design. A negative sign indicates that the threat is not controlled by the design and may still impact the validity of the results.

For example, in the Pretest-Posttest Control Group Design, there is a plus sign next to "maturation." This means that when this design is used, the threat of maturation is controlled. The maturation level of the experimental group and the control group will increase or decrease equally during the experiment, minimizing its impact on the results.

When using a design that controls a specific threat, it is important to mention it in the interpretation of the results. Conversely, if a threat is not controlled by the design, researchers need to address how they handled or mitigated that threat when interpreting the results.

Overall, true experimental designs offer greater control over internal and external threats to the validity of research findings compared to other research designs. By using these well-defined designs, researchers can minimize threats and enhance the reliability of their results.

Lecture 23: Research Designs/Strategies of Inquiry- IV

Topic 101: Qualitative Research Designs: Narrative Research

In the previous modules, we discussed the research designs employed in quantitative research, explored different types of research designs, and examined the threats they control. In this module, and the upcoming ones, we will delve into qualitative research designs. We will begin by discussing **narrative research**.

Narrative research in qualitative research refers to the study of life experiences and stories of individuals. It involves listening to people's narratives and gathering their life stories to interpret their meaning. The aim is to understand how these life stories have shaped people's lives. When you conduct research in this manner, it is known as narrative research. The primary data collected in narrative research are the life stories of individuals, which are obtained through interviews, personal narratives recorded in diaries, or insights gained from their life experiences.

To collect these narratives, you can engage in interviews or chat sessions with individuals, asking them about specific aspects of their lives that you are interested in. You can also request further explanations or ask them to provide relevant documents. These narratives are then recorded, and various methods can be used to analyze them. Thematic analysis is a commonly employed method where you identify and categorize themes within the life experiences. Discourse analysis is another method used to analyze the narratives, aiming to understand the underlying meanings, patterns, and how these life experiences have influenced and shaped individuals' lives.

When you employ this approach to address a research problem, it is referred to as narrative research. Qualitative researchers often utilize this research design when they want to explore different facets of people's lives or gain insights into individuals' experiences through their life stories.

Topic 102: Grounded Theory

In qualitative research, one popular research design is grounded theory. Grounded theory is commonly used when investigating a phenomenon for which there are no existing theoretical explanations. **For example**, when a new phenomenon emerges in society, such as the **impact of social media on personal space**, grounded theory can be employed to understand the underlying factors and develop a theory to explain its occurrence.

The key features of grounded theory can be summarized as follows:

1. **Inductive Approach:** Grounded theory adopts an inductive approach, meaning researchers enter the research process without preconceived assumptions. They collect evidence and synthesize it to develop a theory, moving from specific observations to more general explanations.
2. **Iterative Data Collection:** Data collection in grounded theory is an iterative process. Researchers do not collect data at a single point in time but engage in a continuous cycle of data collection, analysis, and further data collection. This iterative approach allows for a deeper understanding of the phenomenon and ensures that no important aspects are overlooked.
3. **Constant Comparison:** Grounded theory involves constant comparison of data. Researchers analyze data from multiple contexts, communities, and diverse groups of people. By comparing how individuals from different backgrounds and settings experience and interact with the phenomenon, researchers can identify commonalities, differences, and underlying causalities.
4. **Theoretical Saturation:** Grounded theory aims to reach theoretical saturation, which means that researchers collect evidence until no new insights or manifestations of the phenomenon emerge. This occurs when data from various contexts, backgrounds, and communities have been thoroughly examined, and there is a comprehensive understanding of the phenomenon's causations. At this point, researchers can develop a theory based on the synthesis of the empirical evidence collected.

In summary, grounded theory is employed when there is a need to understand and explain a phenomenon for which existing theories are lacking. Researchers adopt an inductive approach, iteratively collect and analyze data, make constant comparisons, and strive to reach theoretical saturation. By following this design, they can develop a theory based on empirical evidence and contribute to the understanding of the phenomenon under investigation.

Topic 103: Phenomenology

Phenomenology is a widely used qualitative research design that focuses on studying individuals' first-hand experiences. Unlike narrative research, which involves gathering and interpreting narratives from others, phenomenology is concerned with researching individuals who have directly experienced a particular phenomenon themselves. For instance, if there was an earthquake, phenomenology would involve studying the experiences of those who personally witnessed and went through the event. The key distinction is that phenomenology centers on capturing the essence of lived experiences.

Description of subjective Experiences: When conducting phenomenological research, the emphasis is on understanding individuals' experiences as they describe them, rather than interpreting or explaining their accounts. These experiences are subjective and unique to each individual. Although two people may have undergone the same experience, their narratives of the event will differ due to the element of subjectivity. It is important to focus on first-person accounts and how individuals describe their experiences.

Suspension of Preconceptions: The primary goal of phenomenological research is to comprehend the essence of individuals' experiences and the meaning they attribute to them. By understanding what individuals experienced and how they interpret their experiences, researchers can extract subjective meanings. It is crucial for the researcher to approach the study without preconceived notions or biases that could influence their understanding of participants' lived experiences.

Bracketing: This involves setting aside personal opinions and preferences, a process known as bracketing or epoché. Bracketing allows researchers to suspend their own preconceptions to gain a better understanding of the phenomenon under investigation.

Intentionality: Intentionality is another important aspect of phenomenological research. Intentionality refers to individuals' intentional acts and the meanings they attribute to their experiences.

Researchers should recognize that when individuals explain their lived experiences, there is intentionality behind their attributions of meaning. It is essential to value, clarify, and rationalize these attributions in the analysis process.

Phenomenological Reduction: Phenomenological reduction is a step within the design that involves examining the layers of meaning within an individual's experience. A person's experience is influenced by their culture, society, community, and personal background, all of which contribute to their process of meaning-making.

Descriptive and interpretive analysis: Analyzing the data in phenomenological research involves studying these layers of meaning, patterns, and structures. By refining these structures, researchers can eliminate personal biases and reach a deeper understanding of the original meaning of the lived experience. In phenomenological analysis, it is important to have a framework that ensures the process remains unbiased and unaffected by the researcher's intentions and reductions. This framework aids in identifying themes, patterns, and structures, ultimately leading to meaningful conclusions. Following the six important steps of phenomenology, researchers can ensure the integrity of the process described above.

By following these six important steps, researchers can conduct phenomenological research that genuinely captures and explores individuals' lived experiences.

Topic 104: Ethnography

In Module 104, we explore ethnography as a qualitative research design. Ethnography involves studying phenomena in their natural settings, without any experimental or artificial arrangements. It focuses on understanding phenomena as they naturally occur within our own culture, society, or community. Ethnographic research involves immersing oneself in the setting and observing the natural flow of events, behaviors, values, and practices.

1. Before collecting data, a crucial prerequisite for ethnographic research is **participant observation**. The researcher spends a considerable amount of time with the people in the natural setting, becoming familiar to them. This ensures that when data collection begins, the researcher's presence does not disrupt the natural setting, and participants don't perceive them as an outsider. By establishing rapport and familiarity, the researcher can collect data effectively without participants being conscious of their presence.

2. **Contextual understanding** is another vital aspect of ethnographic research. Every society and culture has its own norms, values, and practices that shape people's behavior. To comprehend social phenomena within these variables, the researcher's understanding of the context becomes crucial. The closer the research is to the natural setting, the greater the validity of the findings.
3. Ethnographic research emphasizes the need for **multiple perspectives and thick description**. By gathering data through observations and conversations, researchers aim to understand people's experiences comprehensively. It involves collecting data from various situations and contexts, formally and informally, to cover the complexities and nuances of the phenomenon under study. Through thick descriptions, researchers gain a more holistic understanding of the people and their behaviors.
4. Ethnographic research is an **iterative and reflexive process**. It acknowledges that no single data collection method can answer all research questions. After the initial data collection, researchers identify any gaps that need to be addressed. They iterate by collecting additional data from different sources and **triangulating** the information.
5. **Triangulation** involves comparing information from multiple sources to assess its credibility. Information that converges and aligns across multiple sources is considered more trustworthy.

After this process, researchers analyze the filtered information and proceed with report writing and drawing inferences. The information gathered from multiple sources, corroborating with each other, forms the basis for credible conclusions. However, if information from multiple sources about the same phenomenon contradicts or does not align, its reliability diminishes. Researchers use **triangulation** to assess the credibility of different sources and draw meaningful inferences.

By following these steps, ethnographic researchers ensure a rigorous and comprehensive approach to understanding and reporting phenomena in their natural settings.

Topic 105: Case Study Research

The final research design we will discuss in this course is case study research. In case studies, a specific phenomenon is examined, referred to as a "case." A case is characterized by its unique attributes, distinguishing it from other instances of the same phenomenon. While there may be numerous occurrences of the phenomenon, the way it manifests in a particular place, organization, individual, or community sets it apart. The essence of a case study lies in selecting an individual, group, organization,

or community as the focus of investigation. Once a case is chosen, it is studied holistically, exploring all relevant aspects that contribute to its occurrence. The research findings provide insights into the factors that underlie its uniqueness.

The characteristics of the case study design can be summarized as follows:

1. **Focus on a specific case:** The first step in conducting a case study is selecting a specific case. This could be an individual, a group, an organization, a community, an event, or a unique situation. The case's uniqueness is the basis for studying the factors that differentiate it from others.
2. **Holistic perspective:** A case study entails examining the selected case from multiple perspectives. It involves researching its social, cultural, economic, and organizational aspects, among others, to understand its uniqueness in comparison to other cases. This holistic perspective is essential in capturing the complexity and richness of the phenomenon under investigation.
3. **Inductive reasoning:** Given the uniqueness of the case, the case study design typically employs an inductive method. Researchers do not approach the study with preconceived assumptions; instead, they adopt an inductive approach for data collection and analysis. They gather evidence at a basic level and identify patterns, themes, and relationships to gain a comprehensive understanding of the unique case.
4. **Triangulation of data:** As with other research methods, triangulating various sources of information is crucial in case study research. Triangulation enhances the reliability and validity of the collected data. By gathering information from multiple sources and comparing findings, researchers can increase confidence in their understanding of the phenomenon. For example, collecting data on the economic aspect of the case from different sources helps establish credibility.
5. **Unique and contextualized insights:** In a case study, it is essential to contextualize the uniqueness of the case in its various aspects. Researchers explore the reasons behind the case's distinct features and explain the factors contributing to its uniqueness. Identifying and explaining these reasons form the basis of the research findings.

By following these key characteristics, conducting a case study allows researchers to gain a deep understanding of a specific phenomenon. It enables them to explore the unique aspects, factors, and

explanations associated with the case. If a researcher successfully identifies and explains these aspects, they can confidently claim to have conducted a case study.

Lecture 24: Research Designs/Strategies of Inquiry- V

Topic 106: Mixed-Method Research Designs

In the previous modules, we discussed quantitative research designs and qualitative research designs. Continuing in the same vein, we will now explore the third set of research designs known as mixed-method research designs. These designs involve the simultaneous use of both qualitative and quantitative research approaches in various patterns. One such design is the convergent mixed-method design, which we will discuss.

The term "**convergent**" in this context signifies the integration of qualitative and quantitative research designs. There are different ways to employ this design.

1. Simultaneous data collection: One approach is to begin the research by collecting both qualitative and quantitative data simultaneously. In this case, qualitative and quantitative data are collected from the outset, focusing on the same research questions. The qualitative and quantitative data collection processes are not dependent on each other and can be conducted concurrently. By following this approach, researchers aim to gather extensive and rich information from both sources, which will complement and supplement each other.

2. Separate data analysis: In the convergent mixed-method design, the next step is to analyze the qualitative and quantitative data separately using their respective methods. Quantitative data is analyzed using quantitative techniques such as descriptive and inferential analyses, while qualitative data is analyzed by identifying themes and patterns through synthesis. The analytical methods employed in qualitative research, as discussed earlier, can be utilized to analyze the qualitative data. Consequently, researchers perform two distinct types of analyses simultaneously.

3. Comparison and integration: The third characteristic of this design is the integration of the analyzed data from both methods. The goal is to create a cohesive and integrated analysis that combines and reconciles the findings from qualitative and quantitative analyses.

4. Convergent synthesis: Researchers seek to identify points of convergence and divergence in the findings. Even if the findings differ, a synthesis is conducted to determine what conclusions can be

drawn when considering both types of analyses together. This process involves integrating the qualitative and quantitative findings to arrive at a comprehensive and supported conclusion.

In summary, the convergent mixed-method design involves the simultaneous collection of qualitative and quantitative data, followed by separate analyses using appropriate methods. The final step entails integrating the findings from both types of analyses to draw conclusions that are supported by the combined evidence.

Topic 107: Explanatory Sequential Methods

In this module, we will continue our discussion on mixed-method designs and focus on the explanatory sequential method. This design involves conducting the quantitative phase first, followed by the qualitative phase. The term "explanatory" is used because the qualitative phase helps to explain the quantitative findings.

Quantitative phase

In the explanatory sequential method, the researcher begins by completing the quantitative phase of the research. During this phase, data is collected from a relatively large sample. The aim is to collect data on a larger scale and conduct quantitative analysis, which can include statistical analyses to determine differences, relationships, and descriptive information. In this phase, specific research questions are addressed using appropriate quantitative analysis techniques. Once the quantitative phase is completed, the researcher identifies patterns or questions that require further exploration.

Qualitative phase

Once the quantitative phase is completed, the researcher moves on to the qualitative phase. In this phase, a selected sample of participants is chosen based on the patterns or characteristics observed in the quantitative analysis. The qualitative phase is focused on collecting context-specific data from these selected participants. The researcher may utilize tools such as focus groups or in-depth interviews to gather this qualitative data.

It's important to note that the qualitative phase involves different tools and approaches compared to the larger quantitative phase. The data collected in the qualitative phase is then analyzed using qualitative techniques. This analysis typically involves synthesizing themes from the qualitative data, identifying patterns, and analyzing the perspectives and experiences of the participants.

Once the data has been analyzed separately in both the quantitative and qualitative phases, the findings are integrated. The **integration** process involves comparing and contrasting the findings from both phases to identify areas of convergence and divergence. The goal is to understand how the qualitative findings can help explain or further interpret the quantitative findings. The researcher then writes a report, describing the integrated findings, drawing conclusions, and discussing the implications of the research.

In summary, the explanatory sequential method involves conducting the quantitative phase first, followed by the qualitative phase. The findings from both phases are then integrated to provide a comprehensive understanding of the research problem. This method allows for a deeper exploration and explanation of the quantitative results using qualitative data.

Topic 108: Exploratory Sequential Methods

Moving on to the next design in mixed methods research, we have the exploratory sequential method. In this design, the qualitative phase is conducted first, followed by the quantitative phase. The term "exploratory" signifies the aim of exploring a research topic or phenomenon. The sequence in this design is qualitative followed by quantitative.

The exploratory design is typically employed when researchers want to explore a research topic for which there are no well-established theories or understanding of causal relationships. The objective is to gain an in-depth understanding of the phenomenon before using quantitative methods to examine it on a larger scale for generalizability. This design follows an inductive approach, where the qualitative phase precedes the quantitative phase.

Qualitative phase

In the qualitative phase, the researcher gathers information about the phenomenon from various communities, societies, or groups of individuals representing different strata of society. This phase involves collecting rich, ground-level data to explore the nature of the phenomenon. The data collected are then analyzed using coding techniques to identify themes, sub-themes, and patterns that help in understanding the phenomenon and its underlying causalities.

Quantitative phase

Once insights are gained from the qualitative phase, the researcher moves on to the quantitative phase. The purpose of this phase is to collect data from a larger sample to examine the generalizability of the findings. Various quantitative methods can be employed, such as surveys, experiments, correlational studies, or causal-comparative studies. The analysis methods used in the quantitative phase are similar to those used in traditional quantitative research, and the goal is to generalize the findings to a broader population.

Integration and synthesis

After completing both phases, the researcher integrates the findings from the qualitative and quantitative analyses. The integration involves comparing and contrasting the results obtained from both phases, identifying points of convergence and divergence. These connections are then explained in the research report, providing a comprehensive understanding of the research topic. The strength of this design lies in the ability to combine qualitative exploration with quantitative generalizability.

In summary, the exploratory sequential design involves conducting the qualitative phase first to explore a research topic, followed by the quantitative phase to examine the generalizability of the findings. The integration of the findings from both phases helps identify connections and provides a deeper understanding of the research topic.

Topic 109: Action Research

Action research is a type of research that is conducted by researchers or practitioners to address specific problems they encounter in their day-to-day work. Its main purpose is to bridge the gap between theory and practice. When engaging in action research, the researcher focuses on solving practical issues within a specific context, aiming to improve their own practices.

The key characteristic of action research is that it involves active participation from all stakeholders or participants in the context being studied. This includes individuals who are directly affected by the problem and can contribute to its solution. **For example**, if a teacher is conducting action research in their classroom, the students and parents become participants in the research process as they experience and help address the identified problem.

The steps of action research are similar in structure to other types of research. **The first step is problem identification**, where specific practices and processes are examined to identify areas for

improvement within the given context. Unlike traditional research, **the involvement of stakeholders** is crucial in action research to gain a comprehensive understanding of the problem.

Data collection is an important phase in action research. Various research tools such as interviews, surveys, observations, and document analysis can be used to gather data. However, the way data is collected and used differs from other research approaches. In action research, data collection is an ongoing process, and the researcher reflects on the data to gain deeper insights and modify the research process if necessary.

Progressive planning and implementation are integral to action research. It involves developing an action plan that is continuously modified and adjusted based on the evolving needs of the research. **Monitoring the outcomes** and ensuring they align with the desired goals is essential. Action research emphasizes continuous reflection throughout the process, enabling the researcher to develop a deeper understanding of the problem and make necessary adjustments.

Knowledge sharing and reflection play a significant role in action research. The focus is on learning and staying engaged with others involved in the research process. The findings and outcomes are not generalized but modified and tailored to the specific context, with the aim of addressing the problem effectively and continuously. The usefulness of the research output is constantly **evaluated** to ensure it aligns with the intended goals.

In summary, action research follows a similar structure to other research approaches, but its thought process and concepts differ. It is conducted to address specific problems in practice, involving active participation from stakeholders. The steps involve **problem identification, ongoing data collection and reflection, progressive planning and implementation, continuous monitoring, and knowledge sharing**. The emphasis is on improving practice and informing theory by bridging the gap between theory and practice.

Topic 110: Levels of Action Research

Continuing our discussion on action research, we will explore the different levels at which action research can be conducted. By "level" here, we refer to the various tiers or types of interests that can be addressed through action research.

The first level of action research is the **individual or personal level**. At this level, the focus is on personal growth and professional development. **For example**, as a teacher, conducting action research at this level would involve improving one's own practice. The research may target a specific student or a small group of students within a classroom setting. The aim is to resolve issues and enhance teaching methods. Data collection would primarily revolve around student understanding and communication improvement. The analysis of data would be geared towards informing personal skills, knowledge, and practices. The research outcome would contribute to personal and professional development. This level of action research is termed as individual or personal.

Moving on to the next level, we have the **organizational or group level**. At this level, the focus shifts from personal development to addressing issues within an organization or team. For instance, in a school setting, action research at this level may target moral development or training across the entire school rather than a specific class or group. The goal is to improve the organization as a whole, with the school administration and staff working together towards common objectives. The research may involve policy improvements and the resulting changes would have broader implications for the organization. While the impact may not be directly related to the individual conducting the research, it would still affect them indirectly, as the changes occur at an organizational level.

Expanding further, we reach the **community level of action research**. At this level, the focus broadens to address societal issues. Action research conducted at this level may tackle topics such as social justice or community cleanliness. The research is aimed at bringing about social change rather than individual or organizational change. The scope of collaboration in this research extends to community stakeholders and organizations. Engagement with these stakeholders is crucial to identify and address the issues at hand. The research outcomes may lead to policy changes and practices that impact the entire society. This level of action research emphasizes the theory-practice gap and strives to bridge it for the benefit of the community as a whole.

In summary, all levels of action research share a common purpose of addressing the theory-practice gap. The research problems tackled at each level focus on this gap, and the resulting outcomes benefit the respective level at which the research is conducted—whether it is the individual level, the organizational level, or the community level.

Lecture 25: Sampling Techniques-I

Topic 111: Sampling Techniques

Sampling techniques are crucial in research as they allow researchers to study a selected group of individuals or elements from a larger population. In this module, we will focus on the concepts of population and sample, and their significance in research.

The population refers to the entire group of individuals or elements that the researcher wants to study. For example, if the research topic is educational issues in government schools in the province of Punjab, Pakistan, then the population would consist of all the teachers working in government schools in Punjab. The population can be any group of individuals, objects, cases, or entities that fall within the scope of the research.

On the other hand, a sample is a subset of the population that is selected for study. Instead of studying every single individual in the population, researchers select a smaller group that represents the larger population. The goal is to ensure that the sample is representative of the population so that the findings from the sample can be generalized to the entire population.

To illustrate this concept, imagine you have cooked a dish and want to know how it tastes. Rather than eating the entire dish to judge its taste, you take a small portion, a sample, and taste it. Based on the taste of the sample, you can infer the taste of the whole dish. Similarly, in research, the sample is like a smaller version of the population, and by studying the sample, researchers aim to draw conclusions about the entire population.

Topic 112: Research Population

In this module, we will delve deeper into the concepts of population, target population, and accessible population in research. These terms are essential in defining the scope of a study and selecting a representative sample. Let's explore these concepts in more detail.

The population refers to the entire group of individuals or elements that the researcher aims to study. In your example, the population would consist of all teachers who teach in public schools in Punjab or Pakistan. It encompasses all teachers within this context.

However, within this population, you may have specific criteria or characteristics that you want to focus on for your study. **For instance**, you might be interested in studying teachers who have completed 16 years of education and have at least two years of teaching experience. The group of

teachers who meet these criteria becomes your target population. They are the subset of the population that aligns with your specific research objectives.

It's important to note that even within the target population, you may not have access to data or information on every individual. **For example**, your databases may not have records of all teachers with the desired qualifications and experience. Additionally, certain individuals may be inaccessible due to various reasons, such as incomplete records or unavailability for participation. The group of individuals from the target population for whom you can obtain data is called the accessible population. It represents the portion of the target population that you can feasibly reach and include in your study.

Research Sample

Once you have defined your population, target population, and accessible population, the next step is to select a sample from the accessible population. The sample is a smaller group of individuals that you will include in your study and collect data from. The size of the sample is typically smaller than the population or target population, but it should be sufficient to draw meaningful conclusions.

When selecting a sample, there are certain principles to consider. A good sample should be representative of the population or target population. It means that the sample should reflect the characteristics and diversity present in the larger group. **For example**, if the population consists of both male and female teachers in certain proportions, the sample should also include a similar proportion of male and female teachers.

Furthermore, the size of the sample should be adequate to ensure reliable and valid conclusions. The sample should be large enough to capture the necessary variation in the population and provide sufficient statistical power for analysis.

To ensure generalizability, the sample should possess characteristics that allow for the findings to be applicable to the broader population. By selecting a sample that accurately represents the population or target population, you increase the likelihood of generalizing the conclusions from your study.

In summary, the population represents the entire group of individuals or elements of interest in a research study. The target population is a subset of the population that meets specific criteria relevant to the research objectives. The accessible population refers to the individuals within the target population for whom data is available or accessible. From the accessible population, a sample is

selected to draw conclusions and make inferences about the larger population. The sample should be representative, proportional, and of an adequate size to ensure generalizability and reliability.

Topic 113: Probability/ Random Sampling Techniques

Let's delve deeper into the discussion on sampling methods. There are two broad categories of sampling techniques: random sampling and non-random sampling, also known as probability sampling. Random sampling ensures that the sample drawn is unbiased and representative of the population.

There are four main types of random sampling including simple random sampling, stratified random sampling, and cluster random sampling and multi-stage random sampling. To simplify the concept of multi-stage random sampling, it is not considered a distinct type of sampling. Instead, it involves combining different methods in a sequential manner. For example, you could perform statistical analysis at one stage, followed by cluster enumeration at the next stage, and use both methods within the same sampling procedure. This combination is referred to as multi-stage random sampling.

So, actually there are three main types of sampling: simple random sampling, stratified random sampling, and cluster random sampling. When conducting stratified or cluster sampling, you would typically employ simple random sampling as the final step of both methods

Simple Random Sampling: In simple random sampling, the concept of randomness ensures that the probability of selecting any individual is equal. To illustrate this, imagine a group of people depicted in different colors, with some individuals marked with a circle. If you want to select a number of individuals, you would use simple random sampling. In simpler terms, consider a lottery where you place 10 slips with names into a basket without looking at them. You need to select three people, so you randomly draw slips from the basket. The probability of your name being on the selected slips is equal to the probability of anyone else's name being chosen. Each person has a one in ten chance of being selected. This equal opportunity of selection is what defines randomness. When you draw names from the basket in this way, you are conducting a simple random sample.

Topic 114: Simple Random Sampling Techniques

Let's discuss the concept of simple random sampling in more detail using the same example of teachers.

For instance, you are selecting 10 teachers from a population of 100 teachers in Pakistan. In this case,

the population consists of teachers with different ages, genders, and qualifications, but your main criterion is that they are teachers.

Simple random sampling ensures equal opportunity for every individual in the population to be selected. Each teacher has a $1/100$ chance of being chosen in the first draw. For instance, if you were among the teachers, the probability of your name being selected is also $1/100$.

To conduct a simple random sample, you would draw names one by one, using a lottery-like method. After each draw, you note down the selected name, fold the slip, put it back in the basket, and shake it before drawing the next name. This process continues until you have selected a total of 10 teachers.

The advantage of simple random sampling is that it provides equal chances of selection for every individual, ensuring fairness. However, there is a potential drawback. If your research question involves comparing the opinions of male and female teachers, **for example**, and you randomly mix the slips of male and female teachers in the same basket, there is a possibility that one group may be underrepresented in the sample. This can hinder your ability to compare the opinions of male and female teachers accurately.

Thus, simple random sampling is suitable when the population is homogeneous, without distinct subgroups, and when your research question does not involve comparing specific subgroups. It is essential to consider the characteristics and subgroups of the population when selecting the appropriate sampling method. If you need to compare subgroups, you may need to employ other sampling methods that ensure the representation of all relevant groups.

In summary, simple random sampling involves selecting individuals from a population without any specific criteria except that they belong to the population of interest. It is suitable when the population

is homogeneous, and there is no need to compare subgroups. The process involves randomly drawing individuals one by one from the population until the desired sample size is achieved.

Lecture 26: Sampling Techniques-II

Topic 115: Stratified Sampling

In this module, we will discuss another widely used random sampling technique called stratified random sampling. In your case, where you want to compare the opinions of male and female teachers, simple random sampling may not be sufficient because it does not guarantee an adequate representation of both groups. To ensure a balanced representation, you need to stratify or divide the population into **subgroups** based on relevant characteristics such as gender, **qualifications**, or **age**.

The process of stratification involves grouping individuals with similar characteristics together. **For example**, you can create separate groups for male and female teachers. By doing this, you ensure that each subgroup is represented in the sample. It is important to note that the specific criteria for stratification depend on your research questions and hypotheses.

To implement stratified random sampling, you first need to define your population and determine the desired sample size. Let's say you have a population of 100 teachers and you want to select a sample of 30. The next step is to stratify the population based on a relevant characteristic, such as gender. In this case, you would have two strata: male teachers and female teachers.

Once the population is stratified, you can determine the proportionate allocation of the sample to each stratum. Proportionate allocation means that the sample size for each stratum is proportional to the size of that stratum in the population. For instance, if there are 70 female teachers and 30 male teachers in the population, you would allocate a larger portion of the sample to the female teachers' stratum to reflect their higher representation in the population.

After determining the sample sizes for each stratum, you can proceed with random sampling within each stratum. This can be done using simple random sampling within each group, ensuring that individuals are selected randomly from each stratum.

By following this procedure, where the population is divided into strata and the sample is allocated proportionately, you have created a stratified proportionate random sample. This sampling method

allows for a more accurate comparison or analysis of the subgroups identified in your research question.

In summary, stratified random sampling involves dividing the population into subgroups based on relevant characteristics and ensuring proportional representation of each subgroup in the sample. By doing so, you can address specific research questions or compare different groups accurately. The steps involved are defining the population, determining the sample size, stratifying the population, allocating the sample proportionately, and conducting random sampling within each stratum.

Topic 116: Cluster Sampling

Before discussing cluster random sampling, let's take a few minutes to understand the **distinction between stratification and clustering**. Both methods involve dividing a population into subgroups, but it's important to understand the differences to distinguish between stratified random sampling and cluster random sampling.

When we stratify a population, we divide it into more than two subgroups, with the basic principle that all individuals within a subgroup share at least one common characteristic. **For example**, let's consider teachers and create subgroups based on their gender. We would have one subgroup for male teachers and another for female teachers. Although the individuals within a subgroup may differ in other characteristics such as age or qualifications, they share a common characteristic, in this case, their gender.

In contrast, clustering involves grouping individuals or elements based on certain criteria, and all items within a cluster can differ from each other. Clusters are primarily formed based on location. Cluster sampling is useful when the population is widely dispersed across different areas, and selecting individuals from each location individually would be challenging. Instead, we select entire clusters based on their location. Within a cluster, all individuals share similar characteristics, except for the one characteristic that defines the cluster, such as location.

Cluster Sampling

To illustrate cluster sampling, let's consider the **example** of university teachers. Suppose we find that university teachers across different universities are similar in terms of their characteristics and qualifications. Instead of collecting data from two or three teachers from each university, which would

be a difficult and time-consuming task, we can select a few representative universities or clusters. These clusters, in this case, are intact groups with similar characteristics, and we select them to save time and resources.

The steps for cluster sampling are similar to other sampling methods. You first identify the population and decide on the number of clusters to select. Then, you determine the sample size and select the clusters using a random sampling method. **For example**, if you want to select 25 clusters from a total of 200 universities, you can write the names of all the universities on slips of paper and randomly select the required number of slips.

Once you have selected the clusters, you collect data from all the individuals within those clusters. In the case of university teachers, you would gather data from all the teachers within the selected universities. This approach allows you to save time and expenses by working with intact groups that share similar characteristics.

In summary, cluster random sampling involves dividing a population into clusters based on a defining characteristic, such as location, and selecting intact groups as a representative sample. The clusters have similar characteristics, except for the one characteristic that defines the clusters. The steps for cluster sampling are similar to other sampling methods, where you identify the population, select the number of clusters, determine the sample size, and randomly select the clusters.

Topic 117: Multi stage Sampling

In this module, we will discuss multi-stage sampling. Multi-stage sampling refers to the use of multiple techniques in a single sampling procedure. It is not a new technique on its own but rather a combination of different sampling methods. Let's take a closer look at how multi-stage sampling works.

Cluster sampling, which we previously discussed, involves dividing the population into clusters and selecting intact groups as a sample. In multi-stage sampling, we can combine cluster sampling with other techniques in any order that suits our needs.

Let's consider an example where we use a combination of cluster sampling and stratified sampling. Imagine we have a population of universities, and we start by treating each university as a cluster. We divide the universities into different categories based on their disciplines, such as engineering, medical, and general universities. Each category is represented by a different color.

Now, we have created strata within each cluster, with universities of the same discipline grouped together. **For example**, we have a set of medical universities, a set of engineering universities, and so on. We can represent these strata using boxes or segments.

Next, we select a sample from each stratum. The selection can be proportional to the size of the stratum. **For instance**, if there are three medical universities in a stratum, we may select two from them. Similarly, we select a proportionate number of universities from each stratum.

After selecting the universities from each stratum, we move to the next stage. We can then select samples from the selected universities. This could involve selecting teachers or departments, depending on the level of analysis required. Each selected teacher or department can be represented by a **star**.

The process described above is an example of multi-stage sampling. It involves dividing the population into clusters in the first stage. Then, within each cluster, we create strata based on specific criteria. In the second stage, we select samples from each stratum. Finally, in the third stage, we select the final sampling units, such as teachers or departments.

It's important to note that the number of stages in multi-stage sampling can vary depending on the complexity of the sampling design. The stages can be adjusted as needed to obtain the desired sample.

In summary, multi-stage sampling involves the use of multiple techniques in a single sampling procedure. It allows for more flexibility and customization in selecting samples from a population. The stages typically include dividing the population into clusters, creating strata within each cluster, and selecting samples from each stratum.

Topic: Systematic Sampling

In this module, we will discuss the method of systematic sampling and the different perspectives on whether it should be considered random or non-random sampling.

Systematic sampling is a sampling technique where the researcher selects every "kth" element from the population to form the sample. The value of "k" represents the sampling interval, which is determined by dividing the population size by the desired sample size. **For example**, if the population size is 100 and the sample size is 25, the sampling interval would be $100/25 = 4$.

To implement systematic sampling, the researcher starts by selecting a random starting point from the population. Let's say the starting point is the **second element**. The researcher then selects every third element from that point onwards until the desired sample size is reached. In this example, the researcher would select the second, five, eight, eleven and so on, until a total of 25 individuals are chosen.

Now, let's discuss the differing opinions regarding whether systematic sampling is considered random or non-random sampling. Some people argue that systematic sampling can be considered a form of random sampling because the starting point is selected randomly. Once the starting point is chosen, the subsequent selections follow a systematic pattern according to the sampling interval. From this perspective, systematic sampling is seen as a random procedure.

On the other hand, some individuals argue that systematic sampling is not truly random because the researcher has control over the starting point selection. They believe that the choice of the starting point introduces an element of subjectivity and may result in some individuals having a higher or lower chance of being selected. This perspective suggests that systematic sampling should be classified as a non-random sampling method.

Ultimately, whether systematic sampling is considered random or non-random can vary depending on different interpretations and perspectives. Some researchers may view it as random due to the random selection of the starting point, while others may consider it non-random due to the researcher's control over the starting point selection.

In conclusion, systematic sampling is a sampling technique where every "kth" element is selected from the population to form the sample. The differing opinions on whether it is classified as random or non-random sampling stem from the researcher's control over the starting point selection.

Lecture 27: Sampling Techniques-III

Topic 119: Non-Probability Sampling Techniques

Module 119 focuses on non-probability sampling, which is also known as non-random sampling. The use of non-random sampling methods is necessary in certain research scenarios where population data is unavailable, and individuals related to the research topic are unwilling to participate. In such cases, even if a random sample is drawn, the individuals contacted may not agree to participate due to the

basis of their selection. This makes it difficult to obtain representative samples through random sampling, thus necessitating the use of non-random sampling techniques.

In this course, we discuss **four types of non-random sampling techniques**, two of which are commonly used in educational research: **convenience sampling** and **purposive sampling** (also known as judgmental sampling). The other two techniques covered are **snowball sampling** and **quota sampling**.

Let's consider **convenience sampling** as an example. The term "convenient" refers to the population being studied. The circles on the slide represent the population. For instance, let's say I live in a hostel and want to research the quality of the food in the hostel mess. If I were to conduct a simple random draw from the population, it is possible that some individuals who are registered with the hostel mess have never actually used it because they joined the hostel after having their meals elsewhere. Therefore, a random sample may include individuals who are not relevant to my research objective. In this case, I would prefer to use convenience sampling.

To implement convenience sampling, I would position myself at the entrance of the mess during breakfast time and request individuals entering or leaving to answer a few questions about the quality of the food. By doing this, I can target those individuals who actually consume food from the mess. This approach reduces the effort and waste of sampling procedure by focusing on individuals who are more likely to provide useful information.

It is important to note that not everyone has an equal chance of selection in convenience sampling. As I sit there for a few days, only those individuals who use the mess during that time period have a chance to be selected. In this situation, when the selection is based on characteristics and preferences other than equal probability, it is considered a non-probability, non-random sample.

The main risk associated with convenience sampling and other non-probability sampling methods is that the resulting sample may not be representative of the population. One way to mitigate this issue is by increasing the sample size. In statistics and research literature, it is often suggested that a sample size of at least 320 is needed to minimize the problems associated with non-random sampling. By increasing the sample size, the issues related to generalization can be addressed.

To summarize, non-probability sampling, including convenience sampling, is employed in situations where population data is unknown, and individuals are unwilling to participate voluntarily. While non-

random sampling methods may introduce biases, increasing the sample size can help mitigate these issues and improve generalization.

Topic 120 : Purposive sampling

Module 120 focuses on non-probability sampling, specifically purposive sampling, also known as judgmental sampling. Purposive sampling involves selecting participants for a study based on specific criteria that align with the researcher's purpose. **For example**, if a researcher wants to collect data on cricket bowlers, they may choose individuals who are recognized experts in bowling. The criteria for selection are determined by the researcher and are driven by the research question.

In purposive sampling, the researcher sets specific criteria for selecting participants. These criteria are based on the researcher's knowledge and research question. **For instance**, the researcher may require bowlers who have played a certain number of test matches, achieved a particular bowling speed, and taken a minimum number of wickets in international cricket. The researcher's judgment is used to establish these criteria and select participants accordingly.

It is important to note that the terms "purposive sampling" and "judgmental sampling" are often used interchangeably in literature. However, there is a distinction between the two. In purposive sampling, the researcher establishes criteria based on their research question to guide participant selection. On the other hand, judgmental sampling allows the researcher to use their personal judgment or observation as the basis for selecting participants, even without predefined criteria. In judgmental sampling, the researcher's judgment may be rational and based on certain criteria, or it may be subjective and lacking a specific framework for selection.

While some sources consider purposive sampling and judgmental sampling to be the same, it is important to recognize the potential difference between them. Purposive sampling involves establishing criteria based on knowledge and research question, while judgmental sampling allows for selection based on personal judgment and observation, with or without specific criteria. The distinction between these terms may vary in different sources, but acknowledging the potential difference can help clarify their usage.

In summary, purposive sampling (or judgmental sampling) is a non-probability sampling method where participants are selected based on specific criteria established by the researcher. The criteria are driven by the research question and purpose of the study. While these terms are often used interchangeably, there may be a difference between them, with purposive sampling emphasizing criteria-based selection and judgmental sampling allowing for selection based on personal judgment and observation.

Lecture 28: Sampling Techniques-IV

Topic 121: Snowball Sampling

Snowball sampling is a non-probability sampling technique that is commonly used in research when studying populations or phenomena that are difficult to access through random sampling methods. It is particularly useful when researching socially sensitive topics or studying populations that are stigmatized or hard to reach.

In snowball sampling, the researcher starts with a small number of initial participants, often referred to as "seeds" or "key informants." These individuals are selected based on their knowledge, experience, or connection to the research topic. The researcher establishes a personal connection with them, assuring confidentiality and building trust.

Once the initial participants have been identified, the researcher collects data from them through interviews, surveys, or other data collection methods. Afterward, the researcher asks these participants to refer or introduce them to other individuals who may meet the research criteria or have relevant experiences. This process continues iteratively, with each participant providing referrals to other potential participants.

The snowball sampling method relies on the idea that individuals within a specific population or social network tend to know others who share similar characteristics or experiences. Through these referral chains, the sample size gradually increases, resembling a snowball rolling and accumulating more participants as it moves forward.

One advantage of snowball sampling is that it allows access to populations that are otherwise challenging to reach, such as marginalized communities, hidden populations, or individuals engaged in illicit activities. The method leverages existing social connections and trust networks to gain entry into these groups.

However, it's important to note that snowball sampling has limitations. The resulting sample may be biased and not representative of the entire population, as participants are selected based on referrals and shared characteristics. The sample may also be influenced by the researcher's initial selection of seeds, as their connections may reflect their own biases or preferences. Therefore, generalizing findings from a snowball sample to the broader population should be done with caution.

Determining when to stop the snowball sampling process and consider the sample size sufficient is a crucial consideration. Researchers often monitor the point at which new participants' characteristics start to repeat or become redundant. This indicates that data saturation has been reached, and further sampling may not yield substantially different insights.

In summary, snowball sampling is a valuable technique for studying hard-to-reach populations or researching socially sensitive topics. It relies on referrals and connections within a social network to gradually expand the sample size. While it has its limitations, snowball sampling can provide unique insights into populations that would otherwise be challenging to study.

Topic 122: Quota Sampling

Quota sampling is a non-probability sampling technique used when conducting large-scale studies where certain groups within the population are significantly larger or smaller compared to other groups. It is employed to ensure that each group is adequately represented in the sample, even though proportional-to-size sampling might result in underrepresentation of smaller groups and overrepresentation of larger groups.

Here's an example to illustrate how quota sampling works. Let's say we want to collect data from the four provinces of Pakistan to conduct a survey on inflation. The population of Balochistan is relatively small, while the population of Punjab is much larger. If we were to take a percentage of the population from each province, the percentage from Punjab would result in a sample size in the thousands or even millions, while the percentage from Balochistan would be in the hundreds.

To address this issue, we can set a fixed quota for each province. For instance, we may decide to include 1% of the population from each province in the sample. However, if the resulting sample size exceeds 1,000, we will cap it at 3,000 to avoid an overly large sample. On the other hand, if the sample size falls below 1,000, we will ensure that it is at least 1,000 to maintain representation of the smaller group.

The goal of quota sampling is to achieve proportional representation of each group while keeping the overall sample size manageable. By setting specific quotas, we can control the underrepresentation of smaller groups and the overrepresentation of larger groups. The resulting sample can be used for generalizing results and controlling sampling error.

In summary, quota sampling is a non-random sampling method used within the context of random sampling. It involves setting fixed quotas to ensure proportional representation of different groups within the population. This technique helps maintain representation while controlling sample size, making it a useful approach for certain research scenarios.

Topic 123: Sampling Frame

In topic123, we discuss the concept of a sampling frame, which is essential when conducting any type of sampling, whether it's random or non-random. Obtaining this information can be done through databases, such as our national database that contains various types of information about individuals. For example, in Pakistan, we conduct censuses where we collect extensive information about people, including age, occupation, location, gender, income, religion, educational qualifications, and more.

When requesting data from the Bureau of Statistics for a specific district in Punjab province, I cannot expect them to provide me with the entire dataset containing all variables. Instead, I specify the variables (sample universe) I need for my sample. These variables are based on the information I require to understand the characteristics of the individuals I want to include in my sample. This subset of variables that I select from the database is referred to as the sampling frame, and it represents the relevant information about the entire population.

The sampling frame is used to retrieve the necessary information from the main database to select the sample. It serves several important functions, five of which are highlighted here.

1. **First**, the sampling frame should be comprehensive, meaning it should include all the relevant variables. **For example**, if I want income information, the sampling frame should contain data on income for all individuals in the population.
2. Secondly, the information in the sampling frame should be up-to-date. Often, databases are not regularly updated, and even when they are, the quality of the data might not be reliable.

Therefore, it is essential to ensure that you have the latest data to select individuals accurately and to draw correct conclusions and analyses.

3. Thirdly, cross-checks should be performed on the data in the sampling frame. If any inconsistencies or distortions are identified, they should be addressed before using the data in the sampling process. This helps maintain the accuracy and integrity of the sample selection.
4. The fourth point is that the sampling frame assists in selecting a sample that is representative of the entire population. By examining variables such as gender, location (urban or rural), and other relevant criteria, you can ensure that the sample reflects the population's characteristics.
5. Finally, the sampling frame helps identify units or individuals who are eligible for inclusion in the sample. It also allows for excluding incomplete or inaccurately verified information to maintain the integrity of the sampling procedure.

By considering these factors, you can create an authentic sampling frame by gathering data on the relevant variables from a larger dataset. The sampling frame will contain information about all subjects in the population and can be used for various sampling procedures, both probabilistic and non-probabilistic, depending on your research requirements. The sampling frame is a crucial factor in selecting a credible sample.

Topic 124: Determining Sample Size

In Module 124, we discuss how to determine the appropriate sample size when creating a sample. This is a commonly asked question, and various formulas are available for calculating sample size. However, it's important to note that understanding the concept behind sample size determination is more crucial than memorizing formulas. In today's era, there are readily available tools such as Excel files and online calculators that can assist with calculating sample sizes based on population parameters.

The two main considerations when selecting a sample are to eliminate bias and ensure representativeness. Eliminating bias can be achieved using methods discussed in previous modules. The key is to understand how sample size affects bias reduction. The second consideration is ensuring that the sample is representative of the population. As a general rule, the larger the population, the smaller the sample size should be as a percentage of the total population.

For example, if the total population is 100, and I want to draw a sample, I may need to take around 50-70% of the population to ensure representativeness. However, if the population is smaller, say 50,

I might need to include all of them to achieve representation. Similarly, if I have census data with a population of 100, drawing a sample of 30-40 individuals can provide representative information. The goal is to select a sufficient number of cases from the population to ensure representativeness.

When the population size increases, the percentage of the population that needs to be included in the sample decreases. **For example**, if the population is 500, taking 50% of the population (250 individuals) can still provide representative data. Even if the population size is 1 million, a sample size of half a percent (5,000 individuals) can be considered substantial.

It's important to remember that the specific sample size cannot be determined by a fixed number. The goal is to ensure representativeness by using common sense, considering the research questions, and the level of sampling required. It is also essential to consider the number of groups or categories in your analysis. Typically, having a minimum of 30 individuals in each group is recommended. Multiply the number of groups by 30 to obtain the total sample size. Then, determine what percentage of the population the sample size represents.

Another consideration is the redundancy of information. Even if a smaller sample size is taken, it can still provide sufficient data to represent the population. For instance, if the population is 2000, taking only 2% of that population (40 individuals) can still yield meaningful results.

In summary, the general rule is that the larger the population, the smaller the percentage of individuals needed in the sample to ensure representativeness. Use common sense, consider the research needs, and keep in mind the number of groups or categories in your analysis when determining the sample size.

Topic 125: Sampling Error

Sampling error is a concept that refers to the discrepancy between the sample statistic (such as the sample mean) and the population parameter (such as the population mean) that it represents. When you draw a sample from a larger population, there is always a chance that the sample will not perfectly represent the population, leading to sampling error.

To illustrate this concept, let's consider an example. Imagine you want to assess the satisfaction of students in your college regarding library services. The population in this case would be all the students who use the library. When you randomly select a sample from this population, let's say 10 or 3 students,

there is a possibility that none of the selected students were included in previous samples. In subsequent samples, there is a chance that some students may overlap with previous samples, but the majority will be different.

Statistically, if you calculate the average opinion of the 3 students in your sample regarding library services, it should be representative of the average opinion of all the students in the population. However, in practice, it is highly unlikely that the sample mean will be exactly the same as the population mean. This is where sampling error comes into play.

Sampling error can arise from various sources, including the inherent variability within the population and the random nature of selecting a sample. It is important to note that sampling error is not a result of mistakes in the sampling process but rather a natural occurrence when dealing with samples.

One way to manage sampling error is by increasing the sample size. As the sample size increases, the chance of sampling error decreases. This is a general principle in psychology and other fields. By drawing larger samples, you increase the likelihood of overlapping individuals between different samples. When this overlap occurs, the means of the samples tend to be closer to each other.

To illustrate this, let's go back to the **example** of drawing samples from the population of students. Instead of three separate samples of three students each, let's decrease the population size to 6 and have each sample contain 6 students. With this setup, there is a higher probability of individuals being repeated across the samples. As you increase the sample size and have multiple groups with overlapping individuals, the chances of their means being closer to each other increase.

To minimize sampling error, it is recommended to increase the sample size. Various books provide figures indicating how many samples you should draw based on the population size. For example, if you draw a sample of 45 individuals from a population of 100,000 or 10,000,000, the sampling error will remain around 0.01% (or 0.001). If you increase the sample size to 3,000, the sampling error becomes approximately 1%. Similarly, with a sample size of 1,500, the sampling error is around 3%.

When we say that the sampling error is 1% or 3%, it means that the difference between the sample mean and the population mean is close to that percentage. The larger the sample size, the smaller the sampling error becomes. For instance, if you increase the sample size to 4,500, the sampling error will be closer to 0.5% (or half a percent) rather than 1%.

Lecture 29: Research Instruments**Topic 126: Research Instruments; Construct, Variable and Data**

When conducting research, you need to create tools for data collection, known as research instruments. These instruments can take various forms, such as questionnaires, observations, interviews, focus group discussions, or checklists. Before discussing research instruments, it's important to differentiate between constructs, variables, and data.

Constructs

Constructs are concepts that researchers aim to measure in their studies. Not all constructs can be directly measured because they are abstract or latent in nature. For example, intelligence, happiness, anxiety, and personality traits are constructs that cannot be directly observed or quantified. Unlike concrete measurements like height or weight, constructs require indirect measures to assess them.

To measure constructs, researchers use direct measures, which are observable and quantifiable variables that are associated with the construct of interest. These variables serve as proxies or indicators of the underlying construct. **For example**, if you want to measure intelligence, you might use variables such as reasoning ability, critical thinking skills, abstract thinking, or problem-solving capabilities as direct measures of intelligence.

It's worth noting that different researchers may define and operationalize constructs differently based on their theoretical perspectives and research goals. For example, one researcher might define intelligence as the ability to adapt and adjust in society, while another might emphasize reasoning ability or problem-solving skills.

In the field of education and social sciences, there are numerous constructs that researchers aim to measure. However, these constructs do not have concrete or direct measurements. Researchers develop research instruments to indirectly assess these constructs using observable variables.

When developing research instruments, it is crucial to carefully define the constructs of interest. This involves examining existing literature, identifying commonalities across different definitions, and selecting appropriate variables that capture the essence of the construct. It is important to note that constructs are not exhaustive, and different researchers may emphasize different aspects of a construct based on their research objectives.

Historically, intelligence was considered a unitary concept, but as research progressed, it was recognized that intelligence encompasses multiple dimensions. For instance, Guilford proposed a model with around 150 different facets of intelligence. This highlights the complexity and multidimensionality of constructs.

In summary, constructs are abstract or latent variables that cannot be directly measured. Researchers develop research instruments to indirectly measure these constructs using observable variables as proxies. The selection of variables should align with the theoretical understanding of the construct and the research objectives. Defining constructs accurately is crucial for developing effective research instruments.

Topic 127: Research Instruments: Variable

In the previous module, we discussed the concept of constructs as hidden and abstract concepts that researchers aim to measure. Constructs are latent variables that encompass the underlying abstract nature of the concept being studied. To measure constructs, researchers define them through direct measures, which are observable variables associated with the construct.

For example, if we consider the construct of intelligence, we might define it by including variables such as problem-solving ability, reasoning skills, critical thinking, and learning ability. These variables serve as indicators of the construct of intelligence.

The key **distinction between constructs** and variables is that constructs are abstract and cannot be directly measured, while variables are concrete and measurable. Variables are the observable elements used to capture and represent the construct. Researchers select variables that are related to the construct and develop indicators to measure them.

To further illustrate this, let's take the example of measuring problem-solving as a sub-construct of intelligence. Problem-solving can be divided into measurable indicators, such as the ability to solve complex problems, analytical thinking, or creativity. These indicators are directly measurable and provide specific aspects of the problem-solving construct.

It's important to understand that variables are directly measurable, whereas constructs are not. Constructs need to be broken down into variables and indicators to make them measurable. This is because researchers have limitations in mapping and recording data for abstract constructs. By

converting constructs into observable variables and indicators, researchers make them measurable and collect data against them.

Variables can have different characteristics and generate different types of data. **For instance**, categorical variables, like gender, religion, or marital status, classify individuals into distinct groups. The assigned numbers to these categories have no numerical meaning but serve as labels for classification.

On the other hand, continuous variables, such as age or height, have meaningful numerical values. Each increment or decrement in these variables carries significance. **For example**, the difference between a person who is five years old and another who is six years old is meaningful. Continuous variables can have decimal values, representing smaller increments or divisions.

Understanding the relationship between constructs, variables, and indicators is essential for building research tools. Constructs define the abstract concepts, variables represent those constructs, and indicators measure the variables directly. By connecting these three components, researchers can collect data to draw conclusions about the constructs being studied.

In the next module, we will explore the connection between variables and data, as the ultimate purpose of research instruments is data collection.

Topic 128: Research Instruments; Data

In the previous module, we discussed the process of identifying indicators for variables and collecting data against those indicators to gain insights into the underlying construct. Now, let's delve into the methods of nature of data, data collection and the importance of ensuring ethical practices throughout the process.

Data

Data collection involves gathering information against the identified indicators of each variable. This data can be obtained through various methods and tools. Some common methods of data collection include conducting surveys, interviews, observations, experiments, and reviewing existing records or documents. Depending on the research context, data can also be collected from databases or other digital sources.

The collected data can take different forms, such as numerical values, codes, or textual information. It is essential to organize and analyze the data to derive meaningful insights. Data analysis involves processing the collected data to uncover patterns, relationships, and trends that provide information about the construct being studied.

It is crucial to develop a well-defined data collection instrument that aligns with the identified indicators. The instrument should be carefully designed to accurately capture the relevant data. Prior to data collection, researchers should ensure that the instrument is reliable, valid, and suitable for the research context.

Ethical considerations play a vital role in data collection. When collecting data from individuals, it is important to secure the data, prevent data leakage, and avoid any misuse of the collected information. Researchers should prioritize the confidentiality and privacy of participants, ensuring that the data is stored securely and only used for legitimate purposes.

Ethical data handling involves obtaining informed consent from participants, providing clear explanations about the purpose of the research, and ensuring that the data collected is anonymized and cannot be linked back to individuals. Researchers must adhere to ethical guidelines and regulations to protect participants' rights and well-being.

Accuracy is of utmost importance throughout the entire process. If the indicators are not accurately defined, the subsequent steps will be compromised, and the data collected may not provide reliable insights into the construct. Researchers must meticulously follow the defined sequence and ensure that the data collected is accurate, valid, and representative of the construct being studied.

By maintaining accuracy and adhering to ethical considerations, researchers can collect robust data that supports meaningful analysis and contributes to the understanding of the target construct.

Lecture 30: Levels of Measurement

Topic 129: Research Instruments, data, variable; Measurement

In this unit, we will explore research instruments used for data collection. A research instrument refers to the method or tool employed to gather data during research, such as interviews, observations, or checklists. Developing effective research instruments is a crucial task that requires careful consideration.

To understand the concept of levels of measurement, it is important to first grasp the different types of variables. Variables can be classified into two broad categories: qualitative variables and quantitative variables. Let's delve into this topic further.

Quantitative variables involve data that naturally exists in numerical form. **Examples** of quantitative data include age, height, and weight. These variables possess inherent numerical values, such as 10 kg, 20 kg, 50 kg, or 60 kg for weight. In quantitative research, analysis can be performed directly on variables that are naturally expressed in numerical form. However, all variables, regardless of their natural form, need to be converted into numerical form for analysis purposes. This allows for consistent and comprehensive analysis of all variables. Quantitative variables can further be classified as **discrete** or **continuous**.

Discrete variables are characterized by whole numbers and do not possess decimal values. **For instance**, the number of television sets in a household can be 1, 2, 3, or 4. It cannot be 1.2 or any decimal value, as it would be nonsensical to have a fraction of a television set. Therefore, discrete variables only have meaningful whole number values, and all decimal values between two whole numbers are irrelevant.

On the other hand, **continuous variables** involve values that can include decimal points and fractions. **For example**, height can be measured in feet, such as 4, 5, or 6 feet. One foot can also be expressed as a decimal value, such as 4.1 feet. Continuous variables encompass an infinite number of values between any two given points. All these values have meaning and contribute to the variable, making them significant for analysis.

Apart from quantitative variables, there are variables in social sciences and other fields that do not naturally exist in numerical form. These **qualitative variables**, such as gender, marital status, or hair color, need to be coded or represented numerically for analysis purposes. When converting qualitative variables into numerical form, it is essential to ensure that the assigned numerical values accurately represent the characteristics of the variable.

By assigning meaningful numerical codes to qualitative variables, researchers can preserve the inherent characteristics and information associated with each variable. **For instance**, if representing educational qualifications numerically, one could assign the value of 1 to matriculation, 2 to FA, 3 to BA, and 4 to

MA. This coding scheme reflects the hierarchical nature of the qualifications, where higher codes correspond to higher levels of education.

When coding qualitative variables, it is crucial to assign numerical values in a manner that faithfully represents the characteristics of the variable. This ensures that the numerical representation maintains the integrity and distinctiveness of the original variable.

In summary, research instruments serve as tools for data collection in research. Variables can be classified as qualitative or quantitative. **Quantitative variables** can be **discrete or continuous**, while **qualitative variables** need to be coded into **numerical** form for analysis, with careful consideration given to accurately representing their characteristics.

Topic 130: Levels of Measurement: Nominal

Levels of measurement play a crucial role in data analysis. When we collect numerical data, it is important to understand the information contained within those numbers. In many cases, a single numerical value does not convey the same information in every context. To better comprehend the levels of measurement, let's explore them one by one.

Consider the **example** of collecting gender information. Let's say we have a question where respondents can choose either "male" or "female," and we assign the code 1 to male and 2 to female. In this case, the numerical values (1 and 2) represent different categories or groups. The value 2 does not indicate that females are older or superior to males; it simply represents a distinct category. These numerical codes are discrete numbers with no inherent order or numerical relationship between them.

Another example is assessing educational qualifications. Suppose we ask respondents about their qualifications and assign the code 1 for matriculation, 2 for FA, 3 for BA, and 4 for MA. Here, the numerical codes represent an ordered ranking of qualifications. The code 2 indicates a higher qualification than 1 (matriculation) but a lower qualification than 3 (BA). In this case, the numerical value carries information about the relative ranking of qualifications.

These examples demonstrate the different levels of measurement: **nominal**, **ordinal**, interval, and ratio. **Nominal is the lowest level**, while ratio is the highest.

Nominal

At the nominal level, numerical codes are used solely for identification or categorization. They do not possess any numerical relationships or meaningful mathematical operations. **For instance**, when coding for religion, assigning the code 1 to Islam, 2 to Christianity, and 3 to Hinduism simply helps identify the religious affiliation of individuals. The numerical codes in nominal measurement are discrete, lack order, and are mutually exclusive.

To summarize, nominal measurement involves using numerical codes solely for identification, without any inherent numerical relationships or mathematical operations. The codes represent categories, are discrete, lack order, and are mutually exclusive.

Topic 131: Levels of Measurement: Ordinal

In Module 131, we discussed the **second level** of measurement, which is **ordinal**. In this module, we are discussing the characteristics of ordinal data. Ordinal data has an inherent order, meaning that assigning numerical codes to the data reflects the order or ranking of the variable being measured. A larger number indicates a higher degree or presence of the variable.

Let's take the **example** of measuring liking for hockey using a Likert scale. The scale ranges from "Strongly Disagree" to "Strongly Agree," with five points in total. Each point on the scale represents a different level of liking for hockey, with a higher number indicating a higher degree of liking. **For example**, if someone selects four, it means they like hockey more than someone who selects three. The order of the numerical values on the scale reflects the order of liking for the variable being measured.

However, one limitation of ordinal data is that the intervals between the values are not necessarily equal. We cannot assume that the difference in liking between one and two is the same as the difference between two and three. The intervals between the values may vary, and we do not have a guarantee of equal spacing. Therefore, we cannot perform mathematical operations, such as multiplication, on ordinal data.

To summarize, ordinal data has an inherent order, and a higher number indicates a higher degree or presence of the variable being measured. However, the intervals between the values are not necessarily equal, and we cannot perform mathematical operations on ordinal data.

Topic 132: Interval

Continuing our discussion on levels of measurement, we now move on to the third level, which is interval. The context of this discussion is that understanding the levels of measurement is crucial for properly analyzing the data we collect. Each level of measurement requires a different approach and statistical tests to analyze the variation in the data.

Interval data has two characteristics that were also present in nominal and ordinal data. The first characteristic is that numerical values are used to recognize and distinguish between different data points. The second characteristic is the order or ranking, where we can identify which values are higher or lower than others.

However, interval data has an additional attribute that distinguishes it from ordinal data. In interval data, the intervals between consecutive numbers are equal. Let's revisit the **example** of liking for hockey on a Likert scale. While ordinal data only tells us the order of liking (e.g., someone who selects five likes hockey more than someone who selects four), it doesn't guarantee equal intervals between the values. We cannot assume that the difference between one and two is the same as the difference between two and three.

To qualify as interval data, the intervals between consecutive numbers must be equal. For instance, when measuring the temperature of an object in Celsius, Fahrenheit, or Kelvin, we can say that the difference between 30 and 31 degrees is the same as the difference between 31 and 32 degrees. If this equal interval property holds for all consecutive numbers, we can classify the data as interval.

The benefit of having interval data is that we can perform arithmetic operations on it. For example, if the intervals between numbers are equal, we can confidently say that six is two times three because the relationships and calculations are consistent. However, if the intervals are not equal, we cannot apply multiplication or other mathematical operations.

To summarize, interval data has the characteristics of nominal and ordinal data, along with an additional attribute of equal intervals between consecutive numbers. This allows us to recognize and order the variable, and perform calculations using arithmetic operations.

Topic 133: Levels of Measurement: Ratio

In Module 133, we discussed the highest level of measurement, which is ratio. Ratio data possesses all four characteristics of numerical data. Let's review the characteristics of ratio data in comparison to the previous level of measurement interval.

Interval data has the properties of recognition, order, and equal intervals between consecutive numbers. However, it lacks an absolute zero point. An example is temperature measured in Celsius or Fahrenheit.

For instance, let's consider the example of having zero rupees in my pocket. When I say I have zero rupees, it means that there are no rupees in my pocket at all. However, not all zeros imply the absence of a characteristic. There are two **types of zeros: absolute zero** and **arbitrary zero**.

Absolute zero is a reference point that cannot be surpassed in a particular context. It serves as the starting point from which measurements cannot go below and represents the absence of the variable being measured. Any other value in relation to absolute zero can be compared and analyzed.

On the other hand, **arbitrary zero** is a reference point that is chosen for convenience or practical purposes. It does not indicate the absence of the variable being measured, and comparisons cannot be made based solely on this zero. Ratios and meaningful analyses relying on ratios can only be calculated when a data set has an absolute zero.

To determine if a data set has an absolute zero, we need to establish if there is a starting point from which measurements cannot go below. If there is no absolute zero, comparisons and ratio calculations become arbitrary. Only data sets with an absolute zero, such as income, height, and weight, can be considered ratio variables.

Ratio data incorporates all the characteristics of previous three scales and adds an absolute zero point. An absolute zero represents the absence of the variable being measured. Examples of ratio data include income, height, weight, and time.

When a data set has an absolute zero point, it is considered a ratio variable. Ratio data allows for meaningful calculations of ratios and proportions. Analytical techniques that rely on ratios can be

applied to ratio data. However, these techniques cannot be used with data that lacks an absolute zero point.

To summarize, the four levels of measurement can be compared based on their characteristics: recognition, order, equal intervals, and absolute zero. Nominal data has recognition, while ordinal data has recognition and order. Interval data possesses recognition, order, and equal intervals. Finally, ratio data incorporates all four characteristics, including an absolute zero point.

Topic 134: Levels of Measurement- Test Your Knowledge

Let's practice identifying the levels of measurement with some examples. Understanding the characteristics of these levels is crucial for accurate data analysis. Without correctly identifying the levels, you won't be able to appropriately interpret and analyze your data. Here are the examples:

We'll start with the first example, where the professor wants to examine the relationship between class attendance (number of absentees) and grades. The data set includes two variables: the number of absent days and the corresponding grades.

To determine the level of measurement, we'll follow the principle of starting from the highest level and working downwards. We'll first check if the data has an absolute zero, which indicates ratio data. If it has an absolute zero, we don't need to consider other levels. So, the first step is to check if the number of absent days has an absolute zero.

If a student was 100% present, it means their number of absent days is zero, indicating an absolute zero. Therefore, the number of absent days can be categorized as ratio data. This data exhibits the characteristics of ratio measurement.

Now, let's analyze the grades. If a grade of "F" is given when a student didn't take the subject at all, it indicates an absolute zero. However, if there are no absolute zeros, we need to consider the next level.

Since the grades do not have an absolute zero, we move on to the interval level. For it to be interval data, the differences between consecutive values should be equal. If the differences between grades (e.g., A to B, B to C) are not equal, we can conclude that the data is not interval.

If the differences between grades are not equal, the next level to consider is ordinal. If the grades can be ranked in a specific order (e.g., $A > B > C$), then the data can be categorized as ordinal.

If we cannot determine an order among the grades (e.g., we are unsure whether $A > B$ or $B > C$), we move to the lowest level, which is nominal. Nominal data does not have an inherent order, and the categories are simply labels.

Based on this analysis, the grades can be considered as ordinal data. Therefore, the data set contains ratio (number of absent days) and ordinal (grades) measurements.

Now, let's move on to the next example, which involves room temperature and the number of cars produced. We need to determine the level of measurement for these two variables.

If the room temperature has an absolute zero (e.g., 0 degrees Fahrenheit), it would indicate ratio data. However, since there is no absolute zero in this case, we move on to the next level.

To determine if the data is interval, we check if the differences between consecutive temperature values are equal. If they are not equal, we cannot categorize the data as interval.

Since the data is neither ratio nor interval, we consider the next level, which is ordinal. If there is an order or ranking among the temperature values, it can be considered ordinal data. However, temperature values do not have a natural order, so they cannot be treated as ordinal.

If the data does not fit into the ordinal category, we move to the lowest level, nominal. Nominal data represents categories without any inherent order.

Based on this analysis, the room temperature data can be considered as nominal. On the other hand, the number of cars produced can be categorized as ratio data since it has an absolute zero (i.e., 0 cars produced).

Lastly, let's analyze the data set of educational levels, which includes different levels such as junior high school, senior high school, undergraduate, master's, and doctoral.

Since these educational levels have a natural order (e.g., junior high school < senior high school < undergraduate < master's < doctoral), we can categorize them as ordinal data. The levels can be ranked, but the intervals between them are not necessarily equal.

Therefore, the data set of educational levels represents ordinal measurement.

In summary:

Example 1: Contains ratio (number of absent days) and ordinal (grades) measurements.

Example 2: Contains nominal (room temperature) and ratio (number of cars produced) measurements.

Example 3: Represents ordinal measurement.

Lecture 31: Research Tools and tests

Topic 135: Research Tools and tests

Now, let's delve into the topic of research tools and explore its various aspects. Our first sub-topic will focus on the different types of research tools. Research tools are familiar instruments that come to mind when engaging in research. Novice researchers often associate data collection tools, such as surveys and interviews, with the research process. These tools aid in collecting data, which is then analyzed to draw conclusions and generate results. To ensure accuracy, it is crucial to develop and use appropriate research tools.

When creating a research tool, it is essential to define it correctly. If the tool is not properly defined, the questions formulated from it will likely be flawed, leading to inaccurate data. As a result, subsequent analysis and steps in the research process may be compromised. Thus, the accuracy of the research tool is fundamental to the overall accuracy of the research process.

What are research tools?

'Research instruments are used to collect data in systematic and standardized manner during a research study. The research instruments are selected in accordance with the nature of research objectives, the nature of the study and the data needed.'

To ensure accurate data collection, the research tool must be standardized. This means establishing a consistent procedure for data collection across all subjects, so that the researcher's approach does not influence the data collected. There are various types of tools available for data collection in research, such as questionnaires, interviews, observations, and more. The selection of tools should align with the objectives of the study and be based on the nature of the research.

On the slide in front of you, it is mentioned that the selection of a research tool should be driven by your data requirements and should align with the objectives of the study. Depending on the nature of

your research, you may need to choose one or multiple tools. It is worth noting that the examples provided on the slide are commonly used in educational research, particularly at the undergraduate level. However, these five types of tools are not exhaustive, as there are many more available.

A research tool can be any instrument that generates data. The examples provided earlier serve to illustrate the concept of a research tool. **For instance**, structured statements accompanied by scales for response options are commonly used in online forms or email surveys. Interviews allow for more detailed, unstructured information gathering, enabling researchers to ask follow-up questions. Observations are frequently employed in educational research, where researchers observe and record information in a classroom or other relevant setting. Additionally, assessments and tests are used to evaluate student achievement, and psychological scales measure constructs such as attitudes and perceptions.

While these five types of tools are commonly used in undergraduate research, it is important to acknowledge that other tools exist as well. It is crucial to select appropriate tools to ensure relevant and accurate data generation.

Topic 136: Research Tools and tests, Classification of Research Tools

In this module, we will continue our discussion on the classification of research tools, specifically in the context of educational research. These tools are often categorized into different groups for convenience, but it is important to note that there is no universally standard classification scheme. Different individuals have developed their own groupings based on their specific purposes.

To help us understand and organize these tools, let's draw a parallel with the Domains of Learning, specifically Bloom's Taxonomy.

- Cognitive tests
- Affective tests
- Psychomotor tests
- Attitude tests
- Personality tests
- Projective tests

Just as we have **cognitive**, **affective**, and **psychomotor** domains in learning, we can categorize research tools accordingly. Cognitive tests are used to measure the cognitive abilities of students or learners in various subjects. Affective tests, on the other hand, aim to identify people's attitudes towards different aspects of life. Lastly, psychomotor tests assess body movement and physical skills in research settings.

It is important to emphasize that this classification into three broad categories is not the only way to categorize research tools. Different people may have different classifications based on their own perspectives and purposes. When classifying tools, it is crucial to have a meaningful understanding of the classification system being used.

Let's delve deeper into each category. In this module, we will focus on cognitive tests.

Cognitive tests

Cognition refers to the basic mental processes involved in acquiring and using knowledge. It encompasses various aspects such as language acquisition, mental skills, data storage, information processing, and more. If your research is centered around these mental processes, cognitive tests become relevant tools for data collection.

Cognitive tests are used to assess intellectual functioning, identify strengths and weaknesses, and make informed decisions for improvement. They are a broad category of data collection tools that should be considered when studying mental processes. Furthermore, within cognitive tests, there are further classifications based on specific aspects such as data acquisition, data storage, and information processing. Each classification corresponds to different types of tests used in research.

I have provided some examples of cognitive tests that are commonly used. These include intelligence quotient (IQ) tests, memory tests, tests for assessing attention or concentration, and tests for analyzing and organizing data. These examples serve as a starting point to familiarize yourself with cognitive tools. It is important to note that there are many more tests available, and you may come across them in your research.

Topic 137: Research Tools and tests, Classification of Research Tools, Affective Tools

In this module, we will focus on the second classification of research tools, which are affective tests. In the affective domain of learning, we explore the development of emotions, feelings, attitudes,

moods, anxiety, and other related aspects. Affective tests are designed to measure these elements, quantifying emotions, assessing feelings, and evaluating attitudes. They are commonly used by researchers in educational psychology or sociology who study these aspects or a combination of them. These tests provide data that helps us gather information and draw inferences supporting the development of the affective domain.

I have provided a few examples of affective tests here, but it is important to note that these are not the only tests available. There are other classifications and categories as well. However, the examples I mentioned are commonly used in educational research within these categories. **The measurement of attitudes**, for instance, is a broad area of study in educational research, and there is extensive research focused on understanding and measuring attitudes. Scales have been developed for this purpose.

Another significant concept in educational research is **emotional intelligence**. While IQ remains important, there is growing recognition that a person's ability to adapt to society and effectively handle emotions is crucial for learning. Emotional intelligence encompasses the ability to perceive and understand others' emotions and manage one's own emotions. Research in this area aims to examine the development of individuals from this perspective.

Anxiety, specifically test anxiety, is another important area of study in education. Test anxiety refers to the impact of anxiety on students' performance and behavior in test-taking situations. Numerous well-constructed and standardized tests exist to measure different types of anxiety. If your research focuses on anxiety-related topics, these tools can be valuable resources.

Additionally, **empathy scales** are used to assess empathy, which refers to the ability to understand and share others' emotions. Evaluating empathy can provide insights into how well individuals comprehend and respond to the emotions of others.

In summary, affective tests are tools used to measure the affective development of individuals, including emotions, attitudes, and anxiety. They play a significant role in educational research, and the examples provided serve as a starting point for familiarizing yourself with these types of tests.

Lecture 32: Research Tools and tests-II

Topic 138: Research Tools and tests, Classification of Research Tools, Personality Tests

In this module, we will discuss another broad classification of research tools commonly used in educational research, known as personality tests. Personality tests are designed to measure various characteristics and traits associated with an individual's personality. They provide insights into an individual's organizational skills, behavior towards others, and the influence their personality has on those around them. Many educational metrics are connected to these aspects, such as the impact of teachers' personalities on their performance and their role in students' character development.

There are numerous personality tests available, and researchers can choose to use these tools to understand and study different aspects of personality. These tests help in identifying individuals' strengths, weaknesses, thinking patterns, emotional responses, and behavioral tendencies in various situations. When conducting research focused on these areas, personality tests can be valuable tools for data collection and analysis.

Personality inventories and tests have a long history of research, and significant work has been done in this field. Different inventories and tests have been developed, each approaching personality from different angles.

- **Myers-Briggs** Type Indicator (MBTI)
- Sixteen Personality Questionnaire
- Hogan Personality Inventory
- Rorschach inkpot test
- Thematic Appreciation Tests

One commonly used inventory is the **Myers-Briggs** Type Indicator (MBTI), which categorizes personality into different dimensions. For example, it assesses the preference for extroversion versus introversion, sensing versus intuition, thinking versus feeling, and judging versus perceiving. By placing individuals along these dimensions, the MBTI identifies 16 different personality types.

Another widely used tool is the **Personality Factor Questionnaire**, which examines factors like emotional stability, dominance, and sensitivity. By comparing individuals on these factors, researchers can classify personality types or assess the personality of different groups.

It is important to note that the examples provided, such as the MBTI and Personality Factor Questionnaire, are just a few among many personality tests available. Researchers have drawn inspiration from existing literature on personality and developed various tools to measure and assess personality traits. These tests utilize different questionnaires and data analysis techniques to identify individuals' personality based on the collected data.

In summary, personality tests are research tools used to measure and assess various aspects of an individual's personality. They are valuable in educational research as they provide insights into characteristics, behavior, and influence. The examples mentioned in this module represent only a fraction of the available tests, and researchers can explore other tests that fall within this category to measure personality traits.

Topic 139: Research Tools and tests, Classification of Research Tools, Psychomotor Tests

In this module, we will discuss the third classification of research tools mentioned earlier, which are psychomotor tests. Psychomotor development refers to the development of physical and motor skills, with a particular focus on coordination and movement. Research in this area often emphasizes early age development, as psychomotor skills tend to progress rapidly during this time. Psychomotor tests are used to measure an individual's ability to perform tasks requiring physical and motor skills and assess their proficiency in these areas.

- Eye- Hand coordination tests
- Reflection time tests
- Trail making tests
- Handwriting tests
- Driving simulation

These tests aim to understand how different parts of the body work together in coordination, how individuals coordinate movements, and various aspects related to physical and motor skills. Examples of **psychomotor tests** include measuring motor coordination, such as assessing an individual's ability to write by evaluating the coordination of finger movements. These tests are commonly used in studies that involve practical tasks or activities.

If you are conducting research that focuses on the relationship between body movements and physical coordination, psychomotor tests can be valuable tools. They can help you assess an individual's ability to respond to stimuli and convert them into actions efficiently. **For instance**, if you are examining an individual's ability to play football and measure their coordination in passing the ball accurately, psychomotor tests can provide insights into their performance in the psychomotor domain.

Psychomotor tests also play a role in examining the development of specific skills through practice and training. **For example**, if you are interested in improving handwriting or other fine motor skills, there are tests available to assess an individual's proficiency in these areas. These tests help researchers and educators working in language development or handwriting improvement to conduct research and assess progress.

It is important to note that the examples mentioned in this module represent only a few examples of psychomotor tests. There are many different types of inventories and tests available for various aspects of psychomotor development. Researchers also have the option to create their own assessments or tests tailored to their specific research objectives.

Psychomotor tests are particularly relevant in the context of children's development, and many research studies benefit from incorporating these tests. Developing such tests can be time-consuming and labor-intensive, so utilizing existing tools that align with your research objectives can save valuable resources.

In summary, psychomotor tests are research tools used to measure and assess an individual's physical and motor skills, with a focus on coordination and movement. They are valuable in studies related to body movements, physical coordination, and skill development. Researchers can choose from a variety of inventories and tests or create their own assessments in this area to support their research goals.

Topic 140: Projective Tests

Module 140 further explores research tools, and another classification of tests commonly found in the literature is projective tests. These tests are used to explore individuals' unconscious thoughts, feelings, and motivations, which may not be readily accessible through their conscious awareness. Projective tests involve presenting abstract stimuli, such as images, words, or incomplete sentences, that allow for multiple interpretations. The individual's task is to provide responses or complete the stimuli based on their subconscious associations and interpretations.

Types of projective tests

- Sentence Completion Test
- Draw a person test (DAP)
- Children's Apperception test (CAT)
- Word association test

One example of a projective test is the **Sentence Completion** Test. In this test, an incomplete sentence is provided, and the individual is asked to complete it. **For instance**, a blank might be presented after the statement "After seeing a person lying injured in a road accident..." The individual's response can provide insights into their hidden emotions or thoughts related to the scenario.

Another example is the Thematic Apperception Test (TAT). In this test, the individual is shown an ambiguous image and asked to create a story or describe what they see in the picture. By analyzing the content of the story or description, researchers can gain understanding of the individual's subconscious thoughts and emotions.

Drawing tests are also commonly used in projective assessments. Individuals may be asked to draw specific objects or people, such as drawing a picture of their teacher. The dominant features or themes in the drawings can reveal subconscious elements that are present in the individual's mind.

Another type of projective test is word association test which involves presenting a series of words and asking individuals to connect them in different ways. It is important that the words have multiple possible connections to allow for meaningful interpretations. By examining how individuals connect the words and the associations they form, researchers can gain insight into their underlying psychological processes and the way they create new meanings. These tests provide awareness about the subconscious and can help understand people's feelings, emotions, attitudes, and beliefs that may be hidden in their subconscious minds.

The projective tests mentioned here are just a few examples, and there are various other tests available. The key aspect of projective tests is that they provide open-ended stimuli and allow for multiple interpretations. These tests aim to access individuals' hidden emotions, feelings, attitudes, and beliefs that may not be easily expressed or consciously accessible.

In educational research, projective tests can be particularly useful when studying introverted children who may have difficulty expressing themselves in the classroom. These tests can provide insights into their thoughts, feelings, and attitudes that might not be apparent through their verbal or written communication.

It is important to note that the four categories mentioned in this module (sentence completion tests, thematic apperception tests, drawing tests, and word association tests) are just a few examples of projective tests. There are many other variations and types of projective tests that researchers can explore.

Lecture 33: Research Tools and tests-III

Topic 141: Scales of Measurement

In the previous modules, we discussed various research tools and their classifications to understand their scope. These tools help us choose or create new tools related to their respective categories. When using any research tool, it typically consists of two important parts. The first part involves writing statements or questions, to which individuals provide their opinions. The second part involves a scale on which individuals can rate or express their opinions, which can have multiple options.

For example, let's consider the variable of liking cricket. To measure this variable, we can create statements or questions related to different aspects of cricket such as batting, fielding, bowling, cricketers, and cricket grounds. Each of these aspects represents variables within the overall liking of cricket. We can then provide a scale for individuals to indicate their level of liking towards cricket or its specific aspects. The scale can be in the form of a yes/no response, a Likert scale with multiple options, or any other appropriate scale.

It's important to understand that individuals' liking towards cricket is an inherent quality that exists within them. It doesn't change based on the scale used for measurement. The scale simply helps us quantify and differentiate the varying degrees of liking. The choice of scale depends on the level of detail we want to capture. We can use a **dichotomous scale** with two options (e.g., yes/no), a three-point scale, a five-point scale, a ten-point scale, or even a twenty-point scale. The more divisions or options we have, the more detailed information we can gather about individuals' opinions.

When selecting the number of divisions on the scale, we need to strike a balance. We should have a reasonable number of divisions that allow us to differentiate between different levels of liking without

making it too difficult to distinguish between adjacent points on the scale. It's important to choose a scale that is practical and meaningful for the research context.

There are various types of scales that can be used for variable measurement, including the **Dichotomous Scale**, **Semantic Differential Scale**, **Guttman Scale**, **Thurstone Scale**, and Likert Scale, among others. The ones mentioned here are commonly used in educational research at the undergraduate level. It's advisable to explore these scales in more detail to understand their specific characteristics and applications.

Remember, regardless of the type of scale used, the fundamental idea is to measure a variable by dividing it into different parts or categories to gather detailed information. The scale helps us capture individuals' opinions, attitudes, or preferences on a continuum. Choose a scale that allows meaningful differentiation without making it too complex to interpret the results.

Topic 142: Scales of Measurement: Examples of Guttman Scales

In the previous module, we explored the concept of scales and I provided you with three examples of commonly used scales in educational research. I will now elaborate on these scales and their details. Additionally, there are other scales described in the literature that you may find useful. It is important to refine the language and ensure clarity when discussing these scales. Let's proceed with further understanding and exploration of these scales.

1. **Likert scale:** The most commonly used Likert scale is a psychometric scale that measures individuals' attitudes, opinions, or preferences towards a particular statement or construct. It consists of a series of statements or items to which respondents indicate their level of agreement or disagreement using a predefined set of response options.

Typically, the Likert scale employs a five-point response format, ranging from "Strongly Disagree" to "Strongly Agree." The response options are often assigned numerical values, such as 1 to 5, with 1 representing the lowest level of agreement or disagreement and 5 representing the highest. This allows

for the quantification of respondents' responses, enabling researchers to calculate a total score by summing the numerical values assigned to each response.

Here's an example of how you can use the Likert scale to measure liking towards cricket:

Statement: I enjoy watching a cricket match.

Response options:

Strongly Disagree

Disagree

Neutral

Agree

Strongly Agree

Participants would select the response option that best reflects their level of agreement with the statement. You can create multiple statements related to liking cricket and use the Likert scale to measure respondents' opinions on each statement. In this scale, five points are used, and typically the lowest number is assigned to "Strongly Disagree," while the highest number is assigned to "Strongly Agree." This ordering is based on the understanding that if the attribute being measured is less present in an individual, their assigned number on the scale should also be lower. This ensures that higher scores on the total scale indicate a higher level of the attribute being measured. For example, assigning a numerical value of 1 to "Agree" and 5 to "Strongly Agree" follows this pattern

2. Semantic Differential Scale: Regarding the Semantic Differential Scale, it is another type of rating scale used to measure the meaning of concepts or objects. It involves bipolar adjectives placed at

opposite ends of a continuum, and respondents indicate their perception or evaluation by marking a point on the scale between the two adjectives. This scale captures the connotative meaning or interpretation of a concept or object.

For example, to measure the perception of helpfulness towards others, you might use a Semantic Differential Scale with the bipolar adjectives "Never" and "Always," and respondents would mark their position on the scale to indicate their level of agreement or disagreement with the statement "Others should be helped."

3. **Guttman Scale** (cumulative: scale) The Guttman Scale, also known as cumulative or scalogram analysis, is a scaling technique used to assess the intensity or degree of agreement with a series of statements. The statements are arranged in a hierarchical order of increasing intensity or difficulty. Respondents indicate their agreement or disagreement with each statement, and their responses are then analyzed to determine the level or extent of a particular trait or construct.

For instance, in assessing financial literacy, you might create a Guttman Scale with statements of increasing difficulty or complexity, such as "I can calculate my monthly expenses accurately," "I know how to create and follow a budget," and "I can analyze financial statements and make different investment options." Respondents who agree with the more difficult statements would also be assumed to agree with the less difficult statements, reflecting a higher level of financial literacy.

143: Concept of Reliability and Validity

In this module, we will discuss research tools, which are integral to the research process. These tools may include observations, interviews, or other instruments used to collect data from individuals. We will explore the important features of research tools and understand how these features contribute to the accuracy of the collected data. Specifically, we will focus on two important characteristics: **reliability** and **validity**.

To explain these concepts, let's consider an **example of target boards**. In the example, there are three target boards displayed. Each board consists of a black spot in the center, representing the target, and circles around it. Participants, acting as shooters in a shooting game, aim at the target and mark their shots on the board.

Now, imagine you need to hire a shooter for a specific task. The shooter who consistently hits the target or places their shots very close to the target would be the most trusted and reliable. On the contrary, a shooter who consistently misses the target or randomly places shots in different areas would be the least trusted and reliable.

Reliability refers to consistency. Even if a shooter does not hit the actual target, as long as they consistently hit the same spot repeatedly, they can be considered reliable. It means that their aim is consistent, although not accurate according to the desired target location.

Validity, on the other hand, refers to hitting the actual target or being on the intended target consistently. **In our example**, the shooter who places their shots on or very close to the black spot in the center exhibits both reliability and validity. Their shots are both consistent and on target.

Now, let's consider research tools. It is possible for a research tool to be reliable but not valid. This means that the tool consistently measures something, but it may not accurately measure the intended construct or target.

It is important to remember that the example of the target boards illustrates the concepts of reliability and validity in a simplified manner. In research, these concepts are more complex and require rigorous evaluation and evidence to establish.

Lecture 34: Research Tools and Tests: Validity

Topic 144: Validity of Research Instruments

In this module, we will further explore the concept of validity and discuss different types of evidence that can be used to assess the validity of a research instrument. Validity refers to whether a test or instrument measures what it is intended to measure. This is a crucial concept in research.

For example, if you create a tool to measure intelligence, the questions or tasks within that tool should accurately assess intelligence. Similarly, if you design a test to evaluate writing skills, the

questions should effectively measure writing skills. It's important that the tool aligns with its intended purpose.

Now, let's consider the evidence of validity. There are various types of evidence that can support the claim that a research instrument is valid. I will mention a few of them here:

- **Content validity**
- **Construct validity**
- **Criterion validity**
 1. **Concurrent validity**
 2. **Predictive validity**

While there are additional types of validity evidence, experts in research generally agree that a valid research instrument should have at least three types of evidence: **content validity**, **construct validity**, and **criterion validity**. These three are considered essential for establishing validity.

It's important to note that there are other types of validity evidence, such as face validity, convergent validity, divergent validity, ecological validity, internal validity, and external validity. These concepts provide further insights into the validity of a research instrument, but for the purpose of this course, we will focus on the three essential elements mentioned earlier.

When developing a research tool, it is crucial to ensure that it possesses these three characteristics of validity. By incorporating these types of evidence, researchers can have confidence that their instrument accurately measures the intended construct. If you are interested in learning more, you can explore these validity evidences in greater detail.

Topic 145: Validity of Research Instruments: Content Validity

Let's discuss the concept of content validity, which is an important aspect of ensuring the validity of a research tool. Content validity is concerned with whether the items or questions in the tool adequately represent the construct being measured.

To illustrate this concept, **let's consider an example** where we want to develop a tool to measure teaching effectiveness. We start by conducting a literature review to identify the different aspects or content areas that contribute to teaching effectiveness. In this example, let's say we identify four

aspects: **classroom management skills, lesson planning skills, student engagement, and communication skills.** These aspects are not fixed and can vary depending on the literature and the specific research context.

Once we have identified these aspects, our next step is to define each of them. **For example,** we might define student engagement as the methods and strategies used by teachers to involve students in the learning process within the classroom. By defining the scope of each aspect, we establish the content areas that need to be covered by the questions in our tool.

The goal is to convert these content areas into well-crafted questions that accurately capture the intended content. We want to ensure that the questions cover the defined aspects of teaching effectiveness. By doing so, we can confidently claim that our tool has content validity.

During the question development process, it is crucial to seek feedback from experts in the field. We can present the questions to these experts and ask for their evaluation and input. Their expertise can help identify any items that may need revision or removal. By incorporating their opinions, we can align our tool with the content identified from the literature, creating a synergy between the two sources of information.

After developing the initial version of the tool, it is important to analyze the data collected using the tool. This analysis helps determine if the selected items effectively capture and cover the defined content areas. If adjustments are needed, we can repeat the process of refining the items until we are satisfied that the selected items align with the identified content areas and adequately represent the construct of teaching effectiveness.

By following this process and ensuring that the questions cover the content areas identified for the construct, we can achieve content validity. The main focus should not only be on writing questions but on defining the construct and its content areas. The questions should be designed to accurately measure the identified content. By doing so, we increase the likelihood that our tool has content validity.

Topic 146: Validity of Research Instruments: Construct Validity

Construct validity is an essential aspect of research tool development, particularly in social sciences and education, where abstract concepts or constructs are often measured.

Constructs, such as motivation, intelligence, or teaching skills, do not have a physical existence and require indirect measurement through observable indicators or sub-constructs.

To ensure construct validity during tool development, follow these steps:

Define the construct: Begin by reviewing relevant literature and theories to understand different definitions of the construct. Select a definition that aligns with the research context and purpose.

Identify sub-constructs: Break down the main construct into its component parts or sub-constructs. Define each sub-construct based on established theories and empirical evidence.

Develop items or questions: Create items that capture the essence of each sub-construct. The items should be relevant, meaningful, and aligned with the defined sub-constructs. Consider using established measurement scales or adapting existing tools whenever possible.

Pilot testing: Conduct a pilot study to test the initial version of the tool. Evaluate the clarity, comprehensibility, and relevance of the items. Collect feedback from participants and make necessary refinements to improve the tool's quality.

Assess convergent and discriminant validity: To establish convergent validity, compare the scores obtained from your tool with those from an established or widely accepted tool measuring the same construct. A high correlation between the scores indicates that your tool measures the intended construct effectively.

To establish discriminant validity, compare the scores obtained from your tool with scores from a different construct or unrelated measure. If there is a weak or no relationship between the scores, it demonstrates that your tool can differentiate between the intended construct and other constructs.

By following these steps and collecting evidence from convergent and discriminant validity analyses, you can build a strong argument for the construct validity of your research tool.

Remember to continuously refine and improve the tool based on feedback, and consider seeking expert opinions during the development process to enhance the tool's quality and validity.

Topic 147: Criterion Validity

Criterion-related validity, also known as criterion validity, is an important aspect of evaluating the validity of a measurement tool. It assesses whether the tool is capable of measuring what it intends to measure by comparing it to an external criterion. **Let's consider intelligence as an example.** Suppose you have information from another source indicating that Person A and Person B have different levels of intelligence. To establish criterion-related validity for your tool, it should also be able to differentiate between these individuals based on the criterion you've set for intelligence. The sub-constructs you defined within your tool serve as the standard for comparison.

If your tool can effectively distinguish between Person A and Person B, accurately reflecting the difference in their intelligence as indicated by the external criterion, then we can say that your tool has criterion-related validity. In other words, your tool goes beyond merely differentiating between these specific individuals and demonstrates the ability to measure the construct of intelligence more broadly. By successfully capturing and measuring the observed difference in intelligence, your tool meets the criterion for validity.

The criterion-related validity of a tool is established when it can effectively identify and measure differences based on the specific criteria or standards set for the construct being assessed. It allows the tool to provide meaningful information about individuals' intelligence or other characteristics, enabling informed decision-making based on the measured differences.

There are two types of **criterion-related validity: concurrent validity** and **predictive validity**.

Concurrent validity: Concurrent validity determines whether a tool can differentiate between individuals who possess different levels of the construct being measured. It involves comparing the scores obtained from your tool with scores from an established or widely accepted tool that measures the same construct. The comparison is done simultaneously, meaning both tools are administered to the same individuals at the same time. By examining the correlation between the scores of the two tools, you can determine whether your tool is concurrently valid.

Predictive validity: Predictive validity is concerned with the tool's ability to predict future performance or behavior based on the measured construct. It assesses whether the scores obtained from the tool can predict an individual's performance in a future event or criterion. **For example,** if you are developing an admission test for an M.Phil. program, you might include items that measure critical thinking, problem-solving, and creativity. By administering this test to applicants and

subsequently tracking their performance in the M.Phil. program, you can determine if the test scores accurately predict their success in the program. If there is a significant relationship between the test scores and future performance, the tool demonstrates predictive validity.

In summary, criterion-related validity involves comparing the scores obtained from your tool with an external criterion. Concurrent validity examines the tool's ability to differentiate between individuals at the same time, while predictive validity determines its capacity to predict future performance or behavior. Both forms of criterion-related validity provide evidence of the tool's effectiveness in measuring the intended construct and its practical value in various contexts.

Lecture 35: Research Tools and Tests: Reliability

Topic 148: Reliability of Research Tools

The second characteristic of a good test is reliability. Reliability refers to the consistency and stability of the measurements produced by a test. If a test is reliable, it means that it provides consistent results each time it is administered to the same individual or group of individuals.

To illustrate this concept, **let's consider the example** of a research questionnaire that includes a question about the year Pakistan was created. Suppose a person initially responds with the incorrect answer "1847." If this person consistently provides the same incorrect answer upon subsequent administrations of the questionnaire, it suggests that the test is producing reliable results. Although the answer is wrong, the consistency in their responses implies that the test is accurately capturing their consistent understanding or misconception.

Reliability is about obtaining the same or similar answers consistently over multiple administrations of the test. When a test consistently generates similar responses from a particular group of individuals, we can conclude that the test is reliable. One aspect of reliability is stability, which refers to the consistency of responses over time.

Internal consistency is another aspect of reliability. It focuses on whether the items within a test measure the same aspect of the construct being assessed. **For example**, if a test aims to measure intelligence and includes items related to problem-solving, communication skills, and artistic ability, internal consistency ensures that all the items are effectively measuring different aspects of intelligence.

When the items of a test measure different aspects of the same construct, it is considered internally consistent.

There are two primary **types of evidence for reliability**: **stability** and **internal consistency**. Stability refers to the consistency of responses over time, while internal consistency examines whether the items within a test measure different aspect of the same construct.

Reliability values typically range between +1 and -1. A reliability value of +1 indicates 100% reliability, while 0 indicates no reliability at all. The closer the reliability value is to +1, the more reliable the test is considered.

To assess reliability, various methods can be used, as mentioned in the table in front of you. These methods provide different approaches for measuring reliability, and each has its own procedure for evaluation.

Topic 149: Re Test Reliability

Test-retest reliability is a method used to assess the consistency or stability of a test over time. It involves administering the same test to a group of individuals on two separate occasions and examining the correlation between their scores from the two administrations. The time interval between the two administrations can vary depending on the nature of the construct being measured and the purpose of the test.

To illustrate test-retest reliability, let's consider an example. Suppose we have a questionnaire that measures people's attitudes towards cricket. We administer the questionnaire to a group of individuals and record their responses. After a certain period, such as a week, we administer the same questionnaire to the same group of individuals again and record their responses once more.

To assess test-retest reliability, we compare the responses given by each individual in the first administration with their responses in the second administration. If the individuals provide consistent responses on both occasions, it indicates high test-retest reliability. **For example**, if someone rates their liking for cricket as a 4 on a scale of 1 to 5 in the first administration and rates it as a 4 again in the second administration, it suggests that their responses are consistent over time.

The key idea behind test-retest reliability is to determine whether the scores obtained from the test remain stable when the same individuals are tested again. If the scores are consistent, it suggests that the test is reliable in measuring the construct of interest.

Let's consider a group of six students who took a test for the first time. One student scored 78 on the test. Now, let's focus on Student 2. When we compare the scores of Student 2 in the first and second administrations, we notice that although the score in the second administration is 91 (not 93 as mentioned earlier), the rank remains the same. This means that Student 2 maintains their position in the group relative to others.

To assess test-retest reliability, we can assign ranks to the students based on their scores in the first administration. Then, after a certain period, we administer the same test to the students again and assign ranks based on their scores in the second administration. In this case, two of the students had their ranks changed, which we can highlight by marking them in red. The more consistent the rank order is between the two administrations, the stronger the test-retest reliability.

The correlation coefficient is commonly used to quantify the relationship between the scores from the first and second administrations. The correlation coefficient can range from -1 to +1. A value of +1 indicates a perfect positive relationship, meaning that the scores are identical between the two administrations. A value of -1 indicates a perfect negative relationship, meaning that the scores are completely reversed between the two administrations. A correlation coefficient of 0 indicates no relationship between the scores.

In general, a higher correlation coefficient indicates greater test-retest reliability. However, it's important to note that test-retest reliability can be influenced by factors such as the stability of the construct being measured and the time interval between administrations. For constructs that are expected to change over time, such as test anxiety, shorter time intervals between administrations may be more appropriate. For constructs that are relatively stable, such as attitudes, longer time intervals can be used.

To minimize the potential effect of the exposure effect, which refers to the impact of previous exposure to the test on subsequent administrations, it is recommended to increase the time interval between the two test administrations. This helps reduce the influence of memory and potential practice effects.

Topic 150: Parallel Reliability

Alternate or parallel forms reliability is a method used to assess the consistency or stability of a test by creating multiple versions of the same test. The purpose is to ensure that the different versions of the test yield similar results, indicating that the test is reliable and measures the same construct across different forms.

To create alternate forms of a test, you can make changes to the wording of the questions or alter the order of the questions while maintaining the same content and difficulty level. For example, instead of using the statement "I like playing cricket," you can rephrase it as "I enjoy playing cricket." Additionally, you can change the order of questions between the forms to further reduce potential order effects. These changes are made to minimize the potential impact of memory or practice effects that may occur when individuals are exposed to the same test multiple times.

To assess alternate forms reliability, you administer two different versions of the test to a group of individuals. Let's refer to them as Form A and Form B. After completing both forms, you calculate the average scores for each individual on both forms. Then, you examine the correlation between the scores on Form A and Form B.

If there is a high correlation between the scores on the two forms, it indicates strong alternate forms reliability. This means that both forms of the test are consistent in measuring the construct of interest. The high correlation suggests that individuals who perform well on one form are likely to perform well on the other form, and vice versa.

It's important to note that when designing alternate forms, the goal is to maintain the equivalence of the test content and difficulty, even if the wording or order of questions is changed. This ensures that the construct being measured remains the same across the different forms.

By assessing alternate forms reliability, you can control for chance variability and estimate the stability of the test scores. If the correlation between the scores on the two forms is high, it provides evidence that the test is reliable and consistent in measuring the construct.

However, it's worth mentioning that claiming true parallelism between the forms may not be accurate if the changes in wording or question order significantly affect the way individuals respond.

Nonetheless, when there is a significant correlation between the responses on the different forms, it indicates stability in the measurement of the construct.

In summary, alternate or parallel forms reliability is determined by administering different versions of the same test and examining the correlation between the scores obtained. A high correlation suggests that the test forms are consistent in measuring the construct of interest, providing evidence of reliability and stability in the test scores.

Topic151: Split Half Reliability

In the previous discussion, we mentioned two measures of reliability: test-retest reliability and parallel forms reliability (also known as alternate forms reliability). These measures assess the stability of a test over time and the consistency between different versions of the test, respectively.

Now, let's focus on internal consistency, which is another aspect of reliability. Internal consistency examines whether the items within a test are measuring the same underlying construct. A commonly used method to assess internal consistency is split-half reliability.

In split-half reliability, the items of a test are divided into two halves. There are various ways to divide the items, such as separating them into odd-numbered and even-numbered items or dividing them based on their content. **For instance**, if you have 20 questions in your test, you could create two halves of 10 questions each.

To calculate split-half reliability, you administer the test to a group of individuals and obtain their responses. Then, you split the responses into the two halves and calculate the average score for each half. Next, you examine the correlation between the scores of the two halves. If there is a strong correlation between them, it indicates that the items in both halves are internally consistent and measure the same underlying construct.

For example, suppose you divided your test into two halves, with items 1, 3, 5, etc., in the first half, and items 2, 4, 6, etc., in the second half. You collect responses from individuals and calculate their scores on each half. If there is a strong relationship between the scores, such as a correlation of 0.8 or higher, it suggests that the items in both halves are measuring the same main construct, such as motivation to study.

To determine the split-half reliability, you can take the average of the scores from both halves. If the correlation between the two halves is strong, it indicates that your test is internally consistent. In this way, split-half reliability provides evidence of the internal consistency of the test items.

Topic 152: Cronbach's alpha Reliability

In Module 152, we continued the discussion on reliability measures, specifically focusing on split-half reliability as a measure of internal consistency. The key difference between split-half reliability and other reliability measures is in how the test is divided into halves. While there are various methods to split the test, for example, dividing it based on item numbers or content, I used a specific method in my explanation.

In the example, I divided a 20-item test into two halves using a specific approach. I separated the even-numbered items (2, 4, 6, 8, etc.) and the odd-numbered items (1, 3, 5, 7, 9, etc.) to create two halves. However, it's important to note that there are multiple ways to divide the test into halves. **For instance**, you could divide the first 10 items as one half and the remaining 10 items as the other half. The choice of how to divide the test into halves can vary based on your preference.

To assess split-half reliability, you collect responses from individuals and calculate their scores for each half of the test. Then, you examine the relationship (correlation) between the scores of the two halves. A strong correlation indicates that the items within the test are internally consistent and measure the same construct, such as motivation to study.

It's worth mentioning that split-half reliability is just one method to evaluate internal consistency. Another commonly used method is Cronbach's alpha. Cronbach's alpha assesses the average correlation among all items in the test and provides a measure of internal consistency.

When you run a test in SPSS, it generates a table that provides information about the reliability of your test. In this table, you will find the reliability coefficient, often referred to as Cronbach's alpha. The reliability coefficient is a number between zero and one, with higher values indicating greater internal consistency. For example, if the reliability coefficient is 0.905, it means that your test has a high level of internal consistency.

The table also presents information about each item in your test. Let's focus on the columns related to item analysis: "Scale Mean if Item Deleted" and "Corrected Item-Total Correlation." These columns provide insights into the relationship between each item and the rest of the items.

For instance, let's consider item number one. The number next to item number one in the "Corrected Item-Total Correlation" column is 0.833. This value represents the correlation between item one and

the sum of the scores on the remaining items. In other words, if you calculate the correlation between item one and all the other items, the result is 0.833. This correlation value indicates the strength of the relationship between item one and the rest of the items.

A high correlation suggests that item one is internally consistent with the other items, meaning it measures a similar aspect or construct. **For example**, a correlation of 0.833 can be interpreted as 83.3 percent similarity or correspondence between item one and the other items.

Similarly, for each item in the table, you will find its relationship with the other items. The numbers in the "Corrected Item-Total Correlation" column reflect the strength of these relationships. For instance, if item eight has a correlation of only 0.128 (or 12.8 percent) with the rest of the items, it indicates a weak relationship or poor internal consistency. This suggests that item eight does not align well with the other items or measure the same construct.

When analyzing the items, you need to consider their correlations with the other items. Items with low correlations or poor internal consistency may need to be removed or replaced in order to improve the overall reliability of the test. By examining these numbers in the table, you can make informed decisions about which items are internally consistent and which ones are not.

By examining the item correlations, you can assess the internal consistency of each item. Higher correlations indicate better internal consistency, suggesting that the item measures the same construct as the other items. Conversely, low correlations indicate poor internal consistency, suggesting that the item may not align with the overall construct being measured. In such cases, you may consider replacing or removing the item from the test.

Topic 153: Kuder-Richardson (KR) reliability

In Module, we will discuss another measure of reliability called Kuder-Richardson (KR) reliability. While Cronbach's alpha is commonly used for assessing the internal consistency of scales with multiple response options, Kuder-Richardson reliability is specifically designed for **dichotomous or binary scales**. Dichotomous scales have only two response options, such as **yes** or **no**.

When using Kuder-Richardson reliability, there are two formulas to consider: KR20 and KR21. These formulas are similar in concept to what we have previously discussed. They involve splitting the responses of individuals into two parts to assess the internal consistency of the test.

The main difference between KR20 and KR21 lies in the assumption about the difficulty of the test items. KR20 assumes that all the items in the test have the same level of difficulty. This means that the

questions are equally challenging, and their order or arrangement does not significantly impact the results. If you are assuming that all the items are of the same difficulty, you can use KR20.

On the other hand, KR21 allows for variations in item difficulty. If you believe that the test items have different levels of difficulty, you should use KR21. This formula takes into account the potential differences in the difficulty of the items when assessing the internal consistency of the test.

It's important to note that Kuder-Richardson reliability is applicable only to dichotomous data, where responses can be categorized into two options. The interpretation of KR20 and KR21 focuses on the internal consistency of the test, indicating whether the items are measuring the same construct or underlying factor. Similar to Cronbach's alpha or split-half reliability, Kuder-Richardson reliability provides information about the consistency and reliability of the test scores.

Topic 154: Interrater Reliability and Intrarater Reliability

In this module, we will discuss the concepts of interrater reliability and intrarater reliability, which focus on the consistency of observations made by different raters or by the same rater over time. These reliability measures are different from the previous modules where we discussed test stability and internal consistency, which were focused on the reliability of the test itself.

Interrater reliability refers to the agreement or consistency between different raters who are observing and recording data on the same phenomenon. **For example**, if multiple observers are assessing a teacher's teaching style using a standardized observation tool, their recorded observations may not necessarily be the same. Each rater has their own interests, understanding, and biases that can lead to variations in their recorded data. To identify and minimize these biases, interrater reliability is calculated.

To ensure interrater reliability, one approach is to provide training or instructions to the raters before they begin collecting data. **For instance**, before sending observers to the field to observe 100 classes, a video recording of a lesson can be shown to them. They are then asked to independently observe the video, rate the teaching style using their own scales, and record their ratings. Observers whose ratings are close to each other are selected for data collection. Calculating interrater reliability helps to minimize rater bias and ensures that the data collected is reliable.

On the other hand, intrarater reliability focuses on the consistency of observations made by the same rater over time. It assesses whether a rater's own judgments and scoring of the same phenomenon remain consistent when repeated at different points in time. **For example**, if a rater scores a test item as 6 out of 10 marks on one occasion, but on another occasion rates it differently, such as 5 or 10 marks, there may be a lack of consistency in their own judgments. This inconsistency could be due to factors like changes in mindset or mood during different observation sessions.

Intrarater reliability becomes particularly important in longitudinal studies or when repeatedly assessing a specific phenomenon over time. To ensure the reliability of the rater's own scoring, it is necessary to assess their intrarater reliability. This can be done during a pilot phase or trial period by observing and rating the phenomenon multiple times and examining the consistency or relationship between the rater's own scores. A strong relationship between their own ratings indicates good intrarater reliability.

To summarize, interrater reliability focuses on the consistency of observations made by different raters, while intrarater reliability assesses the consistency of observations made by the same rater over time.

Lecture 36: Development and Administration of Research Tools-I

Topic 155: Different Ways to Collect Data

When conducting research, researchers have the option to either build their own research tools or reuse ready-to-use tools. The choice depends on the specific needs of the study and the availability of suitable existing tools. Here are some key points regarding the use of research tools:

Types of Standardized Research Tools: Research tools can include questionnaires, interview schedules, observation sheets, and other instruments used to collect data in the research process.

Using Existing Tools: If a suitable tool already exists for the research, researchers can choose to use it without making any changes. In such cases, researchers accept the established characteristics, **reliability**, and **validity** of the tool as determined by previous users or developers.

Adapting Existing Tools: Researchers may also decide to adapt an existing tool to fit their specific research needs. This adaptation typically involves making changes to the tool's framework, such as translating it into a different language or adjusting it to align with cultural or contextual factors. When researchers adapt a tool, they need to re-evaluate **its reliability** and **validity** based on the modifications made.

Norms of Standardization: Standardizing a tool refers to the process of using or adapting a tool developed by others while maintaining its original structure and attributes. Standardized tools are often created by experts in tool development and have been tested and used across different cultures or contexts. These tools are considered **norm-free**, meaning they can be used in various cultural settings without affecting their attributes.

Culture-Free Constructs: In the field of education, there are constructs that are considered culture-free, meaning they are not influenced by cultural variations. **For example**, constructs like cognitive development or stress are not tied to specific cultural contexts and can be measured using standardized tools.

Examples of Standardized Tools: Some examples of standardized tools include intelligence scales (e.g., Stanford-Binet Intelligence Scale), psychological inventories (e.g., Beck Depression Inventory), personality assessments (e.g., Myers-Briggs Type Indicator), and scholastic assessment tests (e.g., SAT).

Using standardized tools can save researchers time and effort as they don't need to establish the psychometric properties of the tool. However, it is crucial for researchers to carefully consider whether a standardized tool is appropriate for their specific research goals and context. In some cases, researchers may choose to develop their own tools to ensure they align closely with their research objectives and target population.

Topic 156: Self Developed Instruments

In this Module , we continue our discussion on tool development rather tool selection, which is an essential aspect of research. Developing a reliable and valid tool requires careful consideration and adherence to certain characteristics such as reliability, validity, and feasibility. While standardized tools are commonly used, researchers also have the option to develop their own tools.

Developing a tool is a complex process that requires time and effort. It is important to understand the procedures for establishing reliability and validity, as discussed in previous modules, and to apply them effectively during tool development. Undergraduate research provides an opportunity for students to learn about tool development, but it is important to consider time constraints and balance the learning experience with the practicality of the research project.

In today's era, tools can take various forms, including **software for data collection, equipment used to gather data** (e.g., in clinical psychology), **recording devices for fieldwork**, and **survey instruments**. Each tool serves the purpose of collecting data, and it is crucial to ensure that these tools are reliable and valid.

If you choose to develop your own tool, it is recommended to allocate at least three to four months for the process. This allows sufficient time for tasks such as developing the tool framework, writing items or questions, designing the layout, conducting pilot tests, and ensuring reliability and validity. If you have learned about tool development during your coursework, you can adapt existing tools or modify them to suit your research needs. However, if you lack this skill, it is important to take the initiative to create your own tool while conducting your research project. By following the steps outlined in previous modules and ensuring reliability and validity, you will gain valuable experience in tool development.

It is worth noting that many researchers who develop their own tools do not use them again in future research. Additionally, it is less likely for others to use these tools. Despite these challenges, I strongly encourage you to learn about tool development and strive to create reliable and valid tools. Refining the language and ensuring clarity in your tool development process will contribute to its effectiveness.

Topic 157: Naturally occurring data Tools

In Module 157, we explore the concept of utilizing naturally occurring data for research purposes, which can be an alternative to developing a new data collection tool. Naturally occurring data refers to existing data that can be accessed and utilized for research without the need for extensive data collection efforts.

One common source of naturally occurring data is everyday activities and incidents that naturally generate data on specific subjects. These data-generating activities are part of our normal lives and can provide valuable insights for research purposes. When the research objectives and questions are clear, and the available data is suitable for analysis to address those questions, there is no issue with using ready-made data. By leveraging this existing data, researchers can save time and dedicate more effort to data analysis and interpretation.

1. One common source of naturally occurring data is **social media analytics**. Platforms like WhatsApp, Facebook, and Twitter generate vast amounts of data related to people's social

behaviors. Instead of creating a new tool or questionnaire, researchers can leverage this existing data with the permission of the relevant organization. By selecting the relevant data that aligns with their research objectives and questions, researchers can request access to the data and analyze it to derive meaningful insights. It is important to acknowledge the source of the data in the research report, adding credibility to the findings.

2. Apart from social media analytics, other forms of naturally occurring data include **health records**. **For example**, researchers studying the physical development of children in the field of early childhood can access growth records collected by hospitals and healthcare centers. Rather than duplicating the data collection process, researchers can contact these organizations to obtain the relevant data and conduct their research based on the analysis of existing records.
3. **Satellite images** are another valuable source of naturally occurring data. Researchers interested in weather patterns, for instance, can access images recorded by meteorological organizations that capture hundreds of patterns daily. The availability of records across different seasons and months allows researchers to study and analyze these patterns.
4. **Economic indicators**: Government data is also a valuable resource for research purposes. **For instance**, if researchers want to investigate government investments in education or analyze the utilization and efficiency of the education budget, they can access data collected by statistical bureaus or educational institutions. These organizations generate reports, collect periodic data, and store it in databases, making it accessible to researchers.
5. Other **types of surveys**, such as price index surveys, labor surveys, and educational statistics surveys, are conducted by various organizations. Researchers can access these surveys, extract the relevant data, and analyze it to gain insights for their research.
6. In the context of online education, **learning analytics** provide valuable data. As education is delivered through learning management systems (LMS) in online settings, continuous records are collected, including online time spent, student interactions, chatbox conversations, and usage of learning applications. This data can be utilized for research purposes in understanding student learning behaviors and identifying areas that may require improvement in online education.

When using naturally occurring data, it is important to follow certain protocols. Researchers should select data that aligns with their search criteria and identify the organization that holds the original source of the data. They need to establish agreements with the organization, clarifying the rules and

conditions for data usage and specifying where the findings can be used. Adhering to these protocols allows researchers to access and utilize the data for their research purposes.

Overall, utilizing naturally occurring data can be a valuable approach in research, enabling researchers to leverage existing data sources and focus their efforts on data analysis and interpretation.

Topic 158: Tool Framework

In Module 158, we will delve into the concept of a tool framework, which involves the development and administration of research tools. Often, students mistakenly believe that the first step in research is to create a tool and collect data, neglecting other crucial aspects such as the literature review and rationale of the study. However, the research process typically begins with identifying the problem and conducting a literature review before tool development.

When developing a research tool, it is not as simple as formulating a set of questions based on personal intuition. It is not necessary for every developer to create a tool that comprehensively covers all aspects of the construct they aim to measure. Instead, the process starts with defining the concept and constructs to be measured within a tool framework. This framework outlines the components of the construct and serves as a guide for tool development.

For instance, let's consider the example of teaching effectiveness. The definition of teaching effectiveness may vary among individuals, and it is important to establish a consensus on what constitutes effective teaching. To reach this consensus, it is necessary to consult the literature and identify the constructs associated with teaching effectiveness. These constructs can be represented as sub-constructs within a framework, as depicted in the slide.

The next step is to **define the scope** of each sub-construct. **For example**, content expertise can be defined as having knowledge of the subject matter beyond what is provided in textbooks or possessing contemporary knowledge. By clearly defining the scope, researchers ensure that the measurement is focused and relevant to the construct being assessed.

Once the constructs and their scopes are established, researchers can proceed to create questions that align with the framework. It is essential to generate multiple questions for each sub-construct, as a single question may not sufficiently measure a complex construct. In the slide example, three questions

are created for content expertise, four for classroom activities, and five for how teachers respond to students.

The number of questions is not fixed, and researchers should determine an appropriate number based on the comprehensiveness of the construct. The questions should be formulated as statements and can be measured using scales such as Likert scales or dichotomous scales.

After completing these steps, researchers can compile all the items, including the questions and their associated scales, to create a questionnaire. This process ensures that the developed tool covers the intended construct effectively.

Lecture 37: Development and Administration of Research Tools-II

Topic159: Pilot Testing

Module 159 focuses on the development of the research tool and emphasizes the importance of pilot testing. Pilot testing is a crucial step in the tool development process as it helps evaluate the reliability and validity of the tool before collecting data for the actual research study.

Pilot testing involves following steps:

1. Select your participants
2. Administer the research tool, collect data
3. Analyze data, gather feedback
4. Evaluate reliability and validity
5. Revise and refine
6. Repeat if necessary
7. Document everything
8. Finalize research tools

Once the tool framework is established and all the steps are completed, such as defining constructs, creating questions, and selecting appropriate scales, the tool is ready for pilot testing. During pilot testing, the researcher applies the tool to a **smaller sample** that represents the target population.

In the example of measuring teaching effectiveness, the researcher would go to classrooms and collect data by observing teachers. However, it is not feasible to include all types of teachers in the pilot sample

due to the wide variety of qualifications, experience levels, gender, school settings, and grade levels. Instead, the researcher selects a smaller sample that aims to cover different variations of teachers to some extent.

The purpose of pilot testing is not to assess the performance of teachers but to evaluate the accuracy and effectiveness of the research tool itself. The collected pilot data is analyzed to identify any weaknesses or strengths in the tool. This analysis helps determine how well the questions were understood by the participants and whether the observations made with the tool align with the intended constructs.

To conduct pilot testing effectively, it is recommended to have a sufficient number of participants. As a general rule, it is advisable to include at least five participants for each statement or question in the tool. **For example**, if the tool consists of 20 questions, a minimum of 100 participants should be included in the pilot sample. The selected participants should represent various characteristics, such as qualification, gender, experience level, and school setting, to cover the desired range of diversity.

Once the pilot data is collected, it is analyzed to assess the reliability and validity of the tool. This analysis helps identify any flaws or issues in the tool, which can then be addressed through revisions or improvements. If revisions are made, it is important to conduct another round of pilot testing to gather data for the revised statements or questions.

To ensure validity, it is recommended to create alternate statements for each original statement during piloting. This allows for flexibility in discarding or revising statements without compromising the overall validity of the tool.

During pilot testing, it is also important to identify any participants who may not be responding properly, leaving blanks, or displaying response patterns that may affect the data quality. Their data can be eliminated, and the tool can be revised based on the observations and analysis of the pilot data.

In summary, pilot testing is a critical step in refining the research tool. It is not intended for data collection on research constructs but rather for evaluating and improving the tool's reliability and validity. By analyzing the pilot data and making necessary revisions, researchers can develop a reliable and well-performing tool.

Topic 160: Administration of Research Tools

The administration of a research tool is crucial for collecting accurate and reliable data. Even if the tool itself is well-developed, improper administration can lead to inconsistencies and errors in data collection. This is especially important when multiple individuals are involved in data collection across different locations or settings. Develop comprehensive guidelines that outline the proper administration of the research tool. These guidelines should be error-free, easy to understand, and accessible to all administrators. Clear instructions will help ensure that everyone administers the tool in the same way.

Here are some key considerations for proper tool administration:

1. **Prepare your materials:** Prepare clear, well structure and error free protocols. Clearly define the procedures for data collection to ensure consistency among different administrators. This includes specifying how the tool should be used, what observations or ratings should be recorded, and any specific guidelines or instructions that need to be followed.
2. **Select your sample:** Determine your target population and selected a representative sample from that population.
3. **Obtain informed consent:** Ensure that you have obtained informed consent from them before administering the research tool. Before collecting data, obtain informed consent from participants, ensuring they understand the purpose of the research and their rights as participants. This is an important ethical consideration in the research process.
4. **Provide Training:** Train all individuals involved in data collection, including colleagues or other researchers, on the proper use of the research tool. This includes teaching the protocol for using the tool, explaining the timing and locations for data collection, and addressing any questions or concerns.
5. **Schedule appointments or session:** Coordinate with participants to schedule appointments or session for data collection. Ensure that the timings and locations are convenient for them.
6. **Administration of the research tool:** This involves reading questions aloud, providing instructions or conducting experiments according to the predetermined protocol.
7. **Monitor Data Collection:** Continuously monitor the data collection process to ensure adherence to the established guidelines and protocols. Selectively visit the data collection sites

to conduct quality assurance checks and provide support or guidance to administrators if needed.

Proper administration of the research tool is essential to minimize errors and maintain the reliability and validity of the collected data. It is a logistical task that requires careful planning, attention to detail, and adherence to established protocols. By following these steps, researchers can ensure that the data they collect is trustworthy and that the results they present accurately reflect the research findings.

Topic 161: How to Value Research Findings

In this module, we will discuss data analysis and how to evaluate research when reading it or considering its use in your own research project. When assessing the value and credibility of research, there are several factors that people generally consider.

1. **Authorship:** The first factor is **authorship**. It is important to examine who the authors are and their expertise in the field of education related to the research. It is also useful to know if they have published previous work.
2. **The publication venue** is another important aspect to consider. In recent years, there has been a significant increase in the number of research journals, making it challenging to determine the credibility of each journal. Different measures can be used to assess publication value, such as the Higher Education Commission (HEC) repository, which provides a list of reputable journals.
3. **Methodology and research design:** When evaluating research, it is essential to examine the different components of a published paper. Typically, a research paper consists of sections such as an introduction, research questions, methodology (including sampling and data collection procedures), data analysis, results, and discussion. It is important to assess the alignment between these sections. Do the research questions align with the methodology? Are the analysis and findings consistent with the research objectives? Does the discussion align with the results? By examining these alignments, you can gain insights into the quality of the published research.
4. **Sample size,** representativeness of the sample, and the sample selection process are also important considerations. Assess whether the results align with the study design to determine the reliability of the research.

Peer review is another significant factor. Determine if the research was published in a journal that undergoes a rigorous peer review process. While some journals have strict review mechanisms, others may have more automated processes. Considering the level of peer review can help assess the quality of the research.

Examining the references and citations used in the research provides insights into the researcher's knowledge and the sources they considered. If the researcher cites current and relevant publications, it indicates a well-informed approach.

5. **Data collection and analysis** require careful examination. Ensure that the data collected is appropriate for the research objectives and matches the level of measurement. Consider any prerequisites or conditions for the chosen statistical analyses. **For example**, normality is often a requirement for many tests, so verify whether the data meets this assumption and if any necessary treatments were applied. It is essential to ensure the accuracy of the statistical analysis and avoid any manipulation or preconceived outcomes.

These are a few essential factors to consider when judging the quality of research publications. However, there are many more aspects to explore. By thoroughly assessing these factors, you can make informed decisions about including the research in your analyses or report. It is also important to adapt these considerations depending on whether the research is qualitative or quantitative in nature.

Topic 162: Levels of Measurement and Data Analysis

In previous modules, we discussed the concept of levels of measurement and how to determine the appropriate level for any given dataset or variables. Understanding the level of measurement is crucial as it guides the selection of appropriate statistical analyses and helps evaluate the importance of determining the level of measurement for each variable.

There are **four levels of measurement: nominal, ordinal, interval, and ratio**. Each level has its own characteristics, which are briefly outlined on the slide. We have covered these characteristics in detail in previous modules. For brevity, I will provide a quick recap.

When dealing with nominal data, no mathematical operations can be performed on the values generated by the variable. **For example**, if we consider the variable "gender," where 1 represents male and 2 represents female, adding 1 and 2 = 3, does not make sense. It cannot be averaged either. Nominal

data can only be counted and analyzed in terms of frequencies and percentages. **For instance**, you can determine the number of males and females in a sample and express it as a percentage of the total sample. You can calculate mode for the nominal data to find the central value.

Ordinal data allows for a slightly broader range of descriptive statistics. You can calculate the median, range, and percentile ranks. However, the mean, standard deviation, and variance are not meaningful for ordinal data. Extracting the middle point of ordinal data would involve calculating the mode.

Interval and ratio data offer the most flexibility in terms of descriptive statistics. For interval data, you can compute the mean as a measure of central tendency, as well as the standard deviation and variance to examine the data's dispersion. The same measures can be applied to ratio data.

In the table provided, I have included examples of variables corresponding to each level of measurement. **For instance**, a variable like "gender" would generate nominal data, while measuring the temperature of an object on the Celsius or Fahrenheit scale would yield interval-level data. These examples serve as a refresher and complement the previous discussions on levels of measurement.

By understanding the level of measurement of your data, you can determine which descriptive statistics to apply. When analyzing data, consider the level of measurement associated with each variable. For nominal data, the mode is appropriate. For ordinal data, the median is suitable. And for interval or ratio data, the mean can be used. Similarly, for examining data dispersion, the appropriate measures depend on the level of measurement.

I hope this clarifies the relationship between descriptive statistics and the types of variables they are applicable to.

Topic 163: Data Cleaning

In this module, we will discuss an essential step in data analysis called data cleaning. After collecting data, the next step is typically data entry, where the data is entered into software such as SPSS or Excel. Before conducting any analysis, it is crucial to check the entered data for errors and inconsistencies. Performing data cleaning helps ensure the accuracy and reliability of the data.

The first and most important step in **data cleaning is creating a backup** copy of the original data file. This backup ensures that you have a preserved version of the raw data before making any

modifications or corrections. Once you have the backup, you can work on a separate copy of the file for the data cleaning process.

The next step involves **identifying anomalies and errors** in the data. These errors can be typographical, such as a mistyped number or value. You can use search functions to identify such errors or examine frequency tables to detect values that are outside the expected range. Duplicated data is another type of error that needs to be identified. If the same data is entered multiple times for a single case, it can skew the analysis results. Various techniques can be employed to identify and handle duplicated data.

Missing data is a common issue that occurs after data entry. Some respondents may not answer certain questions or leave entire pages blank. Deleting cases with missing data may lead to a significant loss of data. Instead, there are several methods to **handle missing data**, including deletion, substitution with other values, or using statistical techniques to impute missing values. The choice of method depends on the nature and extent of missing data in the dataset.

After addressing missing data, it is essential to **save the cleaned data** file in a format compatible with your analysis software. Formats such as Excel or SPSS are widely accepted and can be used with various software applications. Additionally, it is essential to **check the integrity** of the data by ensuring that all values fall within reasonable ranges. **For example**, if a variable represents test scores, values exceeding 100% or unusually low values should be investigated for potential errors.

Data cleaning is an iterative process that involves multiple checks and revisions. It is recommended to involve multiple individuals in the data cleaning process to minimize errors and ensure the highest data quality. Through iterative cleaning and validation, you can gradually improve the accuracy and reliability of the dataset.

Please note that this module provides an introductory overview of data cleaning. In-depth knowledge and expertise are required to handle specific data cleaning challenges effectively.

Lecture 38: Types of Statistics-I

Topic 164: Types of Statistics

In this module, we will focus on providing a classification of statistics, highlighting the benefits of statistical thinking, and exploring the fields where statistical analysis is commonly applied. We will also discuss the concept of averages and its relationship to statistical analysis.

Statistics is a field that encompasses various techniques used for data analysis. For social sciences students, statistics can be a challenging concept to grasp. Therefore, in this module, we will present a simplified overview of statistics and its practical applications.

Statistical analysis can be broadly classified into two main categories: **descriptive statistics** and **inferential statistics**.

Descriptive statistics focuses on summarizing and describing the characteristics of a dataset. It involves methods to organize, analyze, and present data in a meaningful way. Descriptive statistics aim to provide a concise summary of the data, such as measures of central tendency (e.g., mean, median, mode) and measures of dispersion (e.g., range, variance, standard deviation). These statistics help researchers understand the distribution, variability, and general characteristics of the data.

Inferential statistics, on the other hand, involves making inferences and generalizations about a population based on a sample. It allows researchers to draw conclusions beyond the observed data and make predictions or test hypotheses. Inferential statistics includes techniques such as ANOVA and MANOVA.

Inferential statistics derives its name from the word "inference," which refers to drawing conclusions or making generalizations based on evidence or observations. It allows researchers to make predictions, test hypotheses, and determine the significance of observed differences or relationships in the data. Inferential statistics involve various techniques such as hypothesis testing, modeling, and estimation, which help researchers understand the relationships between different variables and compare different groups or conditions.

ANOVA (Analysis of Variance) and MANOVA (Multivariate Analysis of Variance) are examples of inferential statistical techniques. ANOVA is used to compare means across multiple groups or conditions to determine if there are statistically significant differences between them. It provides insights into whether the observed differences in means are likely due to chance or if they represent true differences in the population.

MANOVA, on the other hand, extends the concept of ANOVA to situations where there are multiple dependent variables. It allows researchers to analyze the differences or relationships between groups while considering multiple outcome variables simultaneously.

In summary, descriptive statistics involves summarizing and presenting data, providing a description of its features. It focuses on data summary and presentation methods. On the other hand, inferential statistics allows researchers to draw conclusions, make generalizations, and compare groups or conditions. It involves techniques such as hypothesis testing, modeling, and estimation to explore relationships and differences between variables.

In the next modules, we will explore specific techniques and methods within descriptive and inferential statistics, providing a more detailed understanding of their applications and interpretation.

Topic 165: What Statistics Can Offer?

In the previous module, we delved into descriptive statistics and explore the measures of central tendency and dispersion. Descriptive statistics involve summarizing and describing data, and it encompasses two main types: **measures of central tendency** and **measures of dispersion**.

Measures of Central Tendency:

Measures of central tendency are used to determine the central or typical value of a dataset. There are three commonly used measures of central tendency:

Mean: The mean is the average value of the data. To calculate the mean, you sum up all the values in the dataset and divide it by the total number of observations. The mean is suitable for data that is of interval or ratio nature.

Median: The median represents the middle value in a dataset when the values are arranged in ascending or descending order. If there is an odd number of observations, the median is the middle value. If there is an even number of observations, the median is the average of the two middle values. The median is appropriate for ordinal data, which has an inherent order.

Mode: The mode is the value that occurs most frequently in a dataset. It is suitable for nominal data, where there are categories without a specific order. The mode helps identify the most common category or characteristic within the dataset.

These measures of central tendency provide insights into the typical value or central tendency of the data.

Measures of Dispersion:

Measures of dispersion quantify the spread or variability of the data points around the measures of central tendency. They provide information on how the data is distributed. Here are some commonly used measures of dispersion:

Range,

Quartiles,

Standard Deviation,

Variance:

These measures of dispersion help assess how spread out or concentrated the data points are around the measures of central tendency. The standard deviation is a commonly used measure of dispersion. There are additional methods for calculating variance, but they are less popular and not discussed here. In summary, descriptive statistics focuses on understanding the central tendency and dispersion of data.

Topic 166: What Statistics Can Offer-II?

In statistics, there are two main components: **descriptive statistics and inferential statistics**. In the previous module, we discussed descriptive statistics, which involves summarizing and describing data. In this module, we focus on a component of inferential statistics, specifically the test of difference.

Inferential statistics can be categorized into tests of difference and tests of association. In this module, we will focus on tests of difference. The tests of difference include chi-square, the t-test, z-test, ANOVA (Analysis of Variance), and its types, such as one-way ANOVA and MANOVA.

The chi-square test is used for nominal data to examine differences between variables. It is a family of tests with different names based on their specific characteristics. Additionally, the non-parametric statistic can be used for nominal data or data from a less representative sample.

Within the test of difference, the t-test is commonly used. There are **three types of t-tests: independent samples t-test, paired samples t-test, and one-sample t-test**. The independent samples t-test is used to compare the means of two independent groups, such as comparing the performance of boys and girls. The paired samples t-test is used when there are two related measurements taken from the same individuals, such as pre-test and post-test scores. The one-sample t-test compares the mean of a single sample to a known population mean.

ANOVA is used when comparing means across more than two groups. Suppose you want to compare the achievement of individuals based on different factors such as gender, religion, or divisions. If you have two groups, such as males and females, you can use the independent samples t-test to examine the difference in their achievements.

However, if you have more than two groups, such as different divisions (first division, second division, third division), you would need to use a different test. In this case, you can use a one-way ANOVA. The one-way ANOVA compares the means of three or more independent groups to determine if there are significant differences among them. It would allow you to assess the difference in performance between the first division, second division, and third division in your course.

It's important to note that there are further statistical methods and techniques beyond what we can cover in this module. The aim here is to introduce you to the terminology and concepts so that you can familiarize yourself with these terms. Further study and exploration in a data analysis course will provide a more comprehensive understanding of these topics.

Topic 167: What Statistics Can Offer-III?

In Module 167, we explore tests of associations in inferential statistics. These tests are used to examine the relationship or association between variables. When the relationship between variables is strong, it can serve as the basis for prediction, and regression-related tests can be applied.

Association refers to examining how two variables are related to each other. **For example**, you might want to determine if a person's height is related to the size of their steps. If there is a positive relationship, an increase in height would correspond to an increase in step size. Conversely, an inverse relationship occurs when an increase in one variable corresponds to a decrease in the other. **For instance**, if you increase the use of medicine, the occurrence of a disease may decrease.

To measure the strength of the relationship between variables, numerical results are typically obtained, ranging from 0 to plus/minus 1. A value closer to 1 indicates a stronger relationship. The strength of the relationship can also be expressed as a percentage, such as 60%, which suggests a strong relationship.

There are various statistical tests used to examine associations based on the nature of the data. For nominal data, the Chi-Square test is commonly used. It assesses the relationship between two nominal variables. Another widely used test is the Pearson Product-Moment Correlation, which examines the relationship between two variables measured on an interval or ratio scale. If the data is ranked or ordinal, the Spearman Rank Order Test is suitable.

When the association between two variables is strong, one variable can act as a predictor of the other. In this case, regression analysis can be applied. Regression analysis includes different forms, such as linear regression and multiple regression. It helps determine if one variable can predict the occurrence of another variable.

It's important to note that regression analysis should only be applied when there is a strong relationship between the variables. The strength of the relationship can be assessed using the tests discussed earlier, with a value closer to 1 indicating a stronger relationship.

Lecture 39: Types of Statistics-II

Topic 168: Frequency Distribution

In this module, we focus on data organization before analyzing it. The first step is to create a frequency distribution, which is similar to the graphs we used to make in mathematics class using graph paper. We represent different types of data on graphs, typically two-dimensional graphs.

For example, let's consider the cultural understanding of individuals and how frequently they travel abroad. We can create a graph with cultural understanding on the X-axis and the number of times they traveled on the Y-axis. Alternatively, we can put the number of visits on the X-axis and their understanding of other cultures on the Y-axis. By plotting the data on these axes, we create a frequency distribution.

An example of a frequency distribution is a bell-shaped curve, also known as a Gaussian or normal distribution. This curve represents the distribution of the data, but how is it actually created? We collect data on a group of teachers and categorize them based on their visits to the library. **For instance**, we may have five teachers who visited the library once, 15 teachers who visited twice, and so on. Each category is represented by a line on the graph, with the vertical axis indicating the number of individuals in that category.

By drawing these lines for each category and connecting them, we obtain a distribution curve. This distribution curve provides insights into the pattern and spread of the data, and it serves as the foundation for understanding data distribution in statistics.

Topic 169: Normal Distribution

In Module 169, we continue discussing the distribution of data, specifically focusing on the normal distribution. We revisit the previous example and analyze the graphical shape of the distribution to determine if it follows a normal distribution.

To determine if data is normally distributed, we perform a visual check by folding the curve in half from its center. If the lines on one side align perfectly with the lines on the other side, we conclude that the data is normally distributed.

The concept of normality is a fundamental principle in statistics. It states that in any population, such as measuring intelligence or height, there will be few extremely intelligent or extremely dull individuals, while the majority fall within the range of average intelligence or height. When plotting a graph of the data, it typically forms a normal distribution.

From a statistical perspective, the normal distribution can be divided into four equal parts known as quartiles. Each quartile represents a deviation from the center. Around 68% of the data falls within one standard deviation from the mean, while approximately 14% falls within each subsequent quartile. The remaining 2% is distributed beyond two standard deviations.

To assess whether data follows a normal distribution, we examine the graph and observe if around 68% of the data falls within the range of plus or minus one standard deviation from the mean. If the data meets this criterion, along with the distribution of data in subsequent quartiles, we consider it to be normally distributed. However, it's important to note that truly perfect normal data is rare in the real world due to the presence of errors and variability. Therefore, data that closely approximates the normal distribution is considered normal, while deviations from this pattern will be discussed in the next module.

Another characteristic of normally distributed data is that the mean, median, and mode will all be the same value. To find the median, you arrange the data in ascending or descending order and identify the number in the middle. If there is an odd number of data points, the median is the middle value. If there is an even number of data points, the median is the average of the two middle values.

For example, if we have the data set: 1, 1, 2, 2, 2, 2, 2, 3, 3 the median would be 2, as it is the value in the middle when arranged in ascending order. In this case, there are four values to the left and four values to the right of the median. +

To calculate the mean, we add up all the values and divide by the total number of data points. Using the same example, the sum of the values is 18 ($1+1+2+2+2+2+2+3+3$), and since there are nine data points, the mean is 18 divided by 9, which equals to 2.

The mode is the value that appears most frequently in the data set. In normally distributed data, there may or may not be a mode. If there is a mode, it will be the same value as the mean and median (2).

If the data is not normally distributed, we will discuss the implications and characteristics of such distributions in the next module.

Please note that these calculations provide a basic understanding of finding the mean, median, and mode. In practice, there may be variations and additional considerations depending on the specific data set and statistical techniques used.

Topic 170: Skewed Distribution

When data is not normally distributed, it may exhibit characteristics of skewness and kurtosis. Skewness refers to the asymmetry of the data distribution, while kurtosis relates to the shape of the distribution's tails.

If the data distribution is skewed, it means that it is not equally distributed on both sides of its center point. There are two possibilities for skewness: positive skewness and negative skewness.

Positive skewness occurs when the data cluster more towards the lower values on the right side of the peak point, and the values gradually increase towards the left side. You can identify positive skewness by drawing a vertical line from the peak point on the x-axis and observing which side of the line has more data. In a positively skewed distribution, there is more data on the left side and less data on the right side, resulting in a longer tail on the right side of the graph.

Negative skewness, on the other hand, occurs when the data cluster more towards the higher values on the left side of the peak point, and the values gradually decrease towards the right side. You can remember negative skewness by visualizing the graph starting from a higher point and sloping downwards, resembling a negative sign. In a negatively skewed distribution, there is less data on the left side and more data on the right side, resulting in a longer tail on the left side of the graph.

It's important to note that skewness does not affect the mean and median in the same way. In positively skewed data, the mode is the smallest value, the median is greater than the mode, and the mean is the largest. Conversely, in negatively skewed data, the mode is the largest value, the median is smaller than the mode, and the mean is the smallest.

These observations provide a way to identify and interpret skewed data. However, it's essential to rely on statistical tests and analysis to make precise determinations of skewness and the nature of the distribution.

Topic 171: Outlier

In Module 171, we discussed the graphical representation of data and introduced the concept of outliers, skewness, and kurtosis. Let's delve further into outliers and their impact on data distribution.

An outlier is an observation that falls outside the general pattern of the distribution. It is represented by a data point that is significantly different from other data points in the dataset. Outliers can occur due to various reasons, such as data entry errors, measurement inaccuracies, or genuinely extreme values in the population.

When we visualize a data distribution, such as through a histogram or bar chart, outliers are noticeable as data points that lie far away from the main cluster. They can disrupt the overall shape and pattern of the distribution. If we remove or add an outlier, the distribution will be affected accordingly.

For example, if we remove an outlier from a distribution, the remaining data points tend to form a more consistent pattern, and the shape of the distribution becomes closer to a normal distribution. On the other hand, if we add an outlier to a distribution, the shape may become skewed, with a lengthened tail on the side where the outlier is located.

It's important to note that the presence of an outlier does not necessarily indicate an error or mistake in the data. Outliers can be valid observations that represent rare or extreme events. However, they can also be influential points that disproportionately affect statistical analyses or modeling outcomes.

During the data cleaning process, it is common practice to identify and handle outliers appropriately. This may involve removing the outlier if it is deemed to be a data entry error or an extreme value that does not reflect the population accurately. Alternatively, in some cases, outliers may be retained if they are genuinely representative of the underlying phenomena.

Statistical tests and techniques can be applied to assess the impact of outliers on data analysis. These tests can help researchers determine whether outliers significantly affect the results and consider appropriate strategies for handling them.

Lecture 40: Types of Statistics-III

Topic 172: Kurtosis

In Module 2, we discussed the concept of data distribution and highlighted two types of distribution errors: skewness and kurtosis.

Skewness refers to the asymmetry of a data distribution. When the data is not evenly distributed on both sides of the central point, it is considered skewed. There are two types of skewness: positive skewness and negative skewness.

Kurtosis, on the other hand, is the degree of peakness of a distribution, usually taken in relation to a normal distribution. Let's consider the likert scale example. You asked a question with five answer options ranging from one to five, representing strongly disagree to strongly agree. You observed that the majority of respondents selected option three, with a few selecting options two and four, and no one selecting option one or five.

Types of kurtosis:

Leptokurtic: A curve having relatively higher peak than the normal curve, is known as leptokurtic.

Platykurtic: If the curve is more flat-topped than the normal curve, it is called Platykurtic.

Mesokurtic: A distribution that falls between these two extremes is known as mesokurtic or normally distributed. It is neither too peaked nor too flat-topped.

When data is not normally distributed, it can have implications for statistical tests and analyses. Therefore, it is important to assess the normality of the data before applying these tests. If the data is found to be skewed or exhibits kurtosis, data normalization techniques can be employed to transform the distribution into a more normal shape.

By normalizing the data, we adjust its distribution to closely resemble a normal distribution, allowing for more accurate interpretation of statistical results. Failure to consider the normality of the data and apply appropriate normalization techniques can lead to invalid or misleading conclusions.

Therefore, it is crucial to understand the normality of the data and employ data cleaning or statistical methods to normalize the data before conducting inferential statistics or data analysis.

Topic 173: Graphs: Graphical Presentation of Data

In this module and the upcoming ones, we will explore graphical representations of data commonly used in research reports. We will discuss the different types of graphs and how to interpret them. Typically, software like SPSS and Excel offer a range of graph options to choose from. We will discuss three out of them here:

Bar Graphs: Bar graphs are used to display categorical data where each category is represented by a bar. The height of the bar corresponds to the frequency or count of observations in each category. The data shows that in 2005, 40 people committed suicide, and in subsequent years, the numbers increased slightly.

It is crucial to use a standard scale when creating graphs to avoid misinterpretations. When different scales are used, the same dataset can give rise to different impressions. This can lead to misleading conclusions if someone focuses solely on the scale of the graph instead of the actual data.

For example, in the case of the suicide rate data, when comparing the two graphs, it may appear that the rate of suicide is increasing significantly. The years 2006 and 2008 might seem to have a substantial difference in the number of people committing suicide. However, when using a standard scale, the actual difference may be just seven individuals.

To ensure accurate and meaningful representations, it is important to select a scale that is standard or appropriate for the data being presented. By doing so, we can avoid exaggeration or distortion in the appearance of the graph. Bar graphs follow the same principles as we discussed earlier for the normal curve. The methodology remains consistent.

Histograms: The second type of graph you mentioned is called a histogram. It is similar to a bar graph, but with a key distinction. In a histogram, the x-axis represents intervals or ranges of values, and there are no gaps between these intervals. **For example**, the intervals on the x-axis could be of 10 points. The y-axis represents the frequency or count of observations falling within each interval.

To create a histogram, you plot the frequencies of the data points within each interval. **For instance**, if you have data on the frequencies of individuals falling within the range of 39.5 and 49.5, you would plot the corresponding frequency on the y-axis for that interval. Then, you move on to the next interval and repeat the process until all intervals are represented.

The resulting shape of the histogram provides valuable insights into the distribution of the data. It allows you to visually identify patterns or characteristics of the data's distribution. Histograms are indeed one of the most commonly used graphical displays for any type of data.

Polygon: The third type of graph is called a polygon. In a polygon, you plot the frequencies as individual dots on the graph and then connect these dots with lines. The first dot represents the

frequency of the first interval, the second dot represents the frequency of the second interval, and so on. By connecting these dots with lines, you create a shape on the graph.

Typically, the lines connecting the dots are straight lines. However, sometimes, we fit a smooth probabilistic line instead of using straight lines. This means that the lines may not precisely intersect each dot, but rather pass near them. By joining the dots with a smooth line, you create a representation that provides a better understanding of the pattern exhibited by the data.

Topic 174: Graphs: Box Plot Charts, Pie Chart, Stacked Bar Charts

In this Module we will discuss additional types of graphical representation, beyond the three types mentioned previously. These include **box plot charts**, **pie chart**, **stacked bar charts**, and **line graphs**.

Box Plot: The box plot, also known as a box-and-whisker plot, displays the distribution of a dataset using quartiles. It consists of a box with a vertical line in the center, representing the median. The box itself represents the interquartile range (IQR), showing the middle 50% of the data. The lines extending from the box, known as whiskers, indicate the minimum and maximum values, excluding outliers.

Pie Chart: A pie chart is a circular chart divided into sectors, representing the proportion or percentage of different categories or sub-categories. Each sector's size corresponds to the relative quantity or percentage of the data it represents. Pie charts are useful for displaying data with a total of 100% and provide a quick visual understanding of the distribution among categories.

Stacked Bar Chart: A stacked bar chart is used to display multiple variables or sub-categories within a category. Each bar in the chart represents the total for the category, and it is divided into segments representing the different sub-categories. This allows for a visual comparison of the sub-categories within each bar, showing the distribution and proportion of each sub-category.

Line Graph: A line graph is used to depict the progression or trend of a variable over time or a continuous scale. It consists of data points connected by lines, allowing for a visual representation of how the variable changes over the specified period. Line graphs are particularly useful for illustrating trends, patterns, or correlations in data.

Topic 175: Characteristics of Qualitative Data

In this module, we will begin analyzing qualitative data and discuss its characteristics, starting with its nature, followed by its quality, and then its analysis. Before we proceed, let's recap what we mean by qualitative research. Qualitative research delves deep into a phenomenon, seeking a comprehensive understanding of the topic. Unlike quantitative research, which often involves surveys or numerical data collection, qualitative research addresses complex, layered, and in-depth issues. It focuses on human experiences, behaviors, and other similar aspects. Qualitative data usually takes a non-numerical form, including text, images, audio recordings, and observations. What is common among these forms is that they allow for multiple interpretations. **For example**, different viewers may interpret an image in various ways. The information obtained from qualitative data tends to be multifaceted. When exploring such topics, qualitative research is generally more appropriate.

Characteristics of qualitative data

Now, let's discuss the characteristics of qualitative data.

1. Firstly, qualitative data is typically **non-numeric in nature**. Unlike quantitative research, where data is often numerical, qualitative research involves data in narrative form, such as text, images, videos, or audios.
2. Another important characteristic of qualitative data is its **descriptiveness**. It requires detailed narratives or elaborate descriptions to understand and explain the data. Unlike quantitative analysis, which often involves data reduction and summarization, qualitative analysis aims to unpack and explore the data.
3. **Subjectivity** is another characteristic of qualitative data. Different individuals or communities may have varying perspectives and interpretations of the same phenomenon, and all of these interpretations can be valid. This subjective nature distinguishes qualitative data from the objective nature of quantitative data, which is independent of context, society, or individual perspectives.
4. Qualitative data is **contextualized**, meaning it is embedded in the specific cultural, social, and environmental contexts from which it is gathered. It cannot be separated from the place or society where the phenomenon occurs.
5. Qualitative data is **rich and diverse in nature**. Multiple interpretations can be drawn from it, as it is collected from various sources and perspectives. **For example**, to understand the

characteristics of a teacher, you might collect data from the teacher, students, and head teachers. This multiplicity of sources enriches the data and provides depth.

6. Additionally, qualitative data allows for **flexibility** and multiple meanings. It can be interpreted in different ways, and each interpretation can yield new insights. The analysis process for qualitative data is iterative, meaning it is revisited multiple times, and each encounter may reveal new aspects or perspectives.

It is crucial to bear these characteristics in mind during the analysis and interpretation of qualitative data. Remember that qualitative data, such as narratives or visuals, offers unique possibilities for interpretation and understanding.

Lecture 41: Collecting and Analyzing Qualitative Data -I

Topic 176: Validity in Qualitative Data-I

When discussing qualitative data, the concepts of validity and reliability are often approached differently compared to quantitative data. However, they still hold value in ensuring the **trustworthiness** and **credibility** of qualitative research.

Validity in qualitative data refers to the extent to which the data accurately represents or captures the phenomenon being studied. Since qualitative research deals with complex and multi-layered phenomena, ensuring validity is a relatively complex task. It involves examining the credibility of the data and whether it aligns with people's common sense and general observations.

Credibility

One approach to enhancing credibility is through participant feedback. After conducting an interview or collecting data, researchers can share their interpretations or findings with the participants and ask for their input. This process helps verify whether the researchers' understanding aligns with the participants' intended meanings.

Another method to ensure validity is spending more time in the field to gain a deeper understanding of the research context. By immersing oneself in the environment and building rapport with participants, the artificiality of the researcher's presence diminishes, allowing for a more accurate representation of the participants' behaviors and experiences.

Triangulation is another technique used to enhance the validity of qualitative data. It involves collecting information from multiple sources or perspectives on the same topic. For example, when examining the teaching characteristics of a teacher, researchers can gather insights from the teacher, students, and headteachers. By comparing and cross-referencing these different perspectives, a more comprehensive understanding can be obtained.

Transferability

Transferability is another aspect related to validity in qualitative research. Unlike generalizability in quantitative research, transferability refers to the extent to which the findings can be applicable or relevant to similar contexts or settings. To enhance transferability, researchers should provide detailed descriptions of the research settings and methods, allowing readers to assess the applicability of the findings in other situations.

In summary, validity in qualitative data involves ensuring the credibility and trustworthiness of the research findings. This can be achieved through participant feedback, spending adequate time in the field, triangulation of data sources, and providing detailed descriptions for transferability. These approaches contribute to establishing the validity of qualitative research in capturing and representing the complexities of the phenomena under investigation.

Topic 177: Validity in Qualitative Data-II

Module 177: Continuing the discussion on the validity of qualitative data, let's explore the concept of conformability.

Conformability

Conformability relates to objectivity, ensuring that the researcher's biases do not influence the interpretation of the data. As a researcher, it is essential to be aware of our preconceptions and biases that may shape our understanding of the data. By engaging in multiple interpretations and examining the data from different angles, we can mitigate the impact of biases and enhance conformability. This process involves self-reflection, acknowledging and addressing our biases, and striving for an objective interpretation of the data. By achieving conformability, the validity of the data and the subsequent analysis is improved.

Rigorous data analysis:

Additionally, data rigor is another important characteristic in qualitative data analysis. While this term does not imply that other types of data analysis are not regressive, it refers to the need for rigorous coding procedures and the establishment of themes. The coding process should be well-defined, allowing for transparency and preventing any biases from influencing the analysis. By ensuring a robust coding mechanism, qualitative analysis maintains its integrity and validity.

Transparency is another crucial aspect of qualitative data analysis. It is essential to document and multiply the processes involved in analysis, ensuring that findings directly address the phenomena for which the data was collected. By avoiding the influence of extraneous factors, the validity of qualitative data is upheld, and the analysis remains focused on the intended objectives.

Saturation

Saturation is a key consideration in qualitative research. It involves collecting a sufficient amount of information to the point where new data no longer significantly contributes to the understanding of the topic. Saturation indicates that the researcher has obtained a comprehensive range of perspectives and information, ensuring the credibility of the data and the validity of the findings. While achieving 100% saturation may be challenging or even impossible, the aim is to reach a point where new data no longer substantially alters the emerging themes and understanding of the phenomenon under investigation.

Topic 178: Reliability in Qualitative Data

In this module, we will discuss the reliability of qualitative data, which refers to its **trustworthiness** and **consistency**. Ensuring reliability involves several steps and considerations.

Consistency in data collection

Firstly, consistency in data collection is crucial. Regardless of changes in settings, data sources, or the point in time when data is collected, the information obtained should exhibit similar characteristics as the initial data. This means that the collected information should be relatively independent of the source, setting, or time. By maintaining consistency in data collection despite these variations, the reliability of the data is enhanced.

Pilot testing

Pilot testing is another technique to ensure reliability. It involves using research instruments and collecting data from different sets of sources. By examining the repeatability of the data collection process, researchers can refine their sources, tools, and core constructs. Pilot testing helps identify and eliminate any attributes or points in the data generation process that hinder repeatability. The revised form of the research tool, informed by pilot testing, will be more reliable and repeatable.

Training and calibration

Biases of the individuals collecting the data can introduce errors that affect reliability. To mitigate this, training and calibration are employed. Training individuals involved in data collection on the basic constructs and indicators helps align their understanding and promotes repeatability. Calibration involves systematically comparing their understanding of constructs and ensuring it aligns with the intended meaning. Through training and calibration, the background and perspective of data collection remain consistent, enhancing reliability.

Inter-coder reliability

Inter-coder reliability is crucial during the coding and theme-drawing process. It ensures that multiple coders involved in the analysis reach a consensus and code the data consistently. Independent coding followed by discussions and consensus-building increases the reliability of the information. Techniques such as **Cohen's kappa coefficient** or percentage agreement can be utilized to assess inter-coder reliability.

Member checking

Another method to reinforce reliability is member checking. It involves presenting the findings or interpretations of the collected information to the participants and verifying if it accurately represents their ideas or experiences. If discrepancies arise, it indicates potential inaccuracies in the data interpretation process.

Triangulation

Triangulation is a technique used to enhance reliability by collecting data on a phenomenon from multiple perspectives or different stakeholders. By gathering information from various sources, researchers can validate and corroborate their findings, increasing the credibility and reliability of the data.

Regular reflection and reflexivity

Regularly engaging in reflection or reflexivity is important in qualitative research. Given the subjective nature of the data, it is possible for researchers to misinterpret information due to their own biases. By taking breaks, revisiting the data, and critically evaluating if personal opinions or choices have influenced the interpretation, researchers can eliminate biases and improve the reliability of the data.

Topic 179: Types of Qualitative Data: Observation

Qualitative data can be collected through various methods, but in this module, we will focus on three major types: **observation**, **interviews**, and **examination of records or archives**.

Observation:

Observation involves directly observing and documenting behaviors, interactions, and phenomena in their natural settings. There are different forms of observation, including:

Participant Observation: The researcher actively participates in the setting or group being observed, immersing themselves in the context to gain an in-depth understanding. The researcher may take field notes, record conversations, or use audiovisual equipment to capture the observations.

Non-Participant Observation: In non-participant observation, the researcher observes from a distance without directly interacting with or being noticed by the individuals being observed. This method may involve the use of hidden cameras or setting up observation points in a separate location, such as a studio or room. The purpose is to observe the natural behaviors and interactions of the participants without intruding or making them aware of the researcher's presence. It allows for unobtrusive data collection and minimizes the potential for influencing the participants' behaviors.

Structured Observation: In structured observation, the researcher defines specific variables or behaviors of interest in advance. The observations are usually systematic and may involve using a predefined scale or checklist to record the presence or frequency of certain behaviors. This structured format provides a specific framework for data collection and allows for easier analysis. However, it is important to note that this approach lacks the richness and depth of unstructured, open-ended narratives.

Un- Structured Observation: Un-structured observation is where the teacher freely expresses their thoughts and experiences, offer a more open-ended and comprehensive perspective. The observer can document these narratives in writing, capturing a wealth of qualitative data. However, this approach poses challenges during the analysis phase. Each observer may have their own biases and interpretations, making it difficult to identify common themes or points of convergence across the data.

Lecture 42: Collecting and Analyzing Qualitative Data -II

Topic 180: Types of Qualitative Data: Interviews

In this module, we will discuss various types of interviews as a method for collecting qualitative data. Interviews are widely used to gather insights, perspectives, beliefs, and experiences from individuals related to a particular topic of interest. There are different formats of interviews:

Structured Interviews:

In structured interviews, the researcher prepares a predetermined list of questions in advance. These questions are typically closed-ended, with fixed-choice answers. The respondent selects an answer and may provide further explanation or clarification. Structured interviews allow for standardized data collection and facilitate easy comparison between participants.

Semi-Structured Interviews:

Semi-structured interviews involve a core set of pre-determined questions, often open-ended and designed to elicit narrative and descriptive responses. The interviewer may follow up with additional questions based on the respondent's answers, allowing for flexibility and in-depth exploration. Semi-structured interviews strike a balance between predetermined questions and the spontaneity of conversation.

Unstructured Interviews:

Another form of interview is the unstructured interview, where no predetermined set of core questions is prepared. Instead, the interviewer begins with a general topic or opening statement and allows the conversation to unfold naturally. The direction and focus of the interview are not predetermined but rather determined by the flow of the conversation. Conducting unstructured interviews requires a high

level of skill. Without proper guidance, there is a risk that the conversation may veer off in directions unrelated to the research focus. Unless the interviewer possesses expertise similar to that of the interviewee, it can be challenging to steer the interview towards the intended research objectives. Lack of familiarity with the interviewee's field or background can pose a significant disadvantage. Analysis of unstructured interviews tends to be more expansive compared to structured interviews.

Group Interviews (Focus Groups):

Additionally, group interviews, such as focus groups, are another form of gathering interview data. In a focus group, multiple individuals with diverse perspectives relevant to the research topic are brought together, and specific questions or topics are discussed. The interactive nature of group interviews provides valuable insights and multiple perspectives on the subject under study such as teachers, parents, students, or administrators, are gathered to discuss specific questions or topics. The group dynamics and interactions generate valuable insights and multiple perspectives on the subject of study.

Topic 181: Types of Qualitative Data- Examining Records

In this module, we will discuss records as a valuable source of qualitative data in research. Records, in the context of qualitative research, refer to existing data that does not require active data collection, unlike observations or interviews. Instead, researchers systematically review and analyze data from various sources. It is essential to identify relevant sources, determine the data's suitability for the research setup, and decide which portions are needed based on the research objectives.

Records can take various forms, including text, visuals, audio, and video.

One important type of record is **archival data**, which consists of historical information preserved over time. Archival data can include documents, visuals, audio recordings, and other artifacts that provide insights into a specific period or historical context. These archives are often found in major libraries and play a significant role in historical studies.

Government documents are another valuable source of records. These can include daily government letters, policies, parliamentary proceedings, and ministry proceedings, depending on the country's specific context.

Additionally, media sources, including **social media platforms** like Twitter, Facebook, and blogs, can contribute to the record. **Personal diaries**, letters, emails, and even WhatsApp group discussions can serve as sources of data.

Legal records encompass **court proceedings**, decisions, and legal documents. **Educational records** involve policies, implementation reports, and research reports on various educational issues, such as literacy rates or out-of-school children. **Cultural artifacts** found in museums and creative works can also be used as records for research. Meteorological departments provide continuous records of **weather data**, including temperature, rainfall, and seasonal changes.

The list of potential sources of records is extensive and not exhaustive. Researchers should explore and identify records relevant to their research topic within social science and educational contexts. By utilizing these records, researchers can access a rich and valuable source of qualitative data.

Topic: 182: Qualitative Data Analysis- Coding

In this module, we will discuss the purpose and process of qualitative data analysis. Qualitative data analysis involves organizing and making sense of text data in a systematic manner.

1. **Data Reduction:** The initial step in this process is coding, which involves categorizing the data into different codes or categories. The primary purposes of qualitative data analysis are data reduction and organization.

Data reduction involves transforming unstructured text data into a more structured format.

2. **Data organization:** By assigning codes to commonalities and similarities found in the data, the raw and unorganized data is organized into a relatively structured form. Codes are allocated to different parts of the data, which helps in categorizing and structuring the information. This process may involve multiple levels of coding, with codes leading to the emergence of themes, and themes further divided into categories. This multi-layered organization helps in synthesizing and structuring the data.
3. **Patterns identification:** After organizing the data, the next step is to identify patterns within it. By examining similarities and common points in the responses, researchers can identify patterns and recurring themes. These patterns are then compared and matched with existing literature to gain a deeper understanding and interpretation of the data. This process allows

researchers to explore variations in people's opinions while also identifying commonalities and recurring themes.

4. **Interpretation and analysis:** Qualitative data analysis also involves interpretation and analysis of the data. Once the data has been categorized and organized, researchers interpret the findings and draw meaning from them. This process helps in checking for congruence between different components of the same data set and understanding similarities or differences within the data. It allows researchers to explain the variations in the data under specific categories and provides a clearer and more nuanced understanding of the subject matter.
5. **Comparative Analysis:** It's important to note that qualitative data analysis does not aim to discard or compromise the variations in people's opinions. Instead, it seeks to bring together similar variations under one category or theme while preserving the richness of the original data. The analysis is performed in a way that maintains the element of subjectivity separate from the interpretation process.

In summary, qualitative data analysis involves organizing, categorizing, and interpreting text data to identify patterns and gain a deeper understanding of the subject matter. The process aims to synthesize knowledge and explain variations while maintaining the richness and diversity of people's opinions.

Lecture 43: Collecting and Analyzing Qualitative Data -III

Topic 183: Qualitative Data Analysis- Steps in Qualitative Data Analysis

In qualitative data analysis, coding involves assigning labels or codes to different segments of data to identify patterns, themes, and relationships within the data (categories).

1. The first step mentioned is **open coding**, where the researcher reads through the data transcriptions and assigns initial codes to different sections or segments based on the meaning found in the text. This process helps in creating a set of individualized codes for the data.
2. The next step mentioned is **axial coding** or excel coding. During this step, the researcher looks for connections or relationships between the initially assigned codes. They may discover that certain codes are similar or related to each other, leading to the creation of broader categories or themes. This process involves revisiting and refining the codes to ensure clarity and distinctiveness.
3. The final step mentioned is **selective coding** (categories). In this step, the researcher identifies core categories or themes that emerge from the analysis. These categories are often aligned

with the research questions or address specific aspects of the research. The researcher connects the themes to create a coherent and meaningful narrative that can be linked to the theoretical framework of the research.

Overall, the process described follows a grounded theory approach, where the analysis starts with raw data and progresses through different levels of coding to generate themes and categories that capture the essence of the data.

Topic 184: Qualitative Data Analysis- Coding: Analyzing Qualitative Data

In qualitative data analysis, there are different approaches to synthesizing data, and I will discuss three commonly used methods. These methods are not necessarily sequential but represent separate ways of coding and analyzing data. The coding process, however, remains the same across these approaches.

Thematic coding: As we discussed earlier, in the coding process, you review the transcript or data and assign codes to different sections. While coding, you also identify related codes and group them together to form themes. It is preferable to let themes emerge during the coding process rather than pre-determining them. Qualitative research does not start with assumptions but instead generates assumptions based on the data.

Pattern coding is an approach where you assume that there will be similarities and distinctions in the data provided by research participants. You look for patterns, commonalities, and convergences in the data segments. This approach involves finding similarities across different parts of the data and explaining the underlying basis for these commonalities while acknowledging variations within them.

Content analysis is another approach that involves coding data with pre-defined categories. This method is typically used when the data comes from documents, media content, or text-based sources. Unlike qualitative research with human subjects, the data in content analysis is not generated through direct interactions. The categories in content analysis are pre-defined based on the nature of the data. However, it is important to note that if the data does not fit into the pre-defined categories, new categories, themes, or codes can be created. It is crucial not to force-fit the data into pre-existing categories but to allow for the emergence of new categories based on the information present.

In content analysis, the coding process starts with pre-defined codes, themes, and categories. However, if necessary, new codes and themes can be created based on the data. Flexibility is important to ensure that the analysis accurately captures the nuances and variations within the data.

Topic 185: Qualitative Data Analysis- Generating Themes

The purpose of generating themes and categories in qualitative data analysis is to identify recurring ideas and meaningful concepts within the data. This process involves synthesizing the information and organizing it into themes and categories.

To generate themes, you need to go through a coding process. Coding involves systematically reviewing the data, segmenting it, and assigning codes to different sections based on their content. This coding process helps in identifying patterns, relationships, and concepts within the data.

The following steps can be followed to draw themes effectively:

1. **Data Familiarization:** Before starting the analysis, read the data multiple times without making any assumptions about the codes or themes. This step helps in understanding the style, sequence, and patterns of the data and ensures that your mind remains neutral.
2. **Initial Coding:** Once you are familiar with the data, start the process of initial coding. Review each interview or data segment and assign relevant codes to different sections. This step involves identifying key ideas, concepts, or recurring elements in the data.
3. **Code review and grouping:** After completing the initial coding, set aside the data for a few days. Then, review the coded segments and assess the appropriateness of the assigned codes. If necessary, revise the codes to ensure they accurately represent the data. Look for redundancies or overlapping codes and consider consolidating or refining them.
4. **Development of higher-level themes:** After the review process, examine the relationships between the codes and group them accordingly. Identify commonalities, connections, and patterns within the codes to form higher-level themes. Themes should be distinct, self-contained, and not overlapping with each other. Consider the context and content of the codes when developing themes.
5. **Naming the themes:** Assign descriptive and meaningful names to each theme which should convey the essence of what the theme represents. Well-chosen name makes it easier to communicate your findings.

6. **Contextualization:** Contextualization involves adding depth to your analysis by connecting the themes to the specific context in which the data was collected. Ensure that the codes within each theme are contextually relevant and meaningful.
7. **Refine and consolidate** themes if needed, based on their alignment with the data and the overall research objectives.
8. **Member checking:** Share your preliminary themes with the participants or stakeholders who provided the data. Seek their feedback and input on the themes and their connections. Consider their perspectives and refine the themes accordingly.
9. **Peer review and consultation:** You can also involve experts or colleagues in peer review to validate and authenticate the themes.
10. **Finalization of themes:** After incorporating feedback and conducting peer review, finalize the themes. Analyze and interpret each theme in-depth, aligning them with your understanding and interpretation of the data.
11. **Reporting:** Report the entire process, including the generated themes, their interpretations, and their alignment with the research objectives.

By following these steps, you can generate themes accurately and ensure that your analysis is comprehensive and meaningful. It is important to use clear and precise language to describe and define the themes, maintaining their relevance to the data and research context.

Topic 186: What is Research Ethics?

Research ethics is a crucial aspect of conducting research in an ethically correct manner. It encompasses principles, guidelines, and standards that govern the research process, ensuring that it is conducted in an ethical and responsible manner. While research ethics might not have received much attention in our context, there are forums in universities that prioritize and ensure research ethics.

In this module, we will discuss nine fundamental aspects of research ethics.

1. **Informed consent:** The first topic is informed consent. When conducting research, it is essential to clearly communicate the purpose of the study, what will be gained from it, and assure participants that their involvement will not cause harm. Researchers should explain the objectives, procedures, potential risks, and benefits of the research to participants. Individuals should have the freedom to participate willingly and the right to withdraw from the study at

any point if they feel uncomfortable or perceive any risks. It is crucial to document participants' consent in writing, outlining their rights and responsibilities, rather than relying solely on verbal agreements. This consent form should be included in the research report.

2. **Protection of participants:** For instance, if your research involves classroom observation and video recording, and the footage includes individuals, such as students in a girls' school, their consent is necessary before recording. Several other scenarios exist where research activities can impact individuals physically, psychologically, or emotionally. In such cases, researchers must minimize any potential discomfort and obtain necessary permissions, ensuring that participants' autonomy and will are respected. Researchers should also protect the identities and rights of participants, and this responsibility should be clearly communicated to them.
3. **Confidentiality and privacy:** Another essential aspect of research ethics is ensuring confidentiality and privacy. Researchers must handle collected data with care, protecting its confidentiality and the privacy of participants. Consider the scenario where you collect income data from individuals for research purposes to understand the relationship between income and a child's education. It would be inappropriate to use that data for other purposes without the participants' consent, such as sharing it with the tax authorities or another institution. Therefore, it is crucial to maintain data confidentiality and use it responsibly.
4. **Data management and sharing:** Researchers should define how data will be managed, stored, and secured. Participants should be informed about data protection measures and how their data will be used. If data usage extends beyond the initial research purpose, boundaries must be established, and participants should be informed and give consent for such usage.

It is essential to document these considerations for respondents or study subjects and for organizational purposes. Putting these guidelines in writing ensures that everyone involved understands that data provided to the organization is voluntary, will only be used for the stated purpose, and will not be utilized without permission. Clear communication of these principles helps build trust and reassures individuals about the responsible use of their data.

Lecture 44: Research Ethics

Topic 187: What is Research Ethics? General Ethical Practices

5. Incentives: One practice you mentioned is the use of incentives to encourage data collection, such as offering gifts or financial rewards. While this practice is becoming more common, it's crucial to ensure that these incentives do not unduly influence participants or change their perspectives. If incentives are provided, it should be done in a way that does not compromise the integrity of the data or the participants' autonomy. It's important to avoid situations where individuals provide data solely for the sake of receiving incentives and to maintain the ethical standards of the research.

5. Honesty and integrity are vital throughout the research process, including data analysis and tool development. Researchers should refrain from manipulating or fabricating data to achieve desired results. Researchers should approach data analysis with objectivity, avoiding preconceived notions or biases. Maintaining integrity and adhering to ethical practices is essential for conducting reliable and trustworthy research.

6. Conflicts of interest should be acknowledged and managed appropriately. Researchers must disclose any conflicts of interest they may have and explain how these conflicts have been addressed to ensure they do not impact the research process or results. Transparency in reporting and discussing conflicts of interest is crucial for maintaining research integrity.

7. Institutional review boards: Ethics committees or review boards play a significant role in research ethics. These committees, usually formed at the department, faculty, or university level, provide oversight and ensure that research projects meet ethical standards. However, it's important to note that the functioning of such committees may vary, and there may be challenges in their proper implementation.

8. Global ethical standards: In the case of globally relevant research, especially multi-country or multi-institutional studies, researchers must adhere to ethical standards specific to their own institution as well as those of other involved institutions. Establishing a consortium or forum comprising representatives from these institutions can help ensure compliance with global ethical standards.

These are some of the important steps and considerations in research ethics. However, research ethics encompasses a broad range of topics and principles. If you want to delve deeper into this subject, there are extensive resources available that cover various ethical aspects of research. While this course focuses on research methods rather than ethics specifically, it is essential to be aware of and adhere to ethical standards in your research endeavors.

Topic 188: Purpose of Research Proposal

Module 188 focuses on the structure and elements of a research proposal. The purpose of a research proposal is to present your research idea to experts or stakeholders in a structured manner. It serves as a document that outlines your research plan and convinces others of the importance and feasibility of your research.

When writing a research proposal, it's crucial to consider its structure and organization. The proposal should effectively communicate your point and demonstrate your capability to conduct the research. You need to convey to the readers that you have a clear understanding of the research topic and its relevance in your field of study.

The research proposal should include a description of your research methodology, which outlines how you will collect and analyze data. This section presents your action plan, highlighting the specific methods and techniques you intend to use. It's important to provide a well-thought-out plan that demonstrates your research competence.

The proposal should also address the assessment of your capability to conduct the research. Readers should be able to evaluate whether you possess the necessary skills, knowledge, and resources to carry out the proposed research successfully. Additionally, the expected outcomes of the research should be clearly stated, along with how they will contribute to the existing body of knowledge.

It's important to understand that research proposals are typically not rejected solely based on the topic. While the topic's suitability may be a consideration, the primary focus is on the researcher's capability, skills, and the likelihood of achieving the stated outcomes. Make sure that your research proposal clearly communicates your abilities and the potential impact of your research.

Throughout the writing process, it's essential to maintain clarity and professionalism in your language. Ensure that your document effectively conveys your intentions, methodology, and expected outcomes.

By following a well-structured research proposal format and addressing the key elements discussed above, you can increase the chances of your proposal being approved and gaining support for your research endeavor.

Lecture 45: Research Proposal and Report Writing

Topic 189: Research Proposal; Rationale and literature review

Module 189 focuses on the elements of a research proposal and discusses the three main sections typically found in such a document. The three main sections of a research proposal are the **rationale and literature review**, **research design and methodology**, and **work plan and timeline**.

Rationale and literature review: The first section, the rationale and literature review, consists of five major components. The first component is the **title of the study**, which should be carefully crafted to reflect the research topic. The second component is the **introduction**, also known as the background and context section. In this section, you provide a brief overview of the topic, explain its significance, and present the research problem or question.

The research problem is an expanded form of the research title and should clearly state the **objectives or research questions**. If the research questions are inferential in nature, hypotheses should be formulated. If the nature of the questions is descriptive, hypotheses are not necessary. It's important to link the research questions to the objectives so that readers can understand the relationship between them.

The next component is the **literature review**, which involves explaining the meaning and usage of variables in your research. You also summarize the current state of research on the topic, highlighting the findings of contemporary studies. Additionally, you identify any gaps in the existing literature.

The conceptual framework is another important aspect of this section. Here, you outline the sequence of activities and present the overall conceptualization of your research. The theoretical framework provides a theoretical justification for the phenomenon being studied.

Topic 190: Elements of a Research Proposal: Design & Methodology

Module 190 focuses on the second section of a research proposal, which is the design and methodology section.

Design and methodology

In this section, you need to explain the research design you have chosen and how it is appropriate for your research topic. You should also provide details about the population and sample (or participants

and context) of your study, as well as the data collection tools and their framework, data analysis methods, ethical considerations, and a timeline for conducting the research.

First, you should describe the research design you have selected, such as quantitative, qualitative, or mixed methods. Explain why this design is appropriate for your research topic and how it will be used in your study. This section should demonstrate that you have carefully considered the design and its alignment with your research objectives.

Next, address the population and sample (or participants and context) of your study. If it's a quantitative study, describe the population and provide details about the sample, including the estimated number of participants and how you will select them. If it's a qualitative study, explain the context of the study and describe how you will select participants, highlighting their characteristics and why they are suitable for your research.

Discuss the data collection tools you will use, such as questionnaires, observation schedules, checklists, or interviews. Describe the structure and framework of each tool, and explain how they will be used to collect data. Provide details on how the data generated by each tool will be obtained and analyzed. This section should demonstrate your understanding of data analysis and your ability to apply appropriate methods.

Address ethical considerations in your research. Ensure that you are not infringing on anyone's rights or beliefs and that you are not involved in data manipulation. Refer back to the ethical considerations or research ethics section that you have previously discussed and outline the steps you will take to ensure ethical conduct throughout your research.

Include a timeline that outlines the different stages of your research and the estimated time required for each stage. Typically, timelines are presented in weeks or months. Clearly indicate when your research will start and when it will be completed.

Finally, include a short section that highlights the potential significance of your study. Explain how the research results will contribute to your field of study and add to existing knowledge. Emphasize that your research aims to make incremental contributions rather than completely overturning existing knowledge. By articulating what was previously unknown and what will be known after your research, you can demonstrate the significance of your study.

Topic 191: Elements of a Research Proposal: work plan, timeline, and budget.

Module 191 focuses on the last section of a research proposal, which includes the work plan, timeline, and budget.

Work plan, timeline, and budget

In this section, you need to provide information about your reference list, funding considerations, relevant documents, and the timeline of your research.

Start by addressing your **reference list**. When writing your research proposal, it is important to consult various documents and sources. You should provide a list of these references at the end of your proposal. In social sciences, the commonly used referencing style is APS (American Psychological Association). Follow the guidelines provided by the APS style to create your reference list. Ensure that your references are authentic, current, and directly related to your research topic. This demonstrates the extent of your reading and understanding of the subject.

Next, discuss the **funding considerations**. In postgraduate studies, research often involves funding from external organizations or sponsors. If your research is being sponsored by your university or another organization, you need to estimate the expenses associated with your research. This includes budget distribution for data collection, software for data analysis, availability of research publications (such as subscriptions), equipment, travel, and other relevant expenses. Providing an estimate of the budget will support your research proposal. However, please note that at the undergraduate or B.Ed level, or even after MPhil or Ph.D., this section may not be applicable as funding sources may be limited. Include this section only if it is relevant to your research.

Additionally, there may be certain documents that are not directly included in the main body of your proposal but are relevant to your research. These documents can provide better understanding or support for your proposal. For example, if you have obtained permission to use a research tool from another researcher, you can attach the email or letter granting permission as an **appendix**. This allows readers who are interested in further details to access the relevant documents.

Include the **timeline** of your research, which was briefly mentioned in the methodology section. Specify when your research will start and outline the different milestones and timelines for each stage of your research. For example, mention when the literature review will be completed, when data

collection will start, the duration of data collection, time required for analysis, and report writing. Provide a clear order of completion for each milestone. This helps readers understand the timeline and progression of your research.

By addressing these components in the work plan, timeline, and budget section, your research proposal will provide a comprehensive overview of your research plan, including the resources, timeline, and estimated expenses needed for successful completion.

Topic 192: Research Report: Elements of a Research Report

In this module, we will discuss the structure of a research report, commonly referred to as a thesis, and its various elements. It's important to note that the specific structure and chapterization may vary depending on the requirements of your institution or university. However, I will provide a general outline that is commonly followed.

1. Preliminary Pages:

Title Page: This page includes the title of your research, your name, the name of the institution where you conducted the research, and the degree for which you are submitting the thesis.

Abstract: On the next page, provide a concise summary of your research. The abstract should highlight the key objectives, methodology, findings, and conclusions of your study.

Table of Contents: Create a table of contents that lists the chapters and sections of your thesis along with their corresponding page numbers.

List of Figures and List of Tables: If your research includes figures (e.g., graphs, charts, illustrations) or tables (e.g., data tables), list them with their respective page numbers.

List of Abbreviations: If you have used any abbreviations in your thesis, provide a list of their expansions and meanings.

Acknowledgments: In this section, acknowledge the individuals, organizations, or institutions that have supported or contributed to your research.

2. **Introduction** (Chapter 1): In the introduction chapter, provide the rationale for your study, clearly state the research objectives, present the research questions or hypotheses, discuss the significance of the study, and provide an overview of the research structure.
3. **Literature Review** (Chapter 2): In the literature review chapter, begin by defining and operationalizing the variables discussed in your study. Then, describe the theoretical framework that underpins your research. Next, synthesize the existing literature and contemporary research related to your topic. Finally, apply the conceptual framework to your study.
4. **Methodology** (Chapter 3): In the methodology chapter, describe the research design, including details about the population and sample. If it is a qualitative study, provide a description of the study context and participants. Discuss the data collection tools and their development or adaptation. Explain the research procedures and data analysis techniques that you employed.
5. **Results, Data Analysis, Conclusions, and Recommendations:** This section can be a separate chapter (Chapter 4) or combined with the methodology chapter, depending on the structure preferred by your institution. Present your findings based on the data analysis, organized according to your research questions or objectives. Interpret the results, draw conclusions, and provide recommendations based on your findings.
6. **References:** Include a list of all the sources you cited in your thesis. Follow the citation style specified by your institution (e.g., APA, MLA).
7. **Appendices:** If you have additional material that supports your research (e.g., research tools, sampling tables), include them in the appendices. If needed, separate appendices can be included for different documents. Make sure to refer to these appendices within the main body of your thesis.
8. **Autobiography or Biographical Information:** Provide a brief biography, including your educational background and any relevant technical information. If your research was funded or supported by an organization, mention the funding details in this section.

Remember to adhere to the guidelines provided by your institution while structuring your thesis.

While there is no specific page limit, it is generally recommended to present your research in a concise and readable manner. Aim for clarity and ensure that your thesis effectively addresses your research objectives.

